

E50598750_1

The equation $x^5 + x + y^3 + 4y + 1 = 0$ has no integer solutions

Size $h = 51$, length $l = 12$

Theorem 1. *The equation*

$$x^5 + x + y^3 + 4y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 47 auxiliary polynomials $H_0, \dots, H_{46}$, 5 of which are constants, and 2 graphs on these vertices with 1 and 204 edges, respectively; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^5 + x + y^3 + 4y + 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$H_0 = x^2 - 2x - 2y - 1,$$

$$H_1 = 2,$$

$$H_2 = -x - y,$$

$$H_3 = -y - 1,$$

$$H_4 = -y + 1,$$

$$H_5 = x - y,$$

$$H_6 = -2x - y,$$

$$H_7 = -x - y - 1,$$

$$H_8 = -x - y + 1,$$

$$H_9 = -y - 2,$$

$$H_{10} = x - y - 1,$$

$$H_{11} = -2x - y - 1,$$

$$H_{12} = -2x - y + 1,$$

$$H_{13} = -x - y - 2,$$

$$H_{14} = -x - y + 2,$$

$$H_{15} = x - y - 2,$$

$$H_{16} = x - y + 2,$$

$$H_{17} = 2x - y - 1,$$

$$H_{18} = 2x - y + 1,$$

$$H_{19} = -2x - y - 2,$$

$$H_{20} = x^2 + y,$$

$$H_{21} = x^2 - y,$$

$$H_{22} = x^2 - x + y,$$

$$H_{23} = x^2 + x + y,$$

$$H_{24} = x^2 - x - y,$$

$$H_{25} = x^2 - y - 1,$$

$$H_{26} = x^2 - y + 1,$$

$$H_{27} = x^2 + x - y,$$

$$H_{28} = x^2 - 2x + y,$$

$$H_{29} = x^2 - x + y - 1,$$

$$H_{30} = x^2 - x + y + 1,$$

$$H_{31} = x^2 + y - 2,$$

$$H_{32} = x^2 + y + 2,$$

$$H_{33} = x^2 + x + y + 1,$$

$$H_{34} = x^2 - 2x - y,$$

$$H_{35} = x^2 - x - y - 1,$$

$$H_{36} = x^2 - x - y + 1,$$

$$H_{37} = x^2 - y - 2,$$

$$\begin{aligned}
H_{38} &= x^2 + x - y - 1, \\
H_{39} &= x^2 + x - y + 1, \\
H_{40} &= x^2 - x - 2y, \\
H_{41} &= x^2 - 2y - 1, \\
H_{42} &= x^2 + 2y + 1, \\
H_{43} &= 1, \\
H_{44} &= -2, \\
H_{45} &= -1, \\
H_{46} &= 3.
\end{aligned}$$

There are 2 graphs on these vertices. The edges of the first graph are $\{0, 1\}$. The edges of the second graph are $\{2, 4\}, \{2, 5\}, \{2, 11\}, \{2, 14\}, \{2, 19\}, \{2, 25\}, \{2, 32\}, \{2, 35\}, \{2, 37\}, \{2, 41\}, \{3, 4\}, \{3, 10\}, \{3, 14\}, \{3, 15\}, \{3, 16\}, \{3, 17\}, \{3, 18\}, \{3, 23\}, \{3, 24\}, \{3, 26\}, \{4, 5\}, \{4, 7\}, \{4, 9\}, \{4, 11\}, \{4, 15\}, \{4, 18\}, \{4, 19\}, \{4, 20\}, \{4, 24\}, \{4, 25\}, \{4, 31\}, \{4, 34\}, \{4, 35\}, \{5, 6\}, \{5, 13\}, \{5, 14\}, \{5, 20\}, \{5, 21\}, \{5, 23\}, \{5, 26\}, \{5, 32\}, \{5, 41\}, \{6, 8\}, \{6, 9\}, \{6, 19\}, \{6, 20\}, \{6, 22\}, \{6, 23\}, \{6, 27\}, \{6, 30\}, \{6, 33\}, \{6, 38\}, \{7, 10\}, \{7, 11\}, \{7, 12\}, \{7, 25\}, \{7, 30\}, \{7, 32\}, \{7, 39\}, \{8, 13\}, \{8, 15\}, \{8, 17\}, \{8, 18\}, \{8, 23\}, \{8, 25\}, \{8, 27\}, \{8, 30\}, \{8, 34\}, \{8, 37\}, \{8, 42\}, \{9, 20\}, \{9, 23\}, \{9, 24\}, \{9, 27\}, \{9, 30\}, \{9, 35\}, \{9, 36\}, \{9, 38\}, \{9, 39\}, \{9, 41\}, \{10, 12\}, \{10, 23\}, \{10, 26\}, \{10, 27\}, \{10, 33\}, \{10, 36\}, \{10, 39\}, \{10, 40\}, \{11, 14\}, \{11, 16\}, \{11, 24\}, \{11, 25\}, \{11, 26\}, \{11, 28\}, \{11, 33\}, \{11, 36\}, \{11, 41\}, \{12, 13\}, \{12, 17\}, \{12, 23\}, \{12, 27\}, \{12, 35\}, \{12, 41\}, \{13, 26\}, \{13, 27\}, \{13, 28\}, \{13, 31\}, \{13, 35\}, \{13, 39\}, \{14, 28\}, \{14, 38\}, \{14, 39\}, \{14, 40\}, \{14, 42\}, \{15, 17\}, \{15, 26\}, \{15, 30\}, \{15, 33\}, \{15, 34\}, \{16, 17\}, \{16, 23\}, \{16, 28\}, \{16, 33\}, \{17, 20\}, \{17, 23\}, \{17, 31\}, \{17, 39\}, \{18, 22\}, \{18, 29\}, \{18, 33\}, \{18, 35\}, \{19, 21\}, \{19, 27\}, \{19, 29\}, \{19, 33\}, \{19, 38\}, \{19, 41\}, \{20, 27\}, \{20, 28\}, \{20, 30\}, \{20, 32\}, \{20, 35\}, \{20, 41\}, \{20, 42\}, \{21, 27\}, \{21, 35\}, \{21, 38\}, \{22, 33\}, \{23, 26\}, \{23, 31\}, \{23, 42\}, \{24, 25\}, \{24, 26\}, \{24, 30\}, \{24, 33\}, \{24, 34\}, \{25, 33\}, \{25, 39\}, \{26, 36\}, \{26, 40\}, \{27, 33\}, \{27, 37\}, \{28, 31\}, \{28, 39\}, \{29, 32\}, \{29, 35\}, \{29, 38\}, \{29, 39\}, \{30, 38\}, \{31, 33\}, \{31, 37\}, \{32, 33\}, \{32, 37\}, \{32, 39\}, \{32, 41\}, \{32, 42\}, \{33, 34\}, \{33, 37\}, \{33, 40\}, \{33, 41\}, \{34, 35\}, \{34, 42\}, \{35, 39\}, \{35, 40\}, \{36, 39\}, \{36, 41\}, \{37, 39\}, \{37, 41\}, \{39, 42\}, \{5, 1\}, \{7, 1\}, \{9, 44\}, \{10, 1\}, \{11, 45\}, \{12, 45\}, \{13, 1\}, \{14, 45\}, \{18, 45\}, \{19, 1\}, \{25, 45\}, \{26, 46\}, \{29, 45\}, \{31, 45\}, \{42, 46\}. For a vertex i , we write $j \sim_g i$ if $\{i, j\}$ is an edge of the g -th graph. For every non-constant vertex i and every graph g in which i has a neighbour, Table 1 gives a positive integer $m_{g,i}$ and polynomials $W_{g,i}$, $A_{g,i}$ and $B_{g,i}$ with integer coefficients such that$

$$m_{g,i}^2 \prod_{j \sim_g i} H_j - W_{g,i}^2 = A_{g,i}F + B_{g,i}H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij}F + \beta_{ij}H_i + \gamma_{ij}H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^{10} - 54x^6 - 54x^5 - 27x^2 - 54x - 283.$$

For $i \in I = \{0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(8 + t)$ and the $\varphi_i(8 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-9 - t)$ and the $\varphi_i(-9 - t)$. Hence, for $x \geq 8$ and for $x \leq -9$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-8 \leq x \leq 7$. Checking the residues of y , we find that for $x \in \{-7, -4, -2, 1, 3, 6\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x \in \{-8, -1, 0, 7\}$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x = -6$ and $x = 5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , for $x = -5$ and $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , for $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , and for $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y . Hence (1) has no solution with $-8 \leq x \leq 7$.

For a place v , put $B_v^{(g)} = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges of the g -th graph. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{1, 43, 46\}$ and $H_i < 0$ for $i \in \{44, 45\}$. The other signs are given in Table 3 for $x \geq 8$ and for $x \leq -9$. In both cases, counting the edges of each graph whose two values are negative gives $(B_\infty^{(g)})_g = (1, 1)$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 17\}$ be the set of primes dividing 2, the $m_{g,i}$, the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p^{(g)} = 1$ for every g . For an edge $\{i, j\}$ of the g -th graph such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$ of the g -th graph, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim_g i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim_g i} H_j$ is not divisible by p . By (2), $m_{g,i}^2 \Pi_i \equiv W_{g,i}^2 \pmod{p}$, and $p \nmid m_{g,i}$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p^{(g)} = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p^{(g)} = (-1)^{N_g}$, where $N_g = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges of the g -th graph. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges of every graph, then $N_g = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so $B_p^{(g)}$ is determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of the vector $(B_p^{(g)})_g$; every integer solution lies in one of them. The result is $(B_2^{(g)})_g \in \{(1, -1), (-1, 1), (-1, -1)\}$, $(B_3^{(g)})_g \in \{(1, 1)\}$, $(B_5^{(g)})_g \in \{(1, 1)\}$, $(B_7^{(g)})_g \in \{(1, 1)\}$, $(B_{11}^{(g)})_g \in \{(1, 1)\}$, $(B_{13}^{(g)})_g \in \{(1, 1)\}$, $(B_{17}^{(g)})_g \in \{(1, 1)\}$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v^{(g)} = 1$ for every g . By Step 3, the factors with $p \notin S$ are 1, so the componentwise product of the vectors $(B_p^{(g)})_g$ with $p \in S$ is $(B_\infty^{(g)})_g = (1, 1)$ by Step 2. But by Step 4, this product is $(1, -1)$, $(-1, 1)$, or $(-1, -1)$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$g = 1, i = 0, m = 48, \text{neighbours } 1$	
W	$-2x^5 - 10x^4 - 24x^3 - 32x^2 - 14x + 66$
A	$-12x^5 - 54x^4 - 196x^3 - 608x^2 + 48xy^2 + 392xy - 904x + 280y^2 + 88y - 250$
B	$8x^8 + 30x^7 + 16x^6y + 68x^6 + 92x^5y + 166x^5 - 16x^4y^2 - 56x^4y + 44x^4$ $+ 12x^3y^3 - 128x^3y^2 + 272x^3y - 982x^3 + 46x^2y^3 - 384x^2y^2$ $+ 792x^2y - 2046x^2 + 24xy^4 + 44xy^3 - 352xy^2 + 676xy - 1998x + 140y^4 - 26y^3$ $+ 448y^2 + 92y - 502$
$g = 2, i = 2, m = 1, \text{neighbours } 4, 5, 11, 14, 19, 25, 32, 35, 37, 41$	
W	$14x^4 + 2x^3 - 12x^2 + 26x - 6$
A	$4x^9y - 4x^9 + 4x^8y^2 - 10x^8y + 6x^8 - 3x^7y^3 - 14x^7y^2 - 3x^7y$ $+ 20x^7 - 4x^6y^4 - 9x^6y^3 + 24x^6y^2 + 9x^6y - 20x^6 - x^5y^5$ $+ 16x^5y^4 + 14x^5y^3 + 18x^5y^2 - 25x^5y - 22x^5 + 17x^4y^5 - x^4y^4$ $- 37x^4y^3 - 37x^4y^2 + 66x^4y - 8x^4 + 4x^3y^6 - 25x^3y^5 + 10x^3y^4$ $+ 25x^3y^3 + 74x^3y^2 - 22x^3y - 262x^3 - 16x^2y^6 + 23x^2y^5$ $+ 101x^2y^4 + 20x^2y^3 - 141x^2y^2 - 67x^2y + 24x^2 + 3xy^6 - 56xy^5$ $- 123xy^4 - 235xy^3 - 92xy^2 + 307xy + 528x + y^8 - 8y^7 - 56y^6 - 177y^5 + 32y^4$ $- 235y^3 + 123y^2 + 292y - 36$
B	$15x^4y^7 + 14x^4y^6 + 7x^4y^5 - 32x^4y^4 - 477x^4y^3 - 32x^4y^2$ $+ 431x^4y + 690x^4 + 4x^3y^8 - 28x^3y^7 - 72x^3y^6 - 220x^3y^5$ $+ 180x^3y^4 - 298x^3y^3 + 26x^3y^2 - 294x^3y - 30x^3 - 20x^2y^8$ $- 10x^2y^7 + 31x^2y^6 + 125x^2y^5 + 836x^2y^4 + 68x^2y^3 - 85x^2y^2$ $- 1207x^2y - 886x^2 - xy^9 + 19xy^8 + 39xy^7 + 142xy^6 - 361xy^5 + 83xy^4$ $- 723xy^3 + 448xy^2 + 1002xy + 1316x + y^{10} - 6y^9 - 39y^8 - 186y^7 - 228y^6$ $- 1131y^5 - 55y^4 - 588y^3 + 349y^2 + 1371y + 164$
$g = 2, i = 3, m = 1, \text{neighbours } 4, 10, 14, 15, 16, 17, 18, 23, 24, 26$	
W	$4x^4 + 4x^3 + 28x - 32$
A	$4x^7y - 4x^7 - 12x^6y^2 + 12x^6 + 9x^5y^3 + 27x^5y^2 - 45x^5y + 9x^5$ $+ 6x^4y^4 - 45x^4y^3 + 37x^4y^2 + 53x^4y - 51x^4 - 12x^3y^5$ $+ 34x^3y^4 - 4x^3y^3 - 52x^3y^2 - 4x^3y + 22x^3 + 6x^2y^6 - 21x^2y^5$ $+ 26x^2y^4 - 44x^2y^3 + 24x^2y^2 + 17x^2y - 40x^2 - xy^7 + 14xy^6$ $- 46xy^5 + 53xy^4 + 124xy^3 - 213xy^2 + 107xy - 54x - 6y^7 + 24y^6 - 23y^5 - 126y^4$ $+ 31y^3 + 249y^2 - 242y - 131$
B	$-x^4y^7 + 9x^4y^6 - 44x^4y^5 + 95x^4y^4 - 262x^4y^3 + 409x^4y^2$ $- 199x^4y - 31x^4 - 8x^3y^7 + 43x^3y^6 - 124x^3y^5 + 190x^3y^4$ $- 29x^3y^3 - 349x^3y^2 + 347x^3y - 262x^3 + 5x^2y^8 - 23x^2y^7$ $+ 62x^2y^6 - 138x^2y^5 + 156x^2y^4 - 251x^2y^3 + 477x^2y^2 - 682x^2y$ $+ 682x^2 - xy^9 + 11xy^8 - 46xy^7 + 146xy^6 - 216xy^5 + 200xy^4 + 327xy^3$ $- 1110xy^2 + 1610xy - 1977x - 5y^9 + 26y^8 - 77y^7 + 59y^6 - 99y^5 - 206y^4 - 24y^3$ $+ 932y^2 - 1659y + 893$
$g = 2, i = 4, m = 1, \text{neighbours } 2, 3, 5, 7, 9, 11, 15, 18, 19, 20, 24, 25, 31, 34, 35$	
W	$-72x^4 + 192x^3 + 48x^2 - 792x - 432$
A	$8x^{14}y^2 + 24x^{14}y + 16x^{14} + 4x^{13}y^3 - 12x^{13}y^2 - 64x^{13}y - 48x^{13}$ $- 18x^{12}y^4 - 110x^{12}y^3 - 250x^{12}y^2 - 250x^{12}y - 92x^{12} - 9x^{11}y^5$ $+ x^{11}y^4 + 123x^{11}y^3 + 387x^{11}y^2 + 594x^{11}y + 320x^{11} + 12x^{10}y^6$ $+ 135x^{10}y^5 + 509x^{10}y^4 + 1127x^{10}y^3 + 1587x^{10}y^2 + 1078x^{10}y$ $+ 232x^{10} + 6x^9y^7 + 21x^9y^6 - 42x^9y^5 - 328x^9y^4 - 956x^9y^3$ $- 1885x^9y^2 - 2020x^9y - 820x^9 - 2x^8y^8 - 57x^8y^7 - 337x^8y^6$ $- 1145x^8y^5 - 2663x^8y^4 - 4246x^8y^3 - 4330x^8y^2 - 2220x^8y - 336x^8$ $- x^7y^9 - 12x^7y^8 - 9x^7y^7 + 36x^7y^6 + 303x^7y^5 + 1196x^7y^4$ $+ 2429x^7y^3 + 3492x^7y^2 + 2882x^7y + 892x^7 + 8x^6y^9 + 69x^6y^8$ $+ 251x^6y^7 + 968x^6y^6 + 2653x^6y^5 + 4298x^6y^4 + 4822x^6y^3$ $+ 3309x^6y^2 + 1014x^6y + 104x^6 + 2x^5y^{10} - 11x^5y^9 - 100x^5y^8$ $- 207x^5y^7 - 547x^5y^6 - 1905x^5y^5 - 3796x^5y^4 - 4783x^5y^3$ $- 3615x^5y^2 - 1254x^5y - 104x^5 - 8x^4y^{10} + 23x^4y^9 + 291x^4y^8$ $+ 902x^4y^7 + 1632x^4y^6 + 2694x^4y^5 + 5266x^4y^4 + 8629x^4y^3$ $+ 8987x^4y^2 + 4564x^4y + 644x^4 + 3x^3y^{11} + 65x^3y^{10} + 339x^3y^9$ $+ 1248x^3y^8 + 3198x^3y^7 + 6286x^3y^6 + 11138x^3y^5 + 15261x^3y^4$ $+ 13892x^3y^3 + 6112x^3y^2 - 34x^3y - 5620x^3 + x^2y^{12} - 4x^2y^{11}$ $- 109x^2y^{10} - 487x^2y^9 - 1998x^2y^8 - 5705x^2y^7 - 10646x^2y^6$ $- 16633x^2y^5 - 22235x^2y^4 - 24651x^2y^3 - 18489x^2y^2 - 6832x^2y$ $+ 26812x^2 - 12xy^{12} - 81xy^{11} - 329xy^{10} - 1595xy^9 - 4717xy^8 - 9064xy^7$

$$\begin{aligned}
& -14287xy^6 - 17973xy^5 - 18181xy^4 - 11271xy^3 - 2106xy^2 + 384xy - 29920x - 2y^{13} \\
& + 21y^{12} + 145y^{11} + 401y^{10} + 1473y^9 + 4919y^8 + 11645y^7 + 20647y^6 \\
& + 26171y^5 + 24268y^4 + 13942y^3 + 972y^2 - 2686y - 133020 \\
B \quad & -9x^4y^{12} + 2x^4y^{11} + 200x^4y^{10} + 1047x^4y^9 + 3490x^4y^8 \\
& + 9274x^4y^7 + 21947x^4y^6 + 46641x^4y^5 + 88857x^4y^4 + 146782x^4y^3 \\
& + 204587x^4y^2 + 237518x^4y + 244592x^4 + 3x^3y^{13} + 60x^3y^{12} \\
& + 362x^3y^{11} + 1768x^3y^{10} + 6200x^3y^9 + 17428x^3y^8 + 41652x^3y^7 \\
& + 82707x^3y^6 + 141135x^3y^5 + 205570x^3y^4 + 254702x^3y^3 \\
& + 262635x^3y^2 + 250074x^3y + 220728x^3 + x^2y^{14} - 5x^2y^{13} \\
& - 115x^2y^{12} - 617x^2y^{11} - 3152x^2y^{10} - 11513x^2y^9 - 32500x^2y^8 \\
& - 77546x^2y^7 - 153978x^2y^6 - 266724x^2y^5 - 407035x^2y^4 \\
& - 552511x^2y^3 - 635561x^2y^2 - 683484x^2y - 582684x^2 - 12xy^{14} - 87xy^{13} \\
& - 426xy^{12} - 2268xy^{11} - 8347xy^{10} - 24229xy^9 - 57998xy^8 - 111689xy^7 \\
& - 183355xy^6 - 259242xy^5 - 325530xy^4 - 364095xy^3 - 399768xy^2 - 399366xy \\
& - 521348x - 2y^{15} + 20y^{14} + 165y^{13} + 669y^{12} + 2747y^{11} + 9330y^{10} + 27100y^9 \\
& + 68799y^8 + 146545y^7 + 265190y^6 + 404543y^5 + 528774y^4 + 606124y^3 \\
& + 490934y^2 + 481162y - 53604 \\
g = 2, i = 5, m = 1, \text{ neighbours } 1, 2, 4, 6, 13, 14, 20, 21, 23, 26, 32, 41 \\
W \quad & 24x^4 - 12x^3 + 72x^2 - 228x - 48 \\
A \quad & -4x^{11}y + 4x^{11} - 14x^{10}y^2 + 10x^{10}y + 4x^{10} - 18x^9y^3 + 8x^9y^2 \\
& + 18x^9y - 8x^9 - 10x^8y^4 + 6x^8y^3 + 8x^8y^2 + 4x^8y - 8x^8 - 2x^7y^5 \\
& + 10x^7y^4 + 8x^7y^3 - 16x^7y^2 + 40x^7y - 40x^7 + 8x^6y^5 + 68x^6y^4 \\
& - 50x^6y^3 + 62x^6y^2 - 44x^6y - 44x^6 + 2x^5y^6 + 100x^5y^5 \\
& - 16x^5y^4 + 32x^5y^3 - 80x^5y^2 - 50x^5y + 12x^5 + 62x^4y^6 \\
& + 2x^4y^5 + 102x^4y^4 - 170x^4y^3 - 118x^4y^2 + 98x^4y + 24x^4 \\
& + 18x^3y^7 - 20x^3y^6 + 112x^3y^5 - 120x^3y^4 - 204x^3y^3 \\
& + 126x^3y^2 + 8x^3y - 496x^3 + 2x^2y^8 - 26x^2y^7 - 52x^2y^6 \\
& - 22x^2y^5 - 152x^2y^4 + 180x^2y^3 - 274x^2y^2 + 228x^2y + 692x^2 \\
& - 12xy^8 - 182xy^7 - 6xy^6 - 218xy^5 + 254xy^4 - 460xy^3 + 242xy^2 + 350xy - 3568x \\
& - 2y^9 - 250y^8 + 50y^7 - 462y^6 + 866y^5 + 1998y^4 + 170y^3 + 5726y^2 - 6800y \\
& - 2304 \\
B \quad & 92x^4y^8 - 70x^4y^7 + 56x^4y^6 - 650x^4y^5 - 2340x^4y^4 \\
& + 220x^4y^3 - 5272x^4y^2 + 6704x^4y + 14940x^4 + 20x^3y^9 - 70x^3y^8 \\
& - 366x^3y^7 - 420x^3y^6 - 2666x^3y^5 + 730x^3y^4 - 4454x^3y^3 \\
& + 6272x^3y^2 + 14738x^3y - 7880x^3 + 2x^2y^{10} - 42x^2y^9 - 550x^2y^8 \\
& - 184x^2y^7 - 2872x^2y^6 + 982x^2y^5 - 3270x^2y^4 + 6224x^2y^3 \\
& + 14822x^2y^2 - 6132x^2y + 31484x^2 - 14xy^{10} - 460xy^9 - 28xy^8 - 2508xy^7 \\
& + 940xy^6 - 1868xy^5 + 5118xy^4 + 15426xy^3 - 7012xy^2 + 28138xy - 42196x - 2y^{11} \\
& - 254y^{10} + 40y^9 - 1436y^8 + 822y^7 + 148y^6 + 3180y^5 + 14600y^4 - 4122y^3 \\
& + 20770y^2 - 21474y - 16016 \\
g = 2, i = 6, m = 1, \text{ neighbours } 5, 8, 9, 19, 20, 22, 23, 27, 30, 33, 38 \\
W \quad & 20x^4 - 100x^3 + 14x^2 - 14x + 2 \\
A \quad & -2x^{12}y - 4x^{12} - x^{11}y^2 - 6x^{11}y - 8x^{11} + 2x^{10}y^3 - 5x^{10}y^2 \\
& - 16x^{10}y + 4x^{10} + x^9y^4 + 4x^9y^3 - 16x^9y^2 - 34x^9y + 12x^9 \\
& + 8x^8y^4 + 5x^8y^3 - 25x^8y^2 - 4x^8y + 4x^8 + 3x^7y^5 + 26x^7y^4 \\
& + x^7y^3 - 50x^7y^2 + 60x^7y + 8x^7 + 11x^6y^5 + 38x^6y^4 + 3x^6y^3 \\
& - 24x^6y^2 + 72x^6y + 8x^6 - x^5y^6 + 36x^5y^5 + 47x^5y^4 - 7x^5y^3 \\
& + 126x^5y^2 + 44x^5y - 8x^5 - x^4y^7 - 4x^4y^6 + 50x^4y^5 + 37x^4y^4 \\
& - 43x^4y^3 + 198x^4y^2 - 10x^4y - 20x^4 - 13x^3y^7 - 8x^3y^6 \\
& + 31x^3y^5 + 4x^3y^4 + 90x^3y^3 + 91x^3y^2 - 78x^3y - 424x^3 \\
& - 3x^2y^8 - 36x^2y^7 - 10x^2y^6 - 38x^2y^5 - 155x^2y^4 + 203x^2y^3 \\
& - 149x^2y^2 - 150x^2y + 3984x^2 - 16xy^8 - 68xy^7 - 67xy^6 - 142xy^5 \\
& - 189xy^4 + 63xy^3 - 411xy^2 - 152xy - 10556x - y^9 - 26y^8 - 17y^7 - 49y^6 \\
& - 171y^5 + 43y^4 - 267y^3 - 1086y^2 - 5292y - 4 \\
B \quad & -x^4y^9 + 9x^4y^8 + 24x^4y^7 + 43x^4y^6 + 93x^4y^5 + 43x^4y^4 \\
& + 72x^4y^3 - 296x^4y^2 - 2641x^4y - 1696x^4 - 7x^3y^9 + 16x^3y^8 \\
& + 46x^3y^6 + 218x^3y^5 - 81x^3y^4 + 567x^3y^3 + 1441x^3y^2 + 762x^3y \\
& + 1316x^3 - 3x^2y^{10} - 11x^2y^9 - 5x^2y^8 - 54x^2y^7 - 30x^2y^6 \\
& + 52x^2y^5 - 239x^2y^4 - 602x^2y^3 - 562x^2y^2 - 1620x^2y + 1384x^2 \\
& - 14xy^{10} - 14xy^9 - 76xy^8 - 120xy^7 - 10xy^6 - 331xy^5 + 468xy^4 + 2108xy^3 \\
& + 230xy^2 + 7125xy - 3160x - y^{11} - 27y^{10} - 27y^9 - 166y^8 - 271y^7 - 161y^6 \\
& - 988y^5 - 1081y^4 - 6317y^3 - 4615y^2 - 22254y - 5308
\end{aligned}$$

$g = 2, i = 7, m = 1$, neighbours 1, 4, 10, 11, 12, 25, 30, 32, 39

$W \quad -4x^4 + 2x^3 + 10x^2 + 4x - 8$

$A \quad -8x^6y + 8x^6 + 8x^5y - 8x^5 + 6x^4y^3 + 2x^4y^2 - 22x^4y + 14x^4$
 $+ 2x^3y^4 - 20x^3y^2 + 32x^3y - 30x^3 + 28x^2y^3 + 28x^2y^2 - 44x^2y$
 $+ 4x^2 + 24xy^4 - 20xy^3 - 12xy^2 + 4xy + 80x - 28y^4 - 4y^3 - 8y^2 + 32y + 80$

$B \quad 8x^4y^5 + 2x^4y^4 - 20x^4y^3 - 20x^4y^2 + 8x^4y + 102x^4 + 2x^3y^6$
 $- 2x^3y^5 - 14x^3y^4 + 36x^3y^3 - 54x^3y^2 - 18x^3y + 54x^3$
 $+ 20x^2y^5 + 66x^2y^4 - 40x^2y^3 + 140x^2y^2 - 208x^2y - 18x^2 + 24xy^6$
 $- 20xy^5 + 68xy^4 - 4xy^3 - 96xy^2 + 192xy - 56x - 2y^7 - 32y^6 + 30y^5 - 132y^4$
 $+ 126y^3 - 100y^2 + 210y + 148$

$g = 2, i = 8, m = 1$, neighbours 6, 13, 15, 17, 18, 23, 25, 27, 30, 34, 37, 42

$W \quad 96x^4 + 18x^3 - 213x^2 + 54x + 234$

$A \quad 8x^{14} - 4x^{13}y - 8x^{13} - 10x^{12}y^2 - 28x^{12}y - 66x^{12} + 5x^{11}y^3$
 $+ 26x^{11}y^2 + 55x^{11}y + 58x^{11} + 2x^{10}y^4 + 3x^{10}y^3 - 2x^{10}y^2$
 $+ 45x^{10}y + 152x^{10} - x^9y^5 - 6x^9y^4 - 9x^9y^3 - 4x^9y^2 - 130x^9y$
 $- 134x^9 + x^8y^5 + 45x^8y^4 + 218x^8y^3 + 509x^8y^2 + 469x^8y + 22x^8$
 $- 13x^7y^5 - 126x^7y^4 - 270x^7y^3 - 348x^7y^2 + 163x^7y + 142x^7$
 $- 14x^6y^6 - 81x^6y^5 - 258x^6y^4 - 734x^6y^3 - 1382x^6y^2 - 1389x^6y$
 $- 422x^6 + 3x^5y^7 + 43x^5y^6 + 129x^5y^5 + 239x^5y^4 + 491x^5y^3$
 $+ 726x^5y^2 - 123x^5y - 44x^5 + x^4y^8 + 6x^4y^7 - 25x^4y^6$
 $- 218x^4y^5 - 556x^4y^4 - 924x^4y^3 - 274x^4y^2 + 544x^4y + 246x^4$
 $- 3x^3y^8 - 27x^3y^7 - 38x^3y^6 - 179x^3y^5 - 828x^3y^4$
 $- 2549x^3y^3 - 3495x^3y^2 - 1665x^3y - 9540x^3 + 25x^2y^8 + 226x^2y^7$
 $+ 694x^2y^6 + 1793x^2y^5 + 2797x^2y^4 + 4187x^2y^3 + 2592x^2y^2$
 $+ 1138x^2y - 3132x^2 + 7xy^9 + 7xy^8 + 74xy^7 + 827xy^6 + 2884xy^5 + 5998xy^4$
 $+ 8505xy^3 + 8194xy^2 + 4142xy + 41206x - 2y^{10} - 40y^9 - 132y^8 - 250y^7 + 69y^6$
 $+ 1823y^5 + 6288y^4 + 9534y^3 + 10782y^2 + 3197y - 40395$

$B \quad x^4y^{10} + 6x^4y^9 + 14x^4y^8 + 113x^4y^7 + 811x^4y^6 + 2757x^4y^5$
 $+ 6929x^4y^4 + 11426x^4y^3 + 10744x^4y^2 + 3530x^4y - 37477x^4$
 $- 3x^3y^{10} - 38x^3y^9 - 179x^3y^8 - 637x^3y^7 - 1809x^3y^6$
 $- 3951x^3y^5 - 6027x^3y^4 - 1834x^3y^3 + 6323x^3y^2 + 41070x^3y$
 $+ 45502x^3 + 13x^2y^{10} + 143x^2y^9 + 418x^2y^8 + 943x^2y^7$
 $+ 423x^2y^6 - 2057x^2y^5 - 13097x^2y^4 - 29395x^2y^3 - 42078x^2y^2$
 $- 43087x^2y + 18250x^2 + 9xy^{11} + 54xy^{10} + 302xy^9 + 1594xy^8 + 5472xy^7$
 $+ 14584xy^6 + 29117xy^5 + 42811xy^4 + 32519xy^3 + 16323xy^2 - 68585xy - 40444x$
 $- 2y^{12} - 40y^{11} - 163y^{10} - 519y^9 - 934y^8 - 207y^7 + 6050y^6 + 22859y^5$
 $+ 60572y^4 + 108185y^3 + 120452y^2 + 144022y - 14361$

$g = 2, i = 9, m = 1$, neighbours 44, 4, 6, 20, 23, 24, 27, 30, 35, 36, 38, 39, 41

$W \quad -30x^4 + 276x^3 - 576x^2 + 798x - 468$

$A \quad -4x^{16}y + 4x^{16} - 2x^{15}y^2 + 2x^{15}y + 20x^{14}y^2 - 4x^{14}y - 16x^{14}$
 $+ 10x^{13}y^3 - 2x^{13}y^2 - 12x^{13}y + 4x^{13} - 24x^{12}y^3 - 30x^{12}y^2$
 $+ 42x^{12}y + 12x^{12} - 8x^{11}y^4 - 18x^{11}y^3 + 64x^{11}y^2 - 18x^{11}y - 20x^{11}$
 $+ 2x^{10}y^5 - 34x^{10}y^4 + 70x^{10}y^3 - 32x^{10}y^2 - 14x^{10}y + 8x^{10}$
 $- 36x^9y^5 + 28x^9y^4 - 150x^9y^3 - 10x^9y^2 + 140x^9y + 28x^9$
 $- 10x^8y^6 + 90x^8y^5 - 90x^8y^4 - 52x^8y^3 + 124x^8y^2 - 62x^8y$
 $+ 68x^7y^6 + 18x^7y^5 + 120x^7y^4 + 196x^7y^3 - 314x^7y^2 - 88x^7y$
 $+ 8x^6y^7 - 6x^6y^6 + 68x^6y^5 + 264x^6y^4 - 462x^6y^3 + 132x^6y^2$
 $- 4x^6 - 2x^5y^8 + 22x^5y^7 - 50x^5y^6 + 238x^5y^5 - 346x^5y^4$
 $+ 258x^5y^3 + 90x^5y^2 - 190x^5y - 20x^5 + 36x^4y^8 - 108x^4y^7$
 $+ 222x^4y^6 - 180x^4y^5 + 682x^4y^4 + 280x^4y^3 - 706x^4y^2 - 202x^4y$
 $- 24x^4 + 10x^3y^9 - 130x^3y^8 + 94x^3y^7 - 448x^3y^6 + 162x^3y^5$
 $+ 306x^3y^4 - 610x^3y^3 + 480x^3y^2 + 152x^3y - 916x^3 - 68x^2y^9$
 $+ 46x^2y^8 - 486x^2y^7 - 488x^2y^6 - 206x^2y^5 - 972x^2y^4$
 $+ 1576x^2y^3 + 650x^2y^2 - 56x^2y + 16564x^2 - 8xy^{10} + 38xy^9 - 122xy^8$
 $- 250xy^7 + 290xy^6 - 1434xy^5 + 1804xy^4 + 216xy^3 - 632xy^2 + 70xy - 110708x$
 $+ 2y^{11} - 22y^{10} + 70y^9 - 198y^8 + 714y^7 - 1596y^6 + 1184y^5 - 1170y^4$
 $- 888y^3 + 1382y^2 + 478y + 365876$

$B \quad 36x^4y^{10} - 186x^4y^9 + 690x^4y^8 - 2076x^4y^7 + 5786x^4y^6$
 $- 12302x^4y^5 + 26628x^4y^4 - 51312x^4y^3 + 99442x^4y^2 - 199928x^4y$
 $+ 399706x^4 + 10x^3y^{11} - 150x^3y^{10} + 462x^3y^9 - 1876x^3y^8$
 $+ 4158x^3y^7 - 10322x^3y^6 + 19780x^3y^5 - 37838x^3y^4 + 72722x^3y^3$
 $- 143490x^3y^2 + 288712x^3y - 580992x^3 - 68x^2y^{11} + 196x^2y^{10}$

$$\begin{aligned}
& -1132x^2y^9 + 1898x^2y^8 - 6060x^2y^7 + 8492x^2y^6 - 16416x^2y^5 \\
& + 27938x^2y^4 - 48798x^2y^3 + 118554x^2y^2 - 237314x^2y + 540898x^2 \\
& - 8xy^{12} + 54xy^{11} - 268xy^{10} + 400xy^9 - 864xy^8 - 1004xy^7 + 5408xy^6 \\
& - 17662xy^5 + 41668xy^4 - 81762xy^3 + 49616xy^2 - 98202xy - 245880x + 2y^{13} \\
& - 26y^{12} + 126y^{11} - 538y^{10} + 2060y^9 - 6432y^8 + 16694y^7 - 40234y^6 \\
& + 82724y^5 - 167560y^4 + 330876y^3 - 291236y^2 + 585766y + 292450
\end{aligned}$$

$g = 2, i = 10, m = 1$, neighbours 1, 3, 7, 12, 23, 26, 27, 33, 36, 39, 40

W $-86x^4 - 24x^3 + 272x^2 - 512x + 224$

A $-4x^{11}y - 4x^{11} - 6x^{10}y^2 - 16x^{10}y - 10x^{10} - 2x^9y^3 + 2x^9y^2$
 $-10x^9y - 14x^9 + 20x^8y^3 + 54x^8y^2 + 20x^8y - 14x^8 + 8x^7y^4$
 $+56x^7y^3 + 120x^7y^2 + 68x^7y - 4x^7 + 8x^6y^4 + 42x^6y^3$
 $+156x^6y^2 + 134x^6y + 12x^6 - 78x^5y^4 - 60x^5y^3 + 160x^5y^2$
 $+172x^5y + 30x^5 + 2x^4y^6 - 72x^4y^5 - 188x^4y^4 - 200x^4y^3$
 $+26x^4y^2 + 152x^4y + 40x^4 - 32x^3y^6 - 78x^3y^5 - 284x^3y^4$
 $-496x^3y^3 - 236x^3y^2 + 54x^3y - 736x^3 - 8x^2y^7 + 2x^2y^6$
 $-60x^2y^5 - 420x^2y^4 - 770x^2y^3 - 548x^2y^2 - 122x^2y - 4122x^2$
 $+10xy^7 + 84xy^6 - 24xy^5 - 336xy^4 - 796xy^3 - 834xy^2 - 310xy + 46174x - 2y^8$
 $+224y^7 + 88y^6 + 1172y^5 + 1990y^4 - 4128y^3 + 17284y^2 - 30468y - 40016$

B $2x^4y^8 - 114x^4y^7 + 52x^4y^6 - 876x^4y^5 - 1740x^4y^4$
 $+3624x^4y^3 - 18126x^4y^2 + 30046x^4y - 35060x^4 - 40x^3y^8 + 46x^3y^7$
 $-294x^3y^6 - 1602x^3y^5 + 2958x^3y^4 - 14022x^3y^3 + 11558x^3y^2$
 $-5374x^3y - 87770x^3 - 8x^2y^9 + 28x^2y^8 + 146x^2y^7 - 944x^2y^6$
 $+2904x^2y^5 - 8516x^2y^4 + 7104x^2y^3 + 6782x^2y^2 - 63626x^2y$
 $+212994x^2 + 8xy^9 + 280xy^8 - 386xy^7 + 2874xy^6 - 3294xy^5 + 2540xy^4$
 $+21772xy^3 - 54972xy^2 + 166284xy - 213058x - 2y^{10} + 222y^9 - 138y^8 + 2216y^7$
 $+338y^6 + 298y^5 + 26130y^4 - 71116y^3 + 96104y^2 - 200692y + 10160$

$g = 2, i = 11, m = 1$, neighbours 45, 2, 4, 7, 14, 16, 24, 25, 26, 28, 33, 36, 41

W $-2736x^4 - 9648x^3 - 12336x^2 - 10872x - 3072$

A $-x^{13}y + x^{13} - 2x^{12}y^2 + 4x^{12}y - 2x^{12} + 5x^{11}y^2 - 5x^{11} + 2x^{10}y^4$
 $-x^{10}y^3 + 7x^{10}y^2 - 18x^{10}y + 10x^{10} + x^9y^5 - 10x^9y^4$
 $+25x^9y^3 - 57x^9y^2 + 39x^9y + 2x^9 - 11x^8y^5 + 47x^8y^4$
 $-108x^8y^3 + 104x^8y^2 - 27x^8y - 5x^8 - 4x^7y^6 + 42x^7y^5$
 $-128x^7y^4 + 171x^7y^3 - 51x^7y^2 - 41x^7y + 11x^7 + 17x^6y^6$
 $-96x^6y^5 + 180x^6y^4 - 89x^6y^3 - 53x^6y^2 + 54x^6y - 13x^6$
 $+x^5y^7 - 31x^5y^6 + 87x^5y^5 - 14x^5y^4 - 182x^5y^3 + 202x^5y^2$
 $-52x^5y - 11x^5 - x^4y^8 + 14x^4y^7 - 55x^4y^6 + 181x^4y^5$
 $-345x^4y^4 + 327x^4y^3 - 122x^4y^2 + x^4 + 17x^3y^8 - 104x^3y^7$
 $+315x^3y^6 - 582x^3y^5 + 648x^3y^4 - 336x^3y^3 - 23x^3y^2 + 74x^3y$
 $-7485705x^3 + 4x^2y^9 - 70x^2y^8 + 340x^2y^7 - 748x^2y^6 + 845x^2y^5$
 $-354x^2y^4 - 194x^2y^3 + 176x^2y^2 - 7x^2y - 52793848x^2 - 26xy^9$
 $+198xy^8 - 532xy^7 + 727xy^6 - 413xy^5 - 286xy^4 + 499xy^3 - 222xy^2 + 36xy$
 $-160586477x - 3y^{10} + 38y^9 - 87y^8 - 171y^7 + 2594y^6 - 21010y^5 + 167403y^4$
 $-1350407y^3 + 10907069y^2 - 58152938y - 68242168$

B $-x^4y^{10} - 3x^4y^9 + 70x^4y^8 - 293x^4y^7 + 1351x^4y^6$
 $-10037x^4y^5 + 83001x^4y^4 - 674872x^4y^3 + 5453533x^4y^2$
 $-29076495x^4y + 114642426x^4 + 9x^3y^{10} - 44x^3y^9 + 177x^3y^8$
 $-841x^3y^7 + 5155x^3y^6 - 37564x^3y^5 + 296641x^3y^4 - 2389481x^3y^3$
 $+11811433x^3y^2 - 42782946x^3y + 128322429x^3 + 4x^2y^{11} - 39x^2y^{10}$
 $+131x^2y^9 - 169x^2y^8 - 1096x^2y^7 + 14575x^2y^6 - 128098x^2y^5$
 $+1046026x^2y^4 - 8454161x^2y^3 + 15485944x^2y^2 - 57741114x^2y$
 $+69454661x^2 - 20xy^{11} + 138xy^{10} - 502xy^9 + 1790xy^8 - 8534xy^7 + 58359xy^6$
 $-459322xy^5 + 3703338xy^4 - 29912339xy^3 + 21127554xy^2 - 111444449xy - 44421109x$
 $-3y^{12} + 43y^{11} - 149y^{10} + 125y^9 + 2184y^8 - 23971y^7 + 201546y^6$
 $-1633389y^5 + 13189079y^4 - 76576246y^3 + 50611943y^2 - 272316626y - 58804984$

$g = 2, i = 12, m = 1$, neighbours 45, 7, 10, 13, 17, 23, 27, 35, 41

W $4x^4 - 18x^3 + 22x^2 - 58x + 36$

A $-2x^7 - x^6y - 5x^6 + 3x^5y^2 + 12x^5y + 7x^5 + x^4y^3 + 11x^4y^2$
 $+34x^4y + 24x^4 - x^3y^4 - 13x^3y^3 - 32x^3y^2 - 25x^3y - 15x^3$
 $-4x^2y^4 - 34x^2y^3 - 102x^2y^2 - 101x^2y + 115x^2 + 3xy^5 + 18xy^4$
 $+31xy^3 + 5xy^2 - 14xy - 515x + 10y^5 + 45y^4 - 12y^3 - 56y^2 - 336y + 157$

B $4x^4y^5 + 6x^4y^4 - 63x^4y^3 - 83x^4y^2 - 180x^4y - 544x^4 - x^3y^6$
 $+2x^3y^5 + 32x^3y^4 - 8x^3y^3 + 60x^3y^2 + 210x^3y + 1157x^3$

$$\begin{aligned}
& -7x^2y^6 - 24x^2y^5 + 5x^2y^4 - 92x^2y^3 - 164x^2y^2 - 556x^2y \\
& -1294x^2 + 3xy^7 + 5xy^6 - 30xy^5 - xy^4 - 63xy^3 - 32xy^2 + 548xy + 1628x + 8y^7 \\
& + 40y^6 + 34y^5 + 122y^4 - 251y^3 - 343y^2 - 1745y - 1453
\end{aligned}$$

$g = 2, i = 13, m = 1$, neighbours 1, 5, 8, 12, 26, 27, 28, 31, 35, 39

W $18x^4 + 18x^3 - 48x^2 - 186x - 138$

A $4x^{10} + 2x^9y - 10x^9 - 4x^8y^2 - 8x^8y - 8x^8 - 2x^7y^3 + 4x^7y^2$
 $+ 20x^7y + 22x^7 + 10x^6y^3 + 14x^6y^2 - 4x^6y + 4x^6 + 4x^5y^4$
 $- 26x^5y^3 - 2x^5y^2 - 90x^5y + 14x^5 - 8x^4y^4 - 4x^4y^3 - 16x^4y^2$
 $+ 94x^4y - 2x^4 + 6x^3y^5 + 18x^3y^4 + 48x^3y^3 + 58x^3y^2 + 54x^3y$
 $- 364x^3 + 2x^2y^6 - 8x^2y^5 + 2x^2y^4 + 4x^2y^3 - 234x^2y^2 - 40x^2y$
 $- 678x^2 - 18xy^6 + 16xy^5 - 116xy^4 - 34xy^3 + 30xy^2 + 74xy + 1388x - 4y^7$
 $+ 20y^6 - 14y^5 + 52y^4 + 294y^3 - 100y^2 + 1628y + 3860$

B $-14x^4y^6 + 30x^4y^5 - 166x^4y^4 + 266x^4y^3 - 544x^4y^2 + 1776x^4y$
 $- 4580x^4 + 4x^3y^7 + 18x^3y^6 - 6x^3y^5 + 154x^3y^4 + 130x^3y^3$
 $- 406x^3y^2 + 1216x^3y - 570x^3 + 2x^2y^8 - 2x^2y^7 + 12x^2y^6$
 $+ 44x^2y^5 - 216x^2y^4 + 90x^2y^3 - 372x^2y^2 - 3316x^2y + 12994x^2$
 $- 14xy^8 + 12xy^7 - 104xy^6 - 124xy^5 + 242xy^4 - 1504xy^3 + 3750xy^2 - 9060xy$
 $+ 22566x - 4y^9 + 30y^8 - 100y^7 + 344y^6 - 434y^5 + 948y^4 + 974y^3 + 1802y^2$
 $+ 2808y + 11452$

$g = 2, i = 14, m = 1$, neighbours 45, 2, 3, 5, 11, 28, 38, 39, 40, 42

W $34x^4 + 6x^3 - 130x^2 + 194x - 40$

A $2x^8y + 2x^8 + x^7y^2 - x^7 - 2x^6y^3 - 5x^6y^2 - 8x^6y - 5x^6$
 $- x^5y^4 - x^5y^3 + 3x^5y^2 + x^5y - 2x^5 + 3x^4y^4 + x^4y^3$
 $- 3x^4y^2 - 7x^4y - 6x^4 + x^3y^5 - 9x^3y^4 - 17x^3y^3 - 8x^3y^2$
 $+ 4x^3y - 1151x^3 + 6x^2y^5 + 17x^2y^4 + 36x^2y^3 + 61x^2y^2 + 44x^2y$
 $- 400x^2 + 7xy^6 + 24xy^5 + 7xy^4 - 9xy^3 + 34xy^2 + 45xy + 8816x + y^7 - y^6$
 $- 20y^5 + 10y^4 + 401y^3 + 1183y^2 + 2202y - 11768$

B $7x^4y^6 + 18x^4y^5 + 115x^4y^4 + 504x^4y^3 + 1205x^4y^2 + 2190x^4y$
 $- 143x^4 + x^3y^7 - 18x^3y^6 - 119x^3y^5 - 303x^3y^4 - 199x^3y^3$
 $+ 214x^3y^2 + 4509x^3y + 15067x^3 + 4x^2y^7 + 11x^2y^6 - 56x^2y^5$
 $- 505x^2y^4 - 1782x^2y^3 - 4005x^2y^2 - 10602x^2y - 22337x^2 + 6xy^8$
 $+ 45xy^7 + 181xy^6 + 542xy^5 + 948xy^4 + 291xy^3 + 2859xy^2 - 378xy + 11778x + y^9$
 $- 3y^8 - 32y^7 - 61y^6 + 206y^5 + 1623y^4 + 7064y^3 + 7493y^2 + 24977y + 5084$

$g = 2, i = 15, m = 1$, neighbours 3, 4, 8, 17, 26, 30, 33, 34

W $-13x^4 + 23x^3 + 7x^2 - 61x + 48$

A $-2x^5y^2 + 2x^5 - x^4y^3 + 7x^4y^2 + x^4y - 7x^4 + x^3y^4 + 2x^3y^3$
 $- 12x^3y^2 - 2x^3y - 158x^3 - 2x^2y^4 + 2x^2y^3 + 18x^2y^2 - 2x^2y$
 $+ 582x^2 + 8xy^4 - 4xy^3 - 24xy^2 + 4xy - 331x - 15y^4 + 24y^3 + 79y^2 - 380y - 494$

B $2x^4y^5 - 2x^4y^3 - 51x^4y^2 + 356x^4y - 1427x^4 + x^3y^6 - 4x^3y^5$
 $+ 26x^3y^4 - 89x^3y^3 + 215x^3y^2 - 673x^3y + 1332x^3 - 2x^2y^6$
 $+ 20x^2y^5 - 65x^2y^4 + 251x^2y^3 - 209x^2y^2 + 616x^2y + 875x^2$
 $+ 6xy^6 - 34xy^5 + 132xy^4 - 361xy^3 + 190xy^2 - 222xy - 2887x - y^7 - 14y^6$
 $+ 55y^5 - 88y^4 - 126y^3 + 95y^2 - 1630y + 905$

$g = 2, i = 16, m = 1$, neighbours 3, 11, 17, 23, 28, 33

W $-3x^4 - 3x^3 + 6x^2 + 24x + 30$

A $4x^3y - 5x^3 - 18x^2 - xy^3 + 9xy^2 + xy + 18x - 3y^3 - 7y^2 - 38y + 56$

B $3x^4y^3 + 7x^4y^2 + 26x^4y + 112x^4 + 6x^3y^3 + 17x^3y^2 + 47x^3y$
 $+ 63x^3 + 2x^2y^4 + 13x^2y^3 + 2x^2y^2 - 10x^2y - 208x^2 - xy^5 + 5xy^4$
 $- xy^3 - 13xy^2 - 110xy - 518x - 2y^5 - 7y^4 - 58y^3 - 87y^2 - 332y - 478$

$g = 2, i = 17, m = 1$, neighbours 3, 8, 12, 15, 16, 20, 23, 31, 39

W $-80x^4 - 244x^3 + 432x^2 - 660x + 184$

A $-2x^7y - 2x^7 + x^6y^2 - x^6 + 3x^5y^3 - 3x^5y^2 + 7x^5y + 13x^5$
 $- x^4y^4 + 6x^4y^3 + 11x^4y^2 + 12x^4y + 8x^4 - x^3y^5 + 5x^3y^4$
 $+ 3x^3y^3 + 34x^3y^2 + 18x^3y - 6419x^3 - 4x^2y^5 + 12x^2y^4$
 $+ 24x^2y^3 + 24x^2y^2 - 2x^2y - 39058x^2 - 2xy^6 - 3xy^5 + 20xy^4 + 25xy^3$
 $- 28xy^2 - 42xy + 9574x - y^6 + 27y^5 - 203y^4 - 1084y^3 + 11605y^2 + 6245y + 2251$

B $-3x^4y^6 - 8x^4y^5 + 95x^4y^4 + 549x^4y^3 - 5841x^4y^2 - 3183x^4y$
 $+ 51483x^4 - x^3y^7 - 14x^3y^6 + 54x^3y^5 + 307x^3y^4 - 2698x^3y^3$
 $- 4570x^3y^2 + 24119x^3y - 210681x^3 - 10x^2y^7 + 21x^2y^6 + 146x^2y^5$
 $- 1187x^2y^4 - 528x^2y^3 + 9678x^2y^2 - 80422x^2y + 247410x^2 - 2xy^8$
 $- 6xy^7 + 86xy^6 - 542xy^5 - 936xy^4 + 24026xy^3 - 35357xy^2 + 161632xy - 158841x$

$$-2y^8 + 33y^7 - 240y^6 - 755y^5 + 11596y^4 - 9884y^3 + 57451y^2 - 20858y + 36107$$

$g = 2, i = 18, m = 1$, neighbours 45, 3, 4, 8, 22, 29, 33, 35

W $6x^4 + 84x^3 + 126x^2 + 132x + 48$

A $x^4y^2 - x^4 + x^3y^3 - 3x^3y^2 - x^3y - 33x^3 + x^2y^2 - 1009x^2$
 $+ 2xy^4 - 7xy^3 + 7xy - 8570x - 10y^4 - 64y^3 + 189y^2 + 3562y - 4061$

B $3x^4y^4 + 34x^4y^3 - 92x^4y^2 - 1783x^4y - 9346x^4 + x^3y^5$
 $+ 14x^3y^4 - 62x^3y^3 - 844x^3y^2 - 3782x^3y - 14623x^3 + 6x^2y^5$
 $- 38x^2y^4 - 367x^2y^3 - 1469x^2y^2 - 5354x^2y - 12830x^2 + 2xy^6$
 $- 22xy^5 - 162xy^4 - 46xy^3 - 1945xy^2 - 1723xy - 3556x - 9y^6 - 73y^5 + 73y^4$
 $+ 3369y^3 + 3y^2 + 14440y + 1757$

$g = 2, i = 19, m = 1$, neighbours 1, 2, 4, 6, 21, 27, 29, 33, 38, 41

W $84x^4 + 286x^3 + 506x^2 + 518x + 228$

A $-4x^9y + 4x^9 - 6x^8y^2 - 2x^8y + 8x^8 - 2x^7y^3 + 2x^7y^2 + 8x^7y$
 $- 8x^7 + 14x^6y^3 + 14x^6y^2 - 28x^6 + 6x^5y^4 + 22x^5y^3 - 14x^5y^2$
 $+ 6x^5y - 20x^5 + 12x^4y^4 - 24x^4y^3 - 18x^4y^2 + 26x^4y + 4x^4$
 $+ 6x^3y^5 - 12x^3y^4 - 42x^3y^3 - 2x^3y^2 + 22x^3y - 7028x^3$
 $+ 2x^2y^6 - 30x^2y^5 + 6x^2y^4 - 10x^2y^3 - 46x^2y^2 + 34x^2y$
 $- 48004x^2 - 26xy^6 - 22xy^5 + 44xy^4 - 32xy^3 - 62xy^2 + 58xy - 166764x - 6y^7$
 $- 16y^6 - 2y^5 + 102y^4 - 780y^3 + 7282y^2 - 46100y - 125600$

B $2x^4y^6 - 6x^4y^5 + 74x^4y^4 - 424x^4y^3 + 3662x^4y^2 - 23062x^4y$
 $+ 125422x^4 + 2x^3y^7 - 6x^3y^6 + 8x^3y^5 + 104x^3y^4 - 1486x^3y^3$
 $+ 7902x^3y^2 - 39620x^3y + 166384x^3 + 2x^2y^8 - 12x^2y^7 + 4x^2y^6$
 $- 72x^2y^5 + 564x^2y^4 - 6026x^2y^3 + 11918x^2y^2 - 57602x^2y$
 $+ 133416x^2 - 14xy^8 + 10xy^7 - 40xy^6 - 240xy^5 + 2472xy^4 - 24050xy^3$
 $+ 16896xy^2 - 105068xy + 8730x - 6y^9 - 24y^7 + 72y^6 - 952y^5 + 9596y^4 - 68308y^3$
 $+ 39364y^2 - 255846y - 36808$

$g = 2, i = 20, m = 1$, neighbours 4, 5, 6, 9, 17, 27, 28, 30, 32, 35, 41, 42

W $54x^5 + 16x^4 + 58x^3 - 210x^2 + 28x + 46$

A $-4x^{12}y^2 - 4x^{12}y + 8x^{12} + 4x^{11}y^3 + 18x^{11}y^2 + 6x^{11}y - 28x^{11}$
 $+ x^{10}y^4 - 16x^{10}y^3 - 37x^{10}y^2 + 16x^{10}y + 36x^{10} - x^9y^5$
 $- x^9y^4 + 19x^9y^3 + 35x^9y^2 - 16x^9y - 36x^9 + 4x^8y^5 + 33x^8y^4$
 $+ 50x^8y^3 - 23x^8y^2 - 52x^8y - 12x^8 - x^7y^6 - 25x^7y^5$
 $- 156x^7y^4 - 205x^7y^3 + 95x^7y^2 + 184x^7y + 108x^7 - 10x^6y^6$
 $+ 70x^6y^5 + 343x^6y^4 + 246x^6y^3 - 271x^6y^2 - 274x^6y - 104x^6$
 $+ x^5y^7 + 140x^5y^6 + 425x^5y^5 + 296x^5y^4 + 310x^5y^3$
 $+ 176x^5y^2 - 3088x^5y - 3208x^5 + 5x^4y^8 - 130x^4y^7 - 631x^4y^6$
 $- 829x^4y^5 - 479x^4y^4 + 71x^4y^3 + 3361x^4y^2 + 596x^4y - 1660x^4$
 $+ 2x^3y^8 + 42x^3y^7 + 346x^3y^6 + 776x^3y^5 + 493x^3y^4$
 $- 210x^3y^3 - 733x^3y^2 - 544x^3y - 6692x^3 + x^2y^9 + 34x^2y^8$
 $+ 38x^2y^7 - 514x^2y^6 - 1028x^2y^5 - 61x^2y^4 + 625x^2y^3$
 $+ 715x^2y^2 + 134x^2y + 18604x^2 + xy^9 - 118xy^8 - 236xy^7 + 294xy^6$
 $+ 570xy^5 + 98xy^4 + 515xy^3 + 220xy^2 - 744xy - 460x - 4y^{10} + 151y^9 + 694y^8$
 $+ 527y^7 - 660y^6 - 1015y^5 - 2134y^4 + 1999y^3 + 1066y^2 + 8072y - 2116$

B $4x^8y^7 - 107x^8y^6 - 491x^8y^5 - 617x^8y^4 - 425x^8y^3 + 50x^8y^2$
 $+ 3258x^8y + 360x^8 - 4x^7y^8 + 107x^7y^7 + 491x^7y^6 + 617x^7y^5$
 $+ 425x^7y^4 - 50x^7y^3 - 3258x^7y^2 - 360x^7y - 4x^6y^8 + 107x^6y^7$
 $+ 491x^6y^6 + 617x^6y^5 + 425x^6y^4 - 50x^6y^3 - 3258x^6y^2 - 360x^6y$
 $+ 4x^5y^9 - 107x^5y^8 - 491x^5y^7 - 617x^5y^6 - 425x^5y^5$
 $+ 50x^5y^4 + 3258x^5y^3 + 360x^5y^2 + 2276x^5 + 4x^4y^9 - 107x^4y^8$
 $- 483x^4y^7 - 831x^4y^6 - 1407x^4y^5 - 1184x^4y^4 + 2408x^4y^3$
 $+ 460x^4y^2 + 4240x^4y + 4072x^4 - x^3y^{10} - 148x^3y^9 - 419x^3y^8$
 $- 291x^3y^7 - 515x^3y^6 - 419x^3y^5 + 888x^3y^4 - 92x^3y^3$
 $+ 7665x^3y^2 + 4868x^3y + 25456x^3 - 5x^2y^{11} + 117x^2y^{10} + 628x^2y^9$
 $+ 1455x^2y^8 + 3109x^2y^7 + 3960x^2y^6 - 579x^2y^5 - 2616x^2y^4$
 $- 11003x^2y^3 - 8598x^2y^2 + 2688x^2y - 40452x^2 - xy^{11} + 102xy^{10}$
 $+ 132xy^9 - 218xy^8 - 416xy^7 - 1507xy^6 - 2175xy^5 + 27xy^4 + 784xy^3$
 $- 2926xy^2 + 1690xy - 5488x + 4y^{12} - 147y^{11} - 650y^{10} - 1054y^9 - 2189y^8$
 $- 1787y^7 + 4169y^6 + 2648y^5 + 8457y^4 - 13938y^3 - 4147y^2 - 33354y + 392$

$g = 2, i = 21, m = 1$, neighbours 5, 19, 27, 35, 38

W $x^3 + x^2 + x + 1$

A $-x^5 + x^4y - 2x^3 + x^2y + 3x^2 + xy^2 + 2xy - x - y^3 - y^2 - 3y - 1$

B $x^8 - x^7y + x^6y - x^5y^2 - 7x^5 + x^4y^2 + 7x^4y + 5x^4 + x^3y^3$

$$\begin{aligned}
& -x^3y^2 + 8x^3y + 12x^3 - x^2y^4 - 2x^2y^3 - 14x^2y^2 - 11x^2y + 3x^2 \\
& + xy^4 + 2xy^3 - 2xy^2 - 9x - y^5 - 3y^3 - y^2 - 11y - 7
\end{aligned}$$

$g = 2, i = 22, m = 2$, neighbours 6, 18, 33

$W \quad 2x^5 - 6x^4 - 2x^2 + 6x$

$A \quad -2x^5 + 33x^4 - 36x^3 + 8x^2 + 2xy^2 + 24xy - 27x - 25y^2 + 15y$

$B \quad -2x^8 - 11x^7 + 2x^6y - 11x^6 + 13x^5y - 11x^5 - 4x^4y^2 - 30x^4 + 2x^3y^3$
 $+ 8x^3y^2 + 4x^3y + 11x^3 - 27x^2y^3 + 8x^2y^2 - 98x^2y - 6x^2 - 2xy^4$
 $+ xy^3 + 4xy^2 + 35xy + 22x + 25y^4 - 15y^3 + 104y^2 - 35y - 19$

$g = 2, i = 23, m = 32$, neighbours 3, 5, 6, 8, 9, 10, 12, 16, 17, 26, 31, 42

$W \quad -256x^5 - 3200x^4 + 1792x^3 - 3808x^2 - 5696x + 1600$

$A \quad -512x^8y^6 + 5376x^8y^5 + 23936x^8y^4 + 134464x^8y^3 + 232032x^8y^2$
 $+ 1516880x^8y - 331752x^8 + 12288x^7y^3 + 45056x^7y^2 + 49152x^7y$
 $+ 16384x^7 + 10240x^6y^4 - 14336x^6y^3 - 92160x^6y^2 - 22528x^6y$
 $+ 45056x^6 - 19456x^5y^5 - 17408x^5y^4 + 115712x^5y^3 + 136192x^5y^2$
 $- 59392x^5y + 1311016x^5 + 1536x^4y^8 - 17152x^4y^7 - 67968x^4y^6$
 $- 269632x^4y^5 - 199776x^4y^4 - 3022160x^4y^3 + 6313192x^4y^2$
 $+ 14269096x^4y + 25538922x^4 + 8192x^3y^7 - 28672x^3y^6 - 121856x^3y^5$
 $+ 27648x^3y^4 + 152576x^3y^3 - 187392x^3y^2 - 284672x^3y - 9379840x^3$
 $- 8192x^2y^7 + 6144x^2y^6 + 34816x^2y^5 - 239616x^2y^4 - 452608x^2y^3$
 $+ 53248x^2y^2 + 81920x^2y + 9355264x^2 - 1024xy^9 + 15360xy^8 + 35840xy^7$
 $- 45312xy^6 - 47488xy^5 + 288448xy^4 - 509792xy^3 - 41767052xy^2 + 14919504xy$
 $+ 27333102x - 1024y^{10} + 10240y^9 + 38144y^8 + 136320y^7 + 315840y^6 + 1970464y^5$
 $+ 262896y^4 - 12997984y^3 + 23509310y^2 + 16785902y - 2560000$

$B \quad 512x^{11}y^6 - 5376x^{11}y^5 - 23936x^{11}y^4 - 134464x^{11}y^3$
 $- 240224x^{11}y^2 - 1541456x^{11}y + 315368x^{11} - 512x^{10}y^6 + 5376x^{10}y^5$
 $+ 23936x^{10}y^4 + 134464x^{10}y^3 + 240224x^{10}y^2 + 1541456x^{10}y$
 $- 315368x^{10} - 512x^9y^7 + 5888x^9y^6 + 18560x^9y^5 + 110528x^9y^4$
 $+ 105760x^9y^3 + 1301232x^9y^2 - 1856824x^9y + 315368x^9 + 1024x^8y^7$
 $- 11264x^8y^6 - 42496x^8y^5 - 244992x^8y^4 - 345984x^8y^3$
 $- 2842688x^8y^2 + 2172192x^8y - 13572988x^8 - 1024x^7y^8 + 10240x^7y^7$
 $+ 59136x^7y^6 + 231040x^7y^5 + 339648x^7y^4 + 1776800x^7y^3$
 $- 1851856x^7y^2 - 18042112x^7y - 13178374x^7 + 512x^6y^9 - 5376x^6y^8$
 $- 20864x^6y^7 - 172096x^6y^6 - 327392x^6y^5 - 2096912x^6y^4$
 $+ 285896x^6y^3 - 6726384x^6y^2 + 31335116x^6y + 13493742x^6 + 512x^5y^9$
 $- 4864x^5y^8 - 38272x^5y^7 - 58944x^5y^6 - 12256x^5y^5 + 320112x^5y^4$
 $+ 1565960x^5y^3 + 24768496x^5y^2 - 18156742x^5y - 13493742x^5$
 $- 512x^4y^{10} + 4864x^4y^9 + 25728x^4y^8 + 210368x^4y^7 + 382496x^4y^6$
 $+ 2149488x^4y^5 - 426488x^4y^4 + 6168904x^4y^3 - 14529072x^4y^2$
 $- 10291320x^4y - 57280404x^4 + 1024x^3y^{10} - 10240x^3y^9 - 13824x^3y^8$
 $- 290048x^3y^7 - 539264x^3y^6 - 4039360x^3y^5 - 1988000x^3y^4$
 $- 28593692x^3y^3 + 3441872x^3y^2 - 59923584x^3y - 1021290x^3$
 $- 1024x^2y^{11} + 9216x^2y^{10} + 36096x^2y^9 + 189312x^2y^8$
 $+ 406592x^2y^7 + 2277856x^2y^6 - 536944x^2y^5 - 5659824x^2y^4$
 $- 5351278x^2y^3 - 56882144x^2y^2 + 1414084x^2y + 1204224x^2 - 7168xy^{10}$
 $+ 14592xy^9 + 146816xy^8 + 250432xy^7 + 2354400xy^6 + 2260752xy^5$
 $+ 33750732xy^4 + 12953246xy^3 + 118365280xy^2 + 37104038xy - 50384896x + 1024y^{12}$
 $- 8192y^{11} - 37120y^{10} - 192640y^9 - 457152y^8 - 2508832y^7 - 1708656y^6$
 $+ 4742944y^5 - 26491422y^4 + 34977954y^3 - 78491544y^2 - 90661110y - 6545902$

$g = 2, i = 24, m = 4$, neighbours 3, 4, 9, 11, 25, 26, 30, 33, 34

$W \quad -20x^5 + 24x^4 + 12x^3 + 32x^2 - 40x + 16$

$A \quad -32x^8y^3 - 128x^8y^2 - 696x^8y - 2924x^8 + 32x^6y^3 + 64x^6y^2$
 $- 32x^6y - 64x^6 + 16x^5y^4 - 16x^5y^3 - 112x^5y^2 + 16x^5y - 14126x^5$
 $+ 96x^4y^5 + 384x^4y^4 + 1912x^4y^3 + 9012x^4y^2 - 636x^4y + 19423x^4$
 $- 16x^3y^5 - 32x^3y^4 - 32x^3y^3 - 64x^3y^2 + 48x^3y - 64x^2y^5$
 $- 192x^2y^4 - 128x^2y^3 + 64x^2y^2 + 192x^2y + 832x^2 - 64xy^6 - 48xy^5$
 $+ 64xy^4 - 496xy^3 + 40538xy^2 + 22728xy - 3077x - 64y^7 - 144y^6 - 592y^5$
 $- 3136y^4 + 12792y^3 + 25341y^2 - 3557y - 256$

$B \quad 32x^{11}y^3 + 128x^{11}y^2 + 696x^{11}y + 2924x^{11} + 32x^{10}y^3 + 128x^{10}y^2$
 $+ 696x^{10}y + 2924x^{10} + 32x^9y^4 + 160x^9y^3 + 824x^9y^2 + 3620x^9y$
 $+ 2924x^9 + 64x^8y^4 + 288x^8y^3 + 1520x^8y^2 + 6544x^8y + 16746x^8$
 $- 64x^7y^5 - 224x^7y^4 - 896x^7y^3 - 3680x^7y^2 + 10928x^7y + 1207x^7$
 $+ 32x^6y^6 + 128x^6y^5 + 888x^6y^4 + 4092x^6y^3 + 5776x^6y^2$

$$\begin{aligned}
& +40066x^6y + 4131x^6 - 32x^5y^6 - 96x^5y^5 - 8x^5y^4 + 412x^5y^3 \\
& +16704x^5y^2 + 41273x^5y + 4131x^5 + 32x^4y^7 + 96x^4y^6 + 792x^4y^5 \\
& +4084x^4y^4 + 6428x^4y^3 + 16264x^4y^2 + 22980x^4y + 18118x^4 \\
& +48x^3y^7 + 288x^3y^6 + 1552x^3y^5 + 6960x^3y^4 + 22506x^3y^3 \\
& +23360x^3y^2 + 88080x^3y + 14933x^3 - 64x^2y^8 - 208x^2y^7 \\
& -1168x^2y^6 - 4976x^2y^5 + 6312x^2y^4 - 18229x^2y^3 + 39696x^2y^2 \\
& +28694x^2y - 5418x^2 + 144xy^7 + 800xy^6 + 3088xy^5 + 29634xy^4 + 4939xy^3 \\
& +126752xy^2 + 43131xy - 4074x - 64y^9 - 128y^8 - 768y^7 - 3664y^6 + 10232y^5 \\
& +11965y^4 + 44331y^3 + 113980y^2 + 11225y - 4549
\end{aligned}$$

$g = 2, i = 25, m = 1$, neighbours 45, 2, 4, 7, 8, 11, 24, 33, 39

$W \quad -5x^5 - 11x^4 + 3x^3 + 16x^2 + 2x - 8$

$A \quad -x^5y^3 - 3x^5y^2 - 6x^5y + 16x^5 + x^4y^4 + x^4y^3 + 9x^4y^2$
 $-63x^4y - 27x^4 + 9x^3y^3 - x^3y^2 - 7x^3y - 92x^3 + 5x^2y^4 - 3x^2y^2$
 $-6x^2y + 162x^2 + xy^5 - 4xy^4 - 6xy^3 - 9xy^2 + 66xy + 70x - y^6 - 8y^5 - 19y^4$
 $+24y^3 + 136y^2 - 48y - 132$

$B \quad x^8y^3 + 3x^8y^2 + 8x^8y - 43x^8 - x^7y^4 - x^7y^3 - 2x^7y^2$
 $+59x^7y - 86x^7 + x^6y^4 + 4x^6y^3 + 11x^6y^2 - 35x^6y - 43x^6$
 $-x^5y^5 - 2x^5y^4 - 3x^5y^3 + 57x^5y^2 - 27x^5y - 18x^5 + x^4y^5$
 $+5x^4y^4 + 17x^4y^3 - 18x^4y^2 - 130x^4y + 240x^4 + x^3y^6 + 4x^3y^5$
 $+5x^3y^4 + 18x^3y^3 - 74x^3y^2 + 22x^3y - 5x^3 - x^2y^7 - x^2y^6$
 $-25x^2y^5 + 69x^2y^4 - 16x^2y^3 + 118x^2y^2 + 294x^2y - 87x^2 + xy^7$
 $-11xy^6 + 12xy^5 - 30xy^4 + 41xy^3 + xy^2 + 394xy - 94x - y^8 - 8y^7 - 15y^6$
 $+9y^5 + 43y^4 - 17y^3 + 453y^2 - 508y - 68$

$g = 2, i = 26, m = 1$, neighbours 46, 3, 5, 10, 11, 13, 15, 23, 24, 36, 40

$W \quad -9x^5 - 30x^4 - 9x^3 - 48x^2 - 6x - 36$

$A \quad -6x^8y - 6x^8 + 9x^7y^2 + 24x^7y + 15x^7 + 6x^6y^3 + 33x^6y^2 + 42x^6y$
 $+15x^6 + 6x^5y^5 + 36x^5y^4 + 84x^5y^3 + 282x^5y^2 + 765x^5y + 1383x^5$
 $-6x^4y^6 - 60x^4y^5 - 273x^4y^4 - 831x^4y^3 - 1890x^4y^2 - 3357x^4y$
 $-3546x^4 + 3x^3y^6 + 57x^3y^5 + 243x^3y^4 + 396x^3y^3 + 360x^3y^2$
 $+171x^3y - 1044x^3 - 6x^2y^6 - 48x^2y^5 - 39x^2y^4 + 21x^2y^3$
 $-108x^2y^2 - 231x^2y - 3618x^2 - 9xy^7 - 78xy^6 - 222xy^5 - 342xy^4$
 $-288xy^3 + 132xy^2 + 3039xy - 2340x + 6y^8 + 54y^7 + 189y^6 + 309y^5 + 438y^4$
 $+561y^3 - 1329y^2 - 309y - 135$

$B \quad -6x^8y^5 - 48x^8y^4 - 180x^8y^3 - 477x^8y^2 - 945x^8y - 1533x^8$
 $+6x^7y^6 + 60x^7y^5 + 276x^7y^4 + 837x^7y^3 + 1899x^7y^2 + 3423x^7y$
 $+3066x^7 - 6x^6y^6 - 42x^6y^5 - 132x^6y^4 - 297x^6y^3 - 468x^6y^2$
 $-588x^6y + 1533x^6 + 6x^5y^7 + 54x^5y^6 + 216x^5y^5 + 561x^5y^4$
 $+1062x^5y^3 + 1524x^5y^2 - 357x^5y - 957x^5 - 6x^4y^7 - 36x^4y^6$
 $-102x^4y^5 - 261x^4y^4 - 531x^4y^3 - 1074x^4y^2 - 1878x^4y - 3621x^4$
 $-6x^3y^8 - 39x^3y^7 - 186x^3y^6 - 675x^3y^5 - 1356x^3y^4$
 $-2124x^3y^3 - 1848x^3y^2 - 3078x^3y + 1395x^3 + 6x^2y^9 + 57x^2y^8$
 $+267x^2y^7 + 1029x^2y^6 + 3048x^2y^5 + 6909x^2y^4 + 11364x^2y^3$
 $+14238x^2y^2 + 15603x^2y + 3627x^2 - 9xy^9 - 66xy^8 - 192xy^7 - 537xy^6$
 $-1404xy^5 - 2586xy^4 - 711xy^3 - 2286xy^2 + 8673xy + 2043x + 6y^{10} + 66y^9$
 $+315y^8 + 918y^7 + 2202y^6 + 4134y^5 + 4794y^4 + 7143y^3 + 2253y^2 - 312y - 1161$

$g = 2, i = 27, m = 432$, neighbours 6, 8, 9, 10, 12, 13, 19, 20, 21, 33, 37

$W \quad 6048x^5 + 25920x^4 + 5616x^3 - 55728x^2 - 58320x - 11232$

$A \quad 1492992x^9y + 2985984x^9 - 31104x^8y^5 + 1176768x^8y^4 + 10558944x^8y^3$
 $+67960368x^8y^2 + 338407032x^8y + 1702091372x^8 + 1866240x^7y^3$
 $+5971968x^7y^2 - 1492992x^7y - 11943936x^7 - 2426112x^6y^4$
 $-10264320x^6y^3 - 25380864x^6y^2 - 37324800x^6y - 16422912x^6$
 $-3172608x^5y^5 - 16796160x^5y^4 - 40497408x^5y^3 - 53747712x^5y^2$
 $-25380864x^5y - 26327828746x^5 - 31104x^4y^7 + 243648x^4y^6$
 $-11727072x^4y^5 - 136654704x^4y^4 - 879423000x^4y^3 - 4668801340x^4y^2$
 $-32685005220x^4y + 133402514553x^4 - 186624x^3y^7 - 2239488x^3y^6$
 $-8957952x^3y^5 + 2239488x^3y^4 + 51321600x^3y^3 + 67184640x^3y^2$
 $+49641984x^3y + 142953984x^2y^4 + 181025280x^2y^3 + 111227904x^2y^2$
 $+75396096x^2y + 396389376x^2 + 10264320xy^7 + 60652800xy^6 + 166235328xy^5$
 $+278030880xy^4 + 475285968xy^3 - 24506379586xy^2 + 324047122284xy + 162388509423x$
 $+62208y^9 - 1202688y^8 - 19417536y^7 - 76684896y^6 + 18675408y^5$
 $+1357647944y^4 + 25583936464y^3 - 162127598973y^2 - 163067820783y - 126157824$

$B \quad 31104x^{11}y^5 - 1176768x^{11}y^4 - 10558944x^{11}y^3 - 64227888x^{11}y^2$
 $- 328702584x^{11}y - 1697612396x^{11} - 31104x^{10}y^5 + 1176768x^{10}y^4$
 $+ 10558944x^{10}y^3 + 64227888x^{10}y^2 + 328702584x^{10}y + 1697612396x^{10}$
 $+ 31104x^9y^6 - 1145664x^9y^5 - 11735712x^9y^4 - 74786832x^9y^3$
 $- 392930472x^9y^2 - 2026314980x^9y - 1697612396x^9 - 62208x^8y^6$
 $+ 2322432x^8y^5 + 22294656x^8y^4 + 139014720x^8y^3 + 721633056x^8y^2$
 $+ 3723927376x^8y + 27996327798x^8 + 62208x^7y^7 - 2633472x^7y^6$
 $- 11631168x^7y^5 + 8349984x^7y^4 + 295488336x^7y^3 + 1843188488x^7y^2$
 $+ 26947453296x^7y - 163396546139x^7 + 31104x^6y^8 - 1301184x^6y^7$
 $- 5478624x^6y^6 - 34978032x^6y^5 - 241450392x^6y^4 - 1538938172x^6y^3$
 $+ 501700664x^6y^2 - 6070277666x^6y + 161698933743x^6 + 31104x^5y^8$
 $- 1332288x^5y^7 - 6152544x^5y^6 + 43328016x^5y^5 + 536938728x^5y^4$
 $+ 3382126660x^5y^3 + 26445752632x^5y^2 - 157326268473x^5y - 161698933743x^5$
 $+ 31104x^4y^9 - 1332288x^4y^8 - 4146336x^4y^7 - 28825488x^4y^6$
 $- 283985256x^4y^5 - 2105884484x^4y^4 - 3149679068x^4y^3$
 $- 7855126040x^4y^2 - 4941298500x^4y + 29208556810x^4 - 62208x^3y^9$
 $+ 7174656x^3y^8 + 54953856x^3y^7 + 257769792x^3y^6 + 996867168x^3y^5$
 $+ 4454948272x^3y^4 + 31389507942x^3y^3 + 17436485784x^3y^2$
 $+ 144339351568x^3y - 132366004857x^3 + 62208x^2y^{10} - 1140480x^2y^9$
 $+ 4262976x^2y^8 + 59583456x^2y^7 + 371642160x^2y^6 + 2056464440x^2y^5$
 $+ 28618271208x^2y^4 - 153209533423x^2y^3 + 112962360304x^2y^2$
 $- 616090933062x^2y - 2179954944x^2 + 62208xy^{10} + 10741248xy^9 + 54423360xy^8$
 $+ 159928416xy^7 + 484323888xy^6 + 2134410872xy^5 + 2313261438xy^4$
 $+ 170880978823xy^3 + 31061223456xy^2 + 486510851277xy - 3865542912x + 62208y^{11}$
 $- 1016064y^{10} - 17675712y^9 - 77327712y^8 - 57211440y^7 + 1025705480y^6$
 $+ 25571502256y^5 - 156679824781y^4 - 59368455015y^3 - 623049631268y^2$
 $- 814398882105y - 163572452079$

$g = 2, i = 28, m = 1, \text{neighbours } 11, 13, 14, 16, 20, 31, 39$
 $W \quad -2x^5 + 2x^4 + 4x^3 - 34x - 2$
 $A \quad -4x^3 + 4x^2y - 32x^2 - 4xy^2 - 36xy + 12x + 8y^3 - 72y - 4$
 $B \quad -6x^8 - 3x^7y - 11x^7 + x^6y^2 - 16x^6y - 3x^6 + 3x^5y^3 - 4x^5y^2$
 $- 50x^5y + 45x^5 + x^4y^4 + 8x^4y^3 - 17x^4y^2 - 56x^4y - 38x^4$
 $+ 3x^3y^4 + 6x^3y^3 - 18x^3y^2 - 19x^3y + 42x^3 + x^2y^4 + 16x^2y^3$
 $- 7x^2y^2 + 108x^2y + 308x^2 - 4xy^5 + 9xy^4 + 15xy^3 + 22xy^2 + 248xy + 612x$
 $- y^6 - 4y^5 + y^4 + 20y^3 + 12y^2 + 304y + 72$

$g = 2, i = 29, m = 4, \text{neighbours } 45, 18, 19, 32, 35, 38, 39$
 $W \quad 16x^4 + 48x^3 - 32x^2 - 112x - 48$
 $A \quad 8x^8 + 60x^5 + 8x^4y^2 + 128x^4y + 298x^4 - 16x^3y^2 - 144x^3y - 128x^3$
 $- 16x^2y^2 - 272x^2y - 1568x^2 + 32xy^3 + 28xy^2 - 544xy + 462x - 16y^4 - 56y^3$
 $+ 102y^2 - 1230y + 816$
 $B \quad -8x^{11} - 8x^{10} + 8x^9y - 16x^9 + 16x^8y - 20x^8 - 16x^7y^2 - 88x^7y - 182x^7$
 $- 8x^6y^3 - 32x^6y^2 - 84x^6y - 210x^6 + 8x^5y^3 + 40x^5y^2 + 10x^5y$
 $- 392x^5 + 8x^4y^4 + 32x^4y^3 + 96x^4y^2 + 200x^4y - 2884x^4 + 16x^3y^4$
 $+ 12x^3y^3 - 16x^3y^2 + 648x^3y + 1278x^3 - 16x^2y^5 - 136x^2y^4$
 $- 174x^2y^3 - 32x^2y^2 + 2332x^2y + 8624x^2 - 16xy^5 + 28xy^4 + 398xy^3$
 $+ 1064xy^2 + 3350xy + 8718x + 16y^6 + 88y^5 + 114y^4 + 1616y^3 + 320y^2 + 5026y + 3056$

$g = 2, i = 30, m = 72, \text{neighbours } 6, 7, 8, 9, 15, 20, 24, 38$
 $W \quad -144x^5 + 1440x^3 - 3312x^2 + 2880x - 1152$
 $A \quad -864x^8y^2 + 1872x^8y + 14664x^8 + 10368x^5y - 62940x^5 - 864x^4y^4$
 $- 5040x^4y^3 + 58584x^4y^2 + 98664x^4y - 528666x^4 - 5184x^3y^3$
 $- 72576x^3y^2 - 155520x^3y + 352512x^3 - 15552x^2y^4 - 88128x^2y^3$
 $- 119232x^2y^2 + 62208x^2y - 808704x^2 - 5184xy^5 - 15552xy^4 + 46656xy^3$
 $+ 194964xy^2 - 285576xy + 389058x + 1728y^6 - 13248y^5 - 137904y^4 - 113952y^3$
 $+ 1097298y^2 + 253542y - 503004$
 $B \quad 864x^{11}y^2 - 1872x^{11}y - 14664x^{11} + 864x^{10}y^2 - 1872x^{10}y - 14664x^{10}$
 $- 864x^9y^3 + 1872x^9y^2 + 14664x^9y - 1728x^8y^3 + 2880x^8y^2$
 $+ 31200x^8y + 77604x^8 + 1728x^7y^4 + 2304x^7y^3 - 55824x^7y^2$
 $- 73632x^7y + 550134x^7 + 864x^6y^5 + 1584x^6y^4 - 10056x^6y^3$
 $- 96528x^6y^2 - 242964x^6y + 457866x^6 - 864x^5y^5 - 2448x^5y^4$
 $+ 43464x^5y^3 + 32928x^5y^2 - 719466x^5y - 92268x^5 - 864x^4y^6$
 $- 3312x^4y^5 + 6024x^4y^4 + 150048x^4y^3 + 421104x^4y^2 - 399960x^4y$
 $- 2120412x^4 - 1728x^3y^6 + 28800x^3y^5 + 181536x^3y^4 + 447396x^3y^3$

$$\begin{aligned}
& +13968x^3y^2 + 145440x^3y + 8169570x^3 + 1728x^2y^7 + 12672x^2y^6 \\
& +7248x^2y^5 + 67056x^2y^4 + 341502x^2y^3 - 361920x^2y^2 + 4806228x^2y \\
& -8756136x^2 + 3456xy^7 + 6336xy^6 - 32016xy^5 - 219732xy^4 - 150966xy^3 \\
& +85200xy^2 - 4866990xy + 5925366x - 1728y^8 + 9792y^7 + 95280y^6 + 33648y^5 \\
& -560898y^4 + 942540y^3 - 4694040y^2 + 2582574y - 824100
\end{aligned}$$

$g = 2, i = 31, m = 1$, neighbours 45, 4, 13, 17, 23, 28, 33, 37

W $-2x^5 - 2x^4 + 16x^3 + 8x^2 - 14x - 6$

A $-x^5y^2 - 8x^5y - 19x^5 + x^4y^3 + 9x^4y^2 + 39x^4y + 63x^4 + x^3y^3$
 $-2x^3y^2 + 11x^3y + 50x^3 - 2x^2y^3 + 20x^2y + 58x^2 + 2xy^4 + 2xy^2 + 8xy$
 $-44x - y^5 - 7y^4 - 3y^3 + 57y^2 + 138y - 372$

B $x^8y^2 + 6x^8y + 17x^8 - x^7y^3 - 10x^7y^2 - 41x^7y - 68x^7 - x^6y^3$
 $-4x^6y^2 - 5x^6y + 34x^6 + x^5y^4 + 8x^5y^3 + 21x^5y^2 - 14x^5y$
 $-116x^5 + x^4y^4 + 2x^4y^3 - x^4y^2 - 52x^4y - 146x^4 + x^3y^5$
 $+9x^3y^4 + 22x^3y^3 + 21x^3y^2 - 29x^3y - 216x^3 - x^2y^6 - 10x^2y^5$
 $-44x^2y^4 - 102x^2y^3 - 130x^2y^2 - 258x^2y - 23x^2 - 2xy^6 - 3xy^5$
 $-9xy^4 - 12xy^3 + 13xy^2 - 75xy - 124x + y^7 + 8y^6 + 18y^5 + y^4 - 116y^3 - 77y^2$
 $-759y - 168$

$g = 2, i = 32, m = 1$, neighbours 2, 5, 7, 20, 29, 33, 37, 39, 41, 42

W $-12x^5 + 84x^4 + 84x^3 + 258x^2 + 108x + 270$

A $x^{12} + x^{11}y + 2x^{11} - x^{10}y^2 + 2x^{10}y - x^{10} - x^9y^3 - x^9y^2 + x^9y$
 $-7x^9 - 2x^8y^3 - 3x^8y^2 - 8x^8y - 11x^8 - x^7y^4 - 8x^7y^3$
 $-8x^7y^2 - 23x^7y - 6x^7 + 3x^6y^4 - 4x^6y^3 - 15x^6y + 10x^6$
 $+20x^5y^5 - 37x^5y^4 + 138x^5y^3 - 288x^5y^2 + 1047x^5y - 1411x^5$
 $-12x^4y^6 + 107x^4y^5 - 292x^4y^4 + 831x^4y^3 - 2183x^4y^2$
 $+5444x^4y - 3129x^4 + 8x^3y^6 + 23x^3y^5 + 22x^3y^4 + 45x^3y^3$
 $+67x^3y^2 + 102x^3y - 5020x^3 + x^2y^7 + 9x^2y^6 + 8x^2y^5$
 $+22x^2y^4 + 26x^2y^3 + 74x^2y^2 + 36x^2y - 1930x^2 - 12xy^7 - 8xy^6$
 $-27xy^5 - 27xy^4 + 90xy^3 - 317xy^2 - 5100xy - 5456x + 13y^8 - 44y^7 - 38y^6$
 $-18y^5 - 5y^4 + 194y^3 - 5743y^2 - 4738y + 12884$

B $-13x^8y^5 + 44x^8y^4 - 125x^8y^3 + 301x^8y^2 - 1015x^8y + 1295x^8$
 $+13x^7y^6 - 96x^7y^5 + 301x^7y^4 - 801x^7y^3 + 2219x^7y^2 - 5355x^7y$
 $+5180x^7 + 13x^6y^6 - 18x^6y^5 + 37x^6y^4 - 51x^6y^3 + 413x^6y^2$
 $+735x^6y - 2590x^6 - 13x^5y^7 + 70x^5y^6 - 109x^5y^5 + 199x^5y^4$
 $-617x^5y^3 + 917x^5y^2 + 5530x^5y - 16364x^5 - 13x^4y^7 - 8x^4y^6$
 $-27x^4y^5 + 65x^4y^4 - 561x^4y^3 - 959x^4y^2 + 5094x^4y - 35784x^4$
 $-20x^3y^8 + 29x^3y^7 - 211x^3y^6 + 336x^3y^5 - 1000x^3y^4$
 $+1711x^3y^3 - 3411x^3y^2 + 9149x^3y - 30632x^3 + 12x^2y^9 - 103x^2y^8$
 $+348x^2y^7 - 1260x^2y^6 + 3222x^2y^5 - 7984x^2y^4 + 13067x^2y^3$
 $-22888x^2y^2 + 32544x^2y - 50353x^2 + 12xy^9 - 16xy^8 + 78xy^7 - 89xy^6$
 $+222xy^5 + 66xy^4 + 4676xy^3 - 2904xy^2 + 32268xy - 32874x - 13y^{10} + 74y^9$
 $-150y^8 + 482y^7 - 785y^6 + 1472y^5 + 2845y^4 - 1714y^3 + 13324y^2 - 1953y - 42892$

$g = 2, i = 33, m = 128$, neighbours 6, 10, 11, 15, 16, 18, 19, 22, 24, 25, 27, 31, 32, 34, 37, 40, 41

W $-39936x^5 + 473856x^4 + 1877760x^3 + 2046720x^2 + 1516800x + 96768$

A $-262144x^{22} + 524288x^{21}y + 786432x^{21} + 393216x^{20}y + 2293760x^{20}$
 $-458752x^{19}y^3 - 3801088x^{19}y^2 - 7471104x^{19}y - 8388608x^{19}$
 $+81920x^{18}y^4 + 1409024x^{18}y^3 + 3784704x^{18}y^2 - 5931008x^{18}y$
 $-5177344x^{18} + 147456x^{17}y^5 + 2834432x^{17}y^4 + 11026432x^{17}y^3$
 $+28622848x^{17}y^2 + 48201728x^{17}y + 31588352x^{17} - 16384x^{16}y^6$
 $-1015808x^{16}y^5 - 8830976x^{16}y^4 - 19365888x^{16}y^3 - 12255232x^{16}y^2$
 $+10485760x^{16}y - 7798784x^{16} - 16384x^{15}y^7 - 868352x^{15}y^6$
 $-5537792x^{15}y^5 - 11976704x^{15}y^4 - 43728896x^{15}y^3 - 92012544x^{15}y^2$
 $-125894656x^{15}y - 38207488x^{15} + 163840x^{14}y^7 + 4177920x^{14}y^6$
 $+21446656x^{14}y^5 + 46333952x^{14}y^4 + 71155712x^{14}y^3 + 30949376x^{14}y^2$
 $+72318976x^{14}y + 51445760x^{14} + 98304x^{13}y^8 + 1343488x^{13}y^7$
 $-1654784x^{13}y^6 - 18186240x^{13}y^5 - 29278208x^{13}y^4 - 34013184x^{13}y^3$
 $+35291136x^{13}y^2 - 19791872x^{13}y - 67305472x^{13} - 655360x^{12}y^8$
 $-7290880x^{12}y^7 - 21266432x^{12}y^6 - 61095936x^{12}y^5 - 118112256x^{12}y^4$
 $-135135232x^{12}y^3 - 246169600x^{12}y^2 - 307986432x^{12}y - 69730304x^{12}$
 $-147456x^{11}y^9 + 1884160x^{11}y^8 + 25706496x^{11}y^7 + 84475904x^{11}y^6$
 $+225837056x^{11}y^5 + 438484992x^{11}y^4 + 691601408x^{11}y^3$
 $+831291392x^{11}y^2 + 828047360x^{11}y + 279838720x^{11} + 16384x^{10}y^{10}$
 $+1245184x^{10}y^9 + 2686976x^{10}y^8 - 8339456x^{10}y^7 + 7766016x^{10}y^6$

$$\begin{aligned}
& +6881280x^{10}y^5 + 94519296x^{10}y^4 + 303644672x^{10}y^3 \\
& +716636160x^{10}y^2 + 296943616x^{10}y - 36241408x^{10} - 262144x^9y^{10} \\
& -8994816x^9y^9 - 45858816x^9y^8 - 130760704x^9y^7 - 359268352x^9y^6 \\
& -727465984x^9y^5 - 1602076672x^9y^4 - 2482847744x^9y^3 \\
& -2995290112x^9y^2 - 1846149120x^9y - 348913664x^9 - 98304x^8y^{11} \\
& -475136x^8y^{10} + 12763136x^8y^9 + 57245696x^8y^8 + 176488448x^8y^7 \\
& +485949440x^8y^6 + 751140864x^8y^5 + 812941312x^8y^4 + 138461184x^8y^3 \\
& +178339840x^8y^2 + 717324288x^8y + 239403008x^8 + 1179648x^7y^{11} \\
& +10846208x^7y^{10} + 25182208x^7y^9 + 83787776x^7y^8 + 183123968x^7y^7 \\
& +393723904x^7y^6 + 1348599808x^7y^5 + 3216572416x^7y^4 \\
& +4955504640x^7y^3 + 4595531776x^7y^2 + 1509261312x^7y + 91095040x^7 \\
& +147456x^6y^{12} - 3407872x^6y^{11} - 32227328x^6y^{10} - 112345088x^6y^9 \\
& -461717504x^6y^8 - 1300922368x^6y^7 - 2813050880x^6y^6 \\
& -4290347008x^6y^5 - 5108858880x^6y^4 - 4748722176x^6y^3 \\
& -4892688384x^6y^2 - 2444754944x^6y - 340852736x^6 - 16384x^5y^{13} \\
& -1540096x^5y^{12} + 294912x^5y^{11} + 30982144x^5y^{10} + 70647808x^5y^9 \\
& +289128448x^5y^8 + 487161856x^5y^7 + 270434304x^5y^6 \\
& -1792032768x^5y^5 - 3840475136x^5y^4 - 5334663168x^5y^3 \\
& -2004779008x^5y^2 + 639172608x^5y - 1351811072x^5 + 131072x^4y^{14} \\
& -770048x^4y^{13} - 20635648x^4y^{12} - 135106560x^4y^{11} - 750024704x^4y^{10} \\
& -2958787584x^4y^9 - 9495789056x^4y^8 - 23258333440x^4y^7 \\
& -43101093760x^4y^6 - 53091442752x^4y^5 - 12198367200x^4y^4 \\
& +127105574896x^4y^3 + 344517463048x^4y^2 + 466480175612x^4y \\
& +130604102402x^4 + 98304x^3y^{14} + 49152x^3y^{13} - 18186240x^3y^{12} \\
& -79478784x^3y^{11} - 307216384x^3y^{10} - 1000275968x^3y^9 \\
& -2188623872x^3y^8 - 3662643200x^3y^7 - 3414130688x^3y^6 \\
& -3152330752x^3y^5 - 2019524608x^3y^4 - 3416309760x^3y^3 \\
& -6375309312x^3y^2 - 2844721152x^3y - 74852859904x^3 - 1179648x^2y^{14} \\
& -9584640x^2y^{13} - 22396928x^2y^{12} - 114212864x^2y^{11} \\
& -270958592x^2y^{10} - 749633536x^2y^9 - 2313568256x^2y^8 \\
& -5497683968x^2y^7 - 10820272128x^2y^6 - 16947200000x^2y^5 \\
& -20370915328x^2y^4 - 18696781824x^2y^3 - 5643681792x^2y^2 \\
& +583696384x^2y - 1615856599040x^2 - 147456xy^{15} + 3899392xy^{14} + 33570816xy^{13} \\
& +125681664xy^{12} + 602030080xy^{11} + 1900347392xy^{10} + 5139808256xy^9 \\
& +10044145664xy^8 + 16612196352xy^7 + 22638247936xy^6 + 29191716864xy^5 \\
& +28777054208xy^4 + 29243424768xy^3 + 16444489728xy^2 + 3121807360xy \\
& -1727695385346x - 131072y^{16} - 409600y^{15} + 3088384y^{14} + 21819392y^{13} \\
& +116082688y^{12} + 87145472y^{11} - 2236579328y^{10} - 19164610816y^9 \\
& -89594167168y^8 - 297984745536y^7 - 759241625568y^6 - 1517386782736y^5 \\
& -2324328641528y^4 - 2506728203780y^3 - 1301773765886y^2 - 1631322977026y \\
& -1493082013696 \\
B \quad & -131072x^7y^{14} + 1032192x^7y^{13} + 29646848x^7y^{12} + 177147904x^7y^{11} \\
& +906786816x^7y^{10} + 3559343104x^7y^9 + 11124407808x^7y^8 \\
& +27450376448x^7y^7 + 50686410624x^7y^6 + 65085464640x^7y^5 \\
& +26198544352x^7y^4 - 112221399024x^7y^3 - 334340819976x^7y^2 \\
& -463881771516x^7y - 92606395138x^7 + 131072x^6y^{14} - 1032192x^6y^{13} \\
& -29646848x^6y^{12} - 177147904x^6y^{11} - 906786816x^6y^{10} \\
& -3559343104x^6y^9 - 11124407808x^6y^8 - 27450376448x^6y^7 \\
& -50686410624x^6y^6 - 65085464640x^6y^5 - 26198544352x^6y^4 \\
& +112221399024x^6y^3 + 334340819976x^6y^2 + 463881771516x^6y \\
& +92606395138x^6 + 131072x^5y^{15} - 1032192x^5y^{14} - 29646848x^5y^{13} \\
& -177147904x^5y^{12} - 906786816x^5y^{11} - 3559343104x^5y^{10} \\
& -11124407808x^5y^9 - 27450376448x^5y^8 - 50686410624x^5y^7 \\
& -65085464640x^5y^6 - 26198544352x^5y^5 + 112221399024x^5y^4 \\
& +334340819976x^5y^3 + 463881771516x^5y^2 + 92606395138x^5y \\
& -262144x^4y^{15} + 1933312x^4y^{14} + 60325888x^4y^{13} + 383942656x^4y^{12} \\
& +1990721536x^4y^{11} + 8025473024x^4y^{10} + 25808158720x^4y^9 \\
& +66025160704x^4y^8 + 128823197696x^4y^7 + 180857339904x^4y^6 \\
& +117482553344x^4y^5 - 198244253696x^4y^4 - 780903038976x^4y^3 \\
& -1262104363008x^4y^2 - 649094561792x^4y - 3707756412928x^4 + 16384x^3y^{16} \\
& +3112960x^3y^{15} + 24772608x^3y^{14} + 98942976x^3y^{13} + 530407424x^3y^{12} \\
& +2082295808x^3y^{11} + 8786098176x^3y^{10} + 33531315200x^3y^9 \\
& +111535357440x^3y^8 + 311882465536x^3y^7 + 726329900928x^3y^6
\end{aligned}$$

$$\begin{aligned}
& +1386756481088x^3y^5 + 2112822136800x^3y^4 + 2376946538512x^3y^3 \\
& +1662283887608x^3y^2 + 1726090651140x^3y - 4044672267010x^3 \\
& -131072x^2y^{17} + 1015808x^2y^{16} + 15343616x^2y^{15} + 52760576x^2y^{14} \\
& +300726272x^2y^{13} + 643335168x^2y^{12} + 542334464x^2y^{11} \\
& -6705049344x^2y^{10} - 40541849728x^2y^9 - 148292819904x^2y^8 \\
& -409046741536x^2y^7 - 891661838064x^2y^6 - 1544602061960x^2y^5 \\
& -2093980891580x^2y^4 - 1966213227298x^2y^3 - 1955140478960x^2y^2 \\
& +1644708459512x^2y - 2280466251776x^2 + 16384xy^{17} - 4194304xy^{16} \\
& -25714688xy^{15} - 97554432xy^{14} - 536926208xy^{13} - 1863562240xy^{12} \\
& -7747053056xy^{11} - 27436256512xy^{10} - 90906343808xy^9 - 253182518080xy^8 \\
& -604459732576xy^7 - 1191036582096xy^6 - 1920246932376xy^5 - 2356801400372xy^4 \\
& -1950534281446xy^3 - 1409836184076xy^2 + 975545620742xy + 1443504026370x \\
& +131072y^{18} + 278528y^{17} - 2777088y^{16} - 16846848y^{15} - 110721024y^{14} \\
& -69411840y^{13} + 1812632064y^{12} + 16886757120y^{11} + 81584800896y^{10} \\
& +295316069824y^9 + 841465786656y^8 + 1957437892464y^7 + 3701831511112y^6 \\
& +5633683352028y^5 + 6482791762706y^4 + 7499772670968y^3 + 1707132610052y^2 \\
& +6119933063938y + 1483717967872
\end{aligned}$$

$g = 2, i = 34, m = 6$, neighbours 4, 8, 15, 24, 33, 35, 42

$$\begin{aligned}
W & -36x^5 + 102x^4 - 120x^3 + 84x^2 - 36x + 6 \\
A & -6x^8y - 53x^8 + 36x^5y - 6521x^5 + 30x^4y^3 + 193x^4y^2 - 4218x^4y \\
& +49389x^4 - 36x^3y^3 + 36x^3y^2 + 180x^3y - 19224x^3 + 36x^2y^3 \\
& -252x^2y^2 + 72x^2y + 30672x^2 - 36xy^4 - 108xy^3 + 26521xy^2 - 57912xy \\
& -269052x - 24y^5 - 80y^4 + 5018y^3 - 3141y^2 - 134652y - 36 \\
B & 6x^{11}y + 53x^{11} + 12x^{10}y + 106x^{10} + 6x^9y^2 + 77x^9y + 212x^9 + 24x^8y^2 \\
& +260x^8y + 5613x^8 - 24x^7y^3 - 68x^7y^2 + 4812x^7y - 30622x^7 + 6x^6y^4 \\
& +29x^6y^3 + 148x^6y^2 + 15455x^6y - 61191x^6 - 12x^5y^4 - 10x^5y^3 \\
& +5108x^5y^2 + 288x^5y - 122382x^5 + 6x^4y^5 + 5x^4y^4 + 128x^4y^3 \\
& -274x^4y^2 - 2955x^4y - 3535x^4 + 24x^3y^5 + 188x^3y^4 + 7197x^3y^3 \\
& +2040x^3y^2 + 36482x^3y - 21822x^3 - 24x^2y^6 - 32x^2y^5 + 4638x^2y^4 \\
& -36107x^2y^3 + 17696x^2y^2 - 124053x^2y - 90765x^2 + 12xy^6 + 124xy^5 \\
& +16761xy^4 - 51454xy^3 + 71154xy^2 - 193140xy - 185418x - 24y^7 - 8y^6 + 5102y^5 \\
& -3557y^4 - 115020y^3 - 7654y^2 - 541569y - 134724
\end{aligned}$$

$g = 2, i = 35, m = 16$, neighbours 2, 4, 9, 12, 13, 18, 20, 21, 29, 34, 39, 40

$$\begin{aligned}
W & -1696x^5 - 2496x^4 + 512x^3 + 4448x^2 + 1216x - 512 \\
A & -1024x^{11}y^2 - 1024x^{11}y + 2048x^{11} - 2048x^{10}y^3 - 1024x^{10}y^2 \\
& +5120x^{10}y - 2048x^{10} - 768x^9y^4 + 5888x^9y^3 + 13568x^9y^2 - 7936x^9y \\
& -10752x^9 + 1280x^8y^6 + 4224x^8y^5 + 73024x^8y^4 + 509536x^8y^3 \\
& +5170960x^8y^2 + 47748504x^8y + 457610532x^8 + 256x^7y^6 + 1024x^7y^5 \\
& -6400x^7y^4 - 24832x^7y^3 - 8704x^7y^2 + 27904x^7y + 10752x^7 \\
& -2304x^6y^6 - 4608x^6y^5 - 11008x^6y^4 - 9984x^6y^3 + 20992x^6y^2 \\
& -1792x^6y + 8704x^6 - 768x^5y^7 + 5120x^5y^6 + 256x^5y^5 - 13056x^5y^4 \\
& +14336x^5y^3 - 18688x^5y^2 + 10752x^5y + 4335640790x^5 - 3840x^4y^8 \\
& -11136x^4y^7 - 185152x^4y^6 - 1561824x^4y^5 - 14903184x^4y^4 \\
& -140966616x^4y^3 - 1336048644x^4y^2 + 302708444x^4y - 16553740633x^4 \\
& -512x^3y^8 - 13312x^3y^7 - 13056x^3y^6 + 2816x^3y^5 - 24320x^3y^4 \\
& +79616x^3y^3 + 60416x^3y^2 - 76288x^3y - 4508672x^3 - 256x^2y^9 \\
& +2048x^2y^8 + 16896x^2y^7 + 30208x^2y^6 + 18944x^2y^5 + 97024x^2y^4 \\
& +96512x^2y^3 - 178432x^2y^2 - 65536x^2y + 17626112x^2 + 3840xy^9 \\
& +4096xy^8 + 17920xy^7 + 37248xy^6 + 73536xy^5 + 1166880xy^4 + 8115632xy^3 \\
& -12929973626xy^2 - 7889026528xy + 18350782953x + 2816y^{10} + 7424y^9 + 79488y^8 \\
& +793920y^7 + 7649568y^6 + 72244592y^5 + 686532392y^4 - 2151131824y^3 \\
& -5280267867y^2 + 11621956553y + 14129935708 \\
B & -1280x^{11}y^6 - 3712x^{11}y^5 - 64576x^{11}y^4 - 498528x^{11}y^3 \\
& -5186064x^{11}y^2 - 47760024x^{11}y - 457603876x^{11} - 1280x^{10}y^6 \\
& -3712x^{10}y^5 - 64576x^{10}y^4 - 498528x^{10}y^3 - 5186064x^{10}y^2 \\
& -47760024x^{10}y - 457603876x^{10} - 1280x^9y^7 - 6272x^9y^6 - 72000x^9y^5 \\
& -627680x^9y^4 - 6183120x^9y^3 - 58132152x^9y^2 - 553123924x^9y \\
& -915207752x^9 - 2560x^8y^7 - 11264x^8y^6 - 140288x^8y^5 \\
& -1190784x^8y^4 - 11867712x^8y^3 - 111078240x^8y^2 - 1058487824x^8y \\
& -5711326786x^8 + 2560x^7y^8 + 3584x^7y^7 + 77696x^7y^6 + 686400x^7y^5 \\
& +6842400x^7y^4 + 64258224x^7y^3 + 608566664x^7y^2 - 2877252384x^7y \\
& +9461129131x^7 - 1280x^6y^9 - 3712x^6y^8 - 79936x^6y^7 - 588512x^6y^6
\end{aligned}$$

$$\begin{aligned}
& -6092752x^6y^5 - 56034808x^6y^4 - 537534388x^6y^3 - 757225560x^6y^2 \\
& -11525242522x^6y + 3292198469x^6 + 1280x^5y^9 + 2432x^5y^8 + 1344x^5y^7 \\
& +175584x^5y^6 + 1436048x^5y^5 + 15065816x^5y^4 + 135290500x^5y^3 \\
& -3025911280x^5y^2 - 4941365775x^5y + 12753327600x^5 - 1280x^4y^{10} \\
& -3712x^4y^9 - 81216x^4y^8 - 667104x^4y^7 - 6526800x^4y^6 \\
& -60752760x^4y^5 - 579568884x^4y^4 - 1167695160x^4y^3 \\
& -2378403944x^4y^2 - 5285419908x^4y - 6614839434x^4 - 2048x^3y^{10} \\
& -16128x^3y^9 - 144640x^3y^8 - 1265280x^3y^7 - 12432576x^3y^6 \\
& -114877088x^3y^5 - 1100887856x^3y^4 - 6101794546x^3y^3 \\
& -3729370888x^3y^2 - 26740696992x^3y + 4226431117x^3 + 2560x^2y^{11} \\
& +3328x^2y^{10} + 103808x^2y^9 + 728000x^2y^8 + 7206752x^2y^7 \\
& +67694864x^2y^6 + 641993496x^2y^5 - 2572077896x^2y^4 \\
& +12390963887x^2y^3 - 11268042336x^2y^2 + 27271374042x^2y + 14146255196x^2 \\
& +1024xy^{11} - 2304xy^{10} - 45440xy^9 - 627392xy^8 - 6390240xy^7 - 59743824xy^6 \\
& -573804472xy^5 - 9775029354xy^4 + 2563885749xy^3 - 41737689096xy^2 \\
& +4775284679xy + 18349275625x + 2816y^{12} + 4096y^{11} + 87168y^{10} + 742336y^9 \\
& +7227488y^8 + 68269712y^7 + 649665624y^6 - 2504177704y^5 + 42286045y^4 \\
& +3661675604y^3 - 12803943188y^2 + 54011501533y + 14130197852
\end{aligned}$$

$g = 2, i = 36, m = 4$, neighbours 9, 10, 11, 26, 39, 41

$$\begin{aligned}
W & 8x^3 - 16x^2 + 24x \\
A & -8x^8 - 84x^5 + 24x^4y^2 + 16x^4y + 110x^4 + 32x^3y + 64x^3 - 16x^2y^2 \\
& -16x^2y + 32x^2 - 16xy^3 + 44xy^2 - 184xy + 66x - 16y^4 + 40y^3 + 18y^2 - 182y + 140 \\
B & 8x^{11} + 8x^{10} + 8x^9y + 16x^8y + 76x^8 - 16x^7y^2 - 8x^7y - 26x^7 \\
& +8x^6y^3 + 84x^6y - 94x^6 - 8x^5y^3 + 8x^5y^2 + 66x^5y - 68x^5 \\
& +8x^4y^4 - 16x^4y^3 - 160x^4y^2 - 280x^4y - 20x^4 + 16x^3y^4 \\
& +92x^3y^3 - 128x^3y^2 - 104x^3y + 74x^3 - 16x^2y^5 + 40x^2y^4 \\
& +190x^2y^3 + 256x^2y^2 - 476x^2y - 816x^2 + 52xy^4 - 14xy^3 + 552xy^2 \\
& -394xy - 250x - 16y^6 - 8y^5 - 134y^4 - 140y^3 + 272y^2 - 406y - 108
\end{aligned}$$

$g = 2, i = 37, m = 1$, neighbours 2, 8, 27, 31, 32, 33, 39, 41

$$\begin{aligned}
W & 14x^5 + 34x^4 - 30x^3 - 78x^2 + 8x + 60 \\
A & x^9 + 2x^8y + 2x^8 + x^7y^2 + 4x^7y + x^7 + x^6y^2 + 4x^6y - 2x^6 \\
& +4x^5y^3 + 19x^5y^2 - 119x^5y + 198x^5 - 5x^4y^4 - 13x^4y^3 \\
& +195x^4y^2 - 893x^4y + 647x^4 - 6x^3y^4 - 20x^3y^3 - 26x^3y^2 - 30x^3y \\
& -322x^3 - x^2y^5 - 10x^2y^4 - 29x^2y^3 - 20x^2y^2 - 14x^2y + 4394x^2 \\
& +xy^5 - 14xy^4 - 4xy^3 + 34xy^2 - 250xy - 332x + 6y^6 + 42y^5 - 110y^4 + 5y^3 \\
& +1719y^2 - 3774y - 11424 \\
B & -5x^8y^3 - 22x^8y^2 + 117x^8y - 403x^8 + 5x^7y^4 + 2x^7y^3 \\
& -205x^7y^2 + 871x^7y - 1612x^7 - 5x^6y^4 - 32x^6y^3 + 73x^6y^2 \\
& -169x^6y - 806x^6 + 5x^5y^5 + 12x^5y^4 - 201x^5y^3 + 461x^5y^2 \\
& +130x^5y - 3383x^5 - 5x^4y^5 - 42x^4y^4 - x^4y^3 - 67x^4y^2 - 751x^4y \\
& +2721x^4 - 5x^3y^6 - 29x^3y^5 + 32x^3y^4 - 199x^3y^3 - 356x^3y^2 \\
& +891x^3y - 3083x^3 + 5x^2y^7 + 13x^2y^6 - 218x^2y^5 + 856x^2y^4 \\
& -1452x^2y^3 + 2546x^2y^2 - 444x^2y - 4571x^2 + xy^7 - 14xy^6 + 31xy^5 \\
& -49xy^4 - 279xy^3 + 363xy^2 + 23xy - 5398x + 6y^8 + 32y^7 - 151y^6 + 470y^5 \\
& +386y^4 - 4623y^3 + 4699y^2 - 22779y - 3912
\end{aligned}$$

$g = 2, i = 38, m = 12$, neighbours 6, 9, 14, 19, 21, 29, 30

$$\begin{aligned}
W & -24x^5 - 72x^4 - 96x^3 + 24x^2 + 264x - 96 \\
A & 24x^8y + 196x^8 - 3338x^5 + 24x^4y^3 - 188x^4y^2 - 4228x^4y + 12211x^4 \\
& +1152x^3y^2 + 576x^3y - 13248x^3 + 720x^2y^3 - 576x^2y^2 - 3456x^2y \\
& -11520x^2 + 144xy^4 - 2762xy^2 + 47900xy + 62047x - 48y^5 - 1040y^4 + 2520y^3 \\
& -7695y^2 - 46463y - 32952 \\
B & -24x^{11}y - 196x^{11} + 24x^{10}y + 196x^{10} - 24x^9y^2 - 244x^9y - 392x^9 \\
& +48x^8y^2 + 464x^8y + 3350x^8 - 48x^7y^3 - 128x^7y^2 + 3680x^7y \\
& -18453x^7 - 24x^6y^4 - 100x^6y^3 + 544x^6y^2 - 674x^6y + 21607x^6 \\
& -24x^5y^4 - 76x^5y^3 + 3008x^5y^2 - 14099x^5y - 40060x^5 - 24x^4y^5 \\
& -100x^4y^4 + 520x^4y^3 - 376x^4y^2 - 9988x^4y + 13326x^4 + 48x^3y^5 \\
& +1472x^3y^4 + 3926x^3y^3 - 3984x^3y^2 + 16416x^3y + 5253x^3 - 48x^2y^6 \\
& +160x^2y^5 + 2360x^2y^4 - 17281x^2y^3 + 8416x^2y^2 - 59998x^2y \\
& +46550x^2 + 96xy^5 - 512xy^4 + 2318xy^3 + 27303xy^2 + 2288xy^2 + 142723xy - 45329x \\
& -48y^7 - 848y^6 + 3464y^5 - 16087y^4 - 22776y^3 - 37860y^2 - 154535y - 23736
\end{aligned}$$

$g = 2, i = 39, m = 2$, neighbours 7, 9, 10, 13, 14, 17, 25, 28, 29, 32, 35, 36, 37, 42

$$\begin{aligned}
W & 1808x^5 + 3848x^4 + 3064x^3 + 5840x^2 + 3632x + 4416
\end{aligned}$$

A $8x^{16}y + 16x^{16} + 12x^{15}y^2 - 20x^{15}y - 88x^{15} - 8x^{14}y^3 - 92x^{14}y^2$
 $-108x^{14}y + 88x^{14} - 16x^{13}y^4 - 4x^{13}y^3 + 248x^{13}y^2 + 548x^{13}y$
 $+328x^{13} + 80x^{12}y^4 + 152x^{12}y^3 - 256x^{12}y^2 - 936x^{12}y - 912x^{12}$
 $+4x^{11}y^6 + 4x^{11}y^5 - 240x^{11}y^4 - 620x^{11}y^3 - 292x^{11}y^2$
 $+456x^{11}y + 832x^{11} - 20x^{10}y^6 + 748x^{10}y^4 + 2284x^{10}y^3$
 $+2824x^{10}y^2 + 2044x^{10}y + 376x^{10} + 4x^9y^7 + 140x^9y^6 + 316x^9y^5$
 $-1148x^9y^4 - 5192x^9y^3 - 8288x^9y^2 - 7368x^9y - 3088x^9 - 440x^8y^6$
 $-1772x^8y^5 - 1584x^8y^4 + 3068x^8y^3 + 8848x^8y^2 + 9456x^8y$
 $+4864x^8 - 20x^7y^8 - 132x^7y^7 + 600x^7y^6 + 4216x^7y^5$
 $+9536x^7y^4 + 11196x^7y^3 + 6672x^7y^2 + 1828x^7y - 1304x^7 - 4x^6y^9$
 $+80x^6y^8 + 568x^6y^7 - 136x^6y^6 - 6492x^6y^5 - 19896x^6y^4$
 $-32392x^6y^3 - 31132x^6y^2 - 20052x^6y - 5240x^6 + 8x^5y^9 - 328x^5y^8$
 $-2080x^5y^7 - 3164x^5y^6 + 3352x^5y^5 + 21204x^5y^4 + 41152x^5y^3$
 $+41584x^5y^2 + 26944x^5y - 3259776x^5 + 2x^4y^{11} - 24x^4y^{10}$
 $-38x^4y^9 + 1196x^4y^8 + 6456x^4y^7 + 17528x^4y^6 + 28252x^4y^5$
 $+32438x^4y^4 + 42288x^4y^3 + 89318x^4y^2 - 1467488x^4y - 15968648x^4$
 $+36x^3y^{10} + 640x^3y^9 + 1436x^3y^8 - 2544x^3y^7 - 18612x^3y^6$
 $-48888x^3y^5 - 80052x^3y^4 - 87656x^3y^3 - 64344x^3y^2 - 33200x^3y$
 $-25891648x^3 + 20x^2y^{11} + 24x^2y^{10} - 1304x^2y^9 - 5256x^2y^8$
 $-8316x^2y^7 - 304x^2y^6 + 30928x^2y^5 + 79080x^2y^4 + 112992x^2y^3$
 $+92088x^2y^2 + 51024x^2y - 44686656x^2 + 4xy^{12} - 74xy^{11} - 200xy^{10}$
 $+1402xy^9 + 6884xy^8 + 20436xy^7 + 36664xy^6 + 40616xy^5 + 12554xy^4$
 $-100476xy^3 - 292722xy^2 + 4365864xy - 31794680x - 2y^{13} + 10y^{12} - 22y^{11}$
 $+54y^{10} - 404y^9 - 5084y^8 - 15140y^7 - 28642y^6 - 23930y^5 + 46162y^4$
 $+1843062y^3 + 4792548y^2 + 11833460y - 21692248$

B $-2x^7y^{11} + 20x^7y^{10} - 78x^7y^9 - 852x^7y^8 - 2076x^7y^7$
 $-4940x^7y^6 - 9576x^7y^5 - 18070x^7y^4 - 43320x^7y^3 - 99398x^7y^2$
 $+1458932x^7y + 2048624x^7 + 2x^6y^{11} - 20x^6y^{10} + 78x^6y^9$
 $+852x^6y^8 + 2076x^6y^7 + 4940x^6y^6 + 9576x^6y^5 + 18070x^6y^4$
 $+43320x^6y^3 + 99398x^6y^2 - 1458932x^6y - 2048624x^6 - 2x^5y^{12}$
 $+20x^5y^{11} - 78x^5y^{10} - 852x^5y^9 - 2076x^5y^8 - 4940x^5y^7$
 $-9576x^5y^6 - 18070x^5y^5 - 43320x^5y^4 - 99398x^5y^3$
 $+1458932x^5y^2 + 2048624x^5y + 4x^4y^{12} - 36x^4y^{11} + 116x^4y^{10}$
 $+18064x^4y^9 + 5856x^4y^8 + 14032x^4y^7 + 29032x^4y^6 + 55292x^4y^5$
 $+122780x^4y^4 + 285436x^4y^3 - 2719068x^4y^2 - 7015112x^4y - 30361376x^4$
 $+286x^3y^{11} + 884x^3y^{10} - 422x^3y^9 - 7968x^3y^8 - 33408x^3y^7$
 $-99224x^3y^6 - 236152x^3y^5 - 490714x^3y^4 + 2425556x^3y^3$
 $-1726746x^3y^2 + 7613044x^3y - 8425376x^3 - 2x^2y^{14} + 28x^2y^{13}$
 $-38x^2y^{12} - 1464x^2y^{11} - 6016x^2y^{10} - 17788x^2y^9 - 32752x^2y^8$
 $-34986x^2y^7 + 12372x^2y^6 + 158774x^2y^5 + 2057228x^2y^4$
 $+10279424x^2y^3 + 3311032x^2y^2 + 34429972x^2y - 9698712x^2 + 2xy^{14}$
 $-64xy^{13} - 272xy^{12} + 902xy^{11} + 6262xy^{10} + 25730xy^9 + 66140xy^8$
 $+116418xy^7 + 123108xy^6 - 41252xy^5 + 1215906xy^4 + 11665370xy^3$
 $+6114710xy^2 + 53072760xy + 19218168x - 2y^{15} + 8y^{14} - 30y^{13} - 14y^{12} - 748y^{11}$
 $-5818y^{10} - 21572y^9 - 68374y^8 - 156144y^7 - 241830y^6 + 1472298y^5$
 $+6423332y^4 + 25676450y^3 + 24999376y^2 + 77126596y + 2191192$

$g = 2, i = 40, m = 8, \text{neighbours } 10, 14, 26, 33, 35$

W $-4x^5 - 6x^4 - 4x^3 - 22x^2 + 12x$

A $-16x^5 - 22x^4 - 132x^3 - 32x^2 + 64xy^2 - 118x - 104y^2 - 428y + 128$

B $-26x^7 - 26x^6 - 52x^5y - 26x^5 - 104x^4y - 204x^4 + 16x^3y^3 + 272x^3y$
 $-198x^3 - 26x^2y^3 - 80x^2y^2 - 240x^2y - 688x^2 + 32xy^4 - 22xy^3$
 $+144xy^2 + 84xy + 4x - 84y^4 - 214y^3 - 16y^2 - 844y - 54$

$g = 2, i = 41, m = 8, \text{neighbours } 2, 5, 9, 11, 12, 19, 20, 32, 33, 36, 37$

W $-117558x^5 + 38784x^4 + 30264x^3 - 252996x^2 - 128226x + 127116$

A $-512x^{10}y - 1024x^{10} - 768x^9y^2 - 2048x^9y - 1024x^9 + 128x^8y^3$
 $-768x^8y^2 - 2432x^8y - 768x^8 + 704x^7y^4 + 1024x^7y^3 - 960x^7y^2$
 $-768x^7y - 768x^7 + 384x^6y^5 + 1408x^6y^4 + 3328x^6y^3 + 6272x^6y^2$
 $+6528x^6y + 4352x^6 + 96x^5y^6 + 1120x^5y^5 - 4352x^5y^4$
 $-334832x^5y^3 - 11878448x^5y^2 - 406653944x^5y - 13902935040x^5$
 $-8x^4y^7 + 352x^4y^6 + 3016x^4y^5 + 87660x^4y^4 + 2901392x^4y^3$
 $+98716002x^4y^2 - 80883991x^4y + 9097979801x^4 + 64x^3y^7 - 1984x^3y^6$
 $-5760x^3y^5 - 2368x^3y^4 + 7616x^3y^3 + 17152x^3y^2 + 11904x^3y$

	$+5611356832x^3 - 1600x^2y^7 - 6912x^2y^6 - 15168x^2y^5 - 23232x^2y^4$ $-23552x^2y^3 - 15424x^2y^2 + 1152x^2y - 71372700016x^2 - 512xy^8$ $-4288xy^7 - 16416xy^6 - 36128xy^5 - 68736xy^4 - 419376xy^3 - 11945264xy^2$ $+1978749496xy - 15217520160x + 32y^9 + 480y^8 + 744y^7 + 2768y^6 + 35656y^5$ $+1595596y^4 + 13876736964y^3 - 1455539818y^2 + 77593743507y + 55258341271$
B	$-32x^8y^6 - 160x^8y^5 + 8832x^8y^4 + 346160x^8y^3 + 11896432x^8y^2$ $+406669816x^8y + 83057052x^8 + 8x^7y^7 + 32x^7y^6 - 2248x^7y^5$ $-84332x^7y^4 - 2887568x^7y^3 - 98693346x^7y^2 + 80903191x^7y$ $+20764263x^7 - 64x^6y^7 - 352x^6y^6 + 17504x^6y^5 + 701152x^6y^4$ $+24139024x^6y^3 + 825236064x^6y^2 + 572783920x^6y + 83057052x^6$ $+16x^5y^8 + 72x^5y^7 - 4464x^5y^6 - 170912x^5y^5 - 5859468x^5y^4$ $-200274260x^5y^3 + 63113036x^5y^2 + 122431717x^5y + 9562538279x^5$ $-128x^4y^8 - 768x^4y^7 + 34592x^4y^6 + 1419488x^4y^5$ $+48996864x^4y^4 + 1675303472x^4y^3 + 1994596768x^4y^2 - 833205848x^4y$ $+17764010788x^4 - 96x^3y^9 - 2336x^3y^8 - 9080x^3y^7 - 23120x^3y^6$ $-41208x^3y^5 - 115124x^3y^4 - 2917856x^3y^3 + 3303305518x^3y^2$ $-2246949457x^3y + 14255732935x^3 + 8x^2y^{10} - 736x^2y^9 - 4008x^2y^8$ $-13796x^2y^7 - 24752x^2y^6 + 346630x^2y^5 + 3468771531x^2y^4$ $-3831265281x^2y^3 + 552997610x^2y^2 - 1616230365x^2y - 63051173445x^2$ $-256xy^{10} - 2272xy^9 - 10000xy^8 - 26712xy^7 - 59968xy^6 - 255776xy^5$ $-5975668xy^4 + 992289564xy^3 - 1190637108xy^2 + 3819075005xy + 7441668679x$ $+16y^{11} + 200y^{10} + 80y^9 + 1616y^8 + 18236y^7 + 795580y^6 + 6938045180y^5$ $-4193583223y^4 + 68647934067y^3 - 2667508064y^2 + 155793471137y + 71416818727$
$g = 2, i = 42, m = 8$, neighbours 46, 8, 14, 20, 23, 32, 34, 39	
W	$-198x^5 + 96x^4 + 12x^3 + 516x^2 - 246x + 252$
A	$192x^7 + 384x^6y - 576x^6 + 288x^5y^2 + 528x^5y - 36144x^5 - 24x^4y^3$ $+180x^4y^2 - 1401x^4y + 38109x^4 + 192x^3y^3 - 1728x^3y^2 - 576x^3y$ $-4080x^3 - 1344x^2y^3 - 576x^2y^2 - 2112x^2y + 334896x^2 - 768xy^4$ $-384xy^3 - 6240xy^2 - 24816xy - 69888x + 96y^5 - 1968y^4 + 39084y^3 - 2436y^2$ $+416421y + 270099$
B	$-96x^8y^2 - 912x^8y - 2676x^8 + 24x^7y^3 + 204x^7y^2 + 441x^7y - 669x^7$ $+192x^6y^3 + 1920x^6y^2 + 6264x^6y + 2676x^6 - 48x^5y^4 - 432x^5y^3$ $-1086x^5y^2 + 897x^5y - 130659x^5 - 384x^4y^4 - 4032x^4y^3$ $-14640x^4y^2 + 19392x^4y - 92316x^4 - 288x^3y^5 + 1200x^3y^4$ $-4728x^3y^3 + 4332x^3y^2 - 10167x^3y - 6765x^3 + 24x^2y^6 + 204x^2y^5$ $+10329x^2y^4 - 8373x^2y^3 - 16032x^2y^2 + 19389x^2y - 251985x^2$ $+384xy^6 + 384xy^5 + 3840xy^4 + 11472xy^3 + 24834xy^2 + 40401xy - 76227x - 48y^7$ $+1104y^6 - 20478y^5 + 15057y^4 - 292155y^3 - 4026y^2 - 829611y - 333603$

Table 2: The data of the identities (3).

$\{i, j\} = \{2, 4\}, k_{ij} = 2^2$	
α	1
β	$x^4 - x^3y + x^2y^2 - xy^3 + y^4 + 1$
γ	$-y^4 - y^3 + 3$
$\{i, j\} = \{2, 5\}, k_{ij} = 2$	
α	2
β	$2x^4 - 2x^3y + 2x^2y^2 - 2xy^3 + y^4 + y^2 + 5$
γ	$-y^4 + y^2 + 3$
$\{i, j\} = \{2, 11\}, k_{ij} = 2^2$	
α	1
β	$x^4 - x^3y + x^2y^2 - xy^3 - y^4 - 2y^3 + 7$
γ	$y^4 + y^3 - 3$
$\{i, j\} = \{2, 14\}, k_{ij} = 2$	
α	0
β	-1
γ	1
$\{i, j\} = \{2, 19\}, k_{ij} = 17$	
α	-1
β	$-x^4 + x^3y - x^2y^2 + xy^3 + y^4 + 4y^3 + 6y^2 + 12y + 17$

$$\begin{aligned}
&\gamma \quad -y^4 - 2y^3 - 3y^2 - 6y - 9 \\
&\{i, j\} = \{2, 25\}, k_{ij} = 1 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y - x^2y^2 - xy^2 - xy - 2x + y^3 + y^2 + 2y - 1 \\
&\gamma \quad -y^3 - y^2 - y - 2 \\
&\{i, j\} = \{2, 32\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad x^5 - 3y - 2 \\
&\beta \quad x^9 - x^8y + x^7y^2 - x^6y^2 - 2x^6y + 4x^6 + x^5y^3 + 2x^5y^2 - 4x^5y \\
&\quad + x^5 + 7x^4 - 7x^3y + 7x^2y^2 - 4xy^2 - 8xy + 25x + 4y^3 + 8y^2 - 28y - 2 \\
&\gamma \quad x^5y^3 - x^5y^2 - 2x^5y + 4x^5 + 7y^3 - 4y^2 - 8y + 25 \\
&\{i, j\} = \{2, 35\}, k_{ij} = 2^5 \\
&\alpha \quad -3x^5 + 9y - 4 \\
&\beta \quad -3x^9 + 3x^8y - 3x^7y^2 + 3x^5y^3 - 3x^5 - 7x^4 + 7x^3y - 7x^2y^2 \\
&\quad - 9xy^2 - 3xy - 36x + 16y^3 + 12y^2 + 48y + 32 \\
&\gamma \quad -3x^5y^3 - 7y^3 - 9y^2 - 3y - 36 \\
&\{i, j\} = \{2, 37\}, k_{ij} = 2^2 \cdot 3 \cdot 17 \\
&\alpha \quad -5x^5 - 25y - 122 \\
&\beta \quad -5x^9 + 5x^8y - 5x^7y^2 - 5x^6y^2 - 10x^6y - 20x^6 + 5x^5y^3 \\
&\quad + 10x^5y^2 + 20x^5y - 5x^5 - 87x^4 + 87x^3y - 87x^2y^2 - 62xy^2 - 114xy - 163x \\
&\quad + 62y^3 + 114y^2 + 138y - 122 \\
&\gamma \quad -5x^5y^3 - 5x^5y^2 - 10x^5y - 20x^5 - 87y^3 - 62y^2 - 114y - 163 \\
&\{i, j\} = \{2, 41\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad -7x^5 - 147y - 220 \\
&\beta \quad -7x^9 + 7x^8y - 7x^7y^2 - 14x^6y^2 - 28x^6y - 70x^6 + 14x^5y^3 \\
&\quad + 28x^5y^2 + 70x^5y - 7x^5 - 157x^4 + 157x^3y - 157x^2y^2 - 167xy^2 - 271xy \\
&\quad - 268x + 167y^3 + 271y^2 + 121y - 220 \\
&\gamma \quad -7x^5y^3 - 14x^5y^2 - 28x^5y - 70x^5 - 157y^3 - 167y^2 - 271y - 268 \\
&\{i, j\} = \{3, 4\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{3, 10\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 - y^2 + y - 6 \\
&\gamma \quad x^4 + 1 \\
&\{i, j\} = \{3, 14\}, k_{ij} = 2 \cdot 11^2 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - 3x^3 - 9x^2 - 27x + y^2 - y - 77 \\
&\gamma \quad x^4 + 3x^3 + 9x^2 + 27x + 82 \\
&\{i, j\} = \{3, 15\}, k_{ij} = 2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 - x^3 - x^2 - x - y^2 + y - 7 \\
&\gamma \quad x^4 + x^3 + x^2 + x + 2 \\
&\{i, j\} = \{3, 16\}, k_{ij} = 2 \cdot 5^3 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + 3x^3 - 9x^2 + 27x - y^2 + y - 87 \\
&\gamma \quad x^4 - 3x^3 + 9x^2 - 27x + 82 \\
&\{i, j\} = \{3, 17\}, k_{ij} = 2^3 \\
&\alpha \quad -2 \\
&\beta \quad -x^4 - 2y^2 + 2y - 11 \\
&\gamma \quad x^4 + 1 \\
&\{i, j\} = \{3, 18\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad -2 \\
&\beta \quad -x^4 + x^3 - x^2 + x - 2y^2 + 2y - 12 \\
&\gamma \quad x^4 - x^3 + x^2 - x + 2 \\
&\{i, j\} = \{3, 23\}, k_{ij} = 5 \cdot 11 \\
&\alpha \quad 2x^4 - 17 \\
&\beta \quad -2x^7 + 2x^6 - 4x^5 + 2x^4y^2 - 2x^4y + 16x^4 + 5x^3 + 9x^2 - 4x - 17y^2 + 17y \\
&\quad - 72
\end{aligned}$$

$$\begin{aligned}
&\gamma \quad -2x^7 + 2x^6 - 4x^5 + 6x^4 + 5x^3 + 9x^2 - 4x + 13 \\
&\{i, j\} = \{3, 24\}, k_{ij} = 3 \\
&\alpha \quad -1 \\
&\beta \quad -x^3 - x^2 - y^2 + y - 4 \\
&\gamma \quad x^3 + x^2 - 1 \\
&\{i, j\} = \{3, 26\}, k_{ij} = 2 \cdot 3 \cdot 11 \\
&\alpha \quad -5x - 4 \\
&\beta \quad -5x^4 - 4x^3 + 10x^2 - 5xy^2 + 5xy - 17x - 4y^2 + 4y - 45 \\
&\gamma \quad 5x^4 + 4x^3 - 10x^2 - 8x + 25 \\
&\{i, j\} = \{4, 5\}, k_{ij} = 2^3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 + x^3 + x^2 + x + y^2 + y + 7 \\
&\gamma \quad -x^4 - x^3 - x^2 - x - 2 \\
&\{i, j\} = \{4, 7\}, k_{ij} = 2^2 \cdot 7 \\
&\alpha \quad -1 \\
&\beta \quad x^4 - 2x^3 + 4x^2 - 8x - y^2 - y + 12 \\
&\gamma \quad -x^4 + 2x^3 - 4x^2 + 8x - 17 \\
&\{i, j\} = \{4, 9\}, k_{ij} = 3 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{4, 11\}, k_{ij} = 2^3 \\
&\alpha \quad 2 \\
&\beta \quad -x^4 + x^3 - x^2 + x + 2y^2 + 2y + 8 \\
&\gamma \quad x^4 - x^3 + x^2 - x + 2 \\
&\{i, j\} = \{4, 15\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \\
&\alpha \quad 1 \\
&\beta \quad x^4 + 3x^3 + 9x^2 + 27x + y^2 + y + 87 \\
&\gamma \quad -x^4 - 3x^3 - 9x^2 - 27x - 82 \\
&\{i, j\} = \{4, 18\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 2 \\
&\beta \quad x^4 + 2y^2 + 2y + 11 \\
&\gamma \quad -x^4 - 1 \\
&\{i, j\} = \{4, 19\}, k_{ij} = 3^2 \cdot 11 \\
&\alpha \quad -32 \\
&\beta \quad 16x^4 - 24x^3 + 36x^2 - 54x - 32y^2 - 32y - 63 \\
&\gamma \quad -16x^4 + 24x^3 - 36x^2 + 54x - 97 \\
&\{i, j\} = \{4, 20\}, k_{ij} = 2^2 \cdot 5 \\
&\alpha \quad -x + 3 \\
&\beta \quad x^4 - 3x^3 - x^2 - xy^2 - xy - 2x + 3y^2 + 3y + 17 \\
&\gamma \quad x^4 - 3x^3 - x^2 + 3x + 2 \\
&\{i, j\} = \{4, 24\}, k_{ij} = 3^2 \cdot 11 \\
&\alpha \quad -2x^4 + 19 \\
&\beta \quad -2x^7 - 2x^6 - 4x^5 - 2x^4y^2 - 2x^4y - 16x^4 + 7x^3 - 11x^2 - 4x + 19y^2 + 19y \\
&\quad + 80 \\
&\gamma \quad 2x^7 + 2x^6 + 4x^5 + 6x^4 - 7x^3 + 11x^2 + 4x + 15 \\
&\{i, j\} = \{4, 25\}, k_{ij} = 2 \cdot 7 \\
&\alpha \quad 5x - 6 \\
&\beta \quad 5x^4 - 6x^3 + 10x^2 + 5xy^2 + 5xy + 13x - 6y^2 - 6y - 5 \\
&\gamma \quad -5x^4 + 6x^3 - 10x^2 + 12x - 25 \\
&\{i, j\} = \{4, 31\}, k_{ij} = 2^4 \\
&\alpha \quad -x + 3 \\
&\beta \quad x^4 - 3x^3 + x^2 - xy^2 - xy - 8x + 3y^2 + 3y + 17 \\
&\gamma \quad x^4 - 3x^3 + x^2 - 3x + 2 \\
&\{i, j\} = \{4, 34\}, k_{ij} = 2^4 \cdot 3^2 \cdot 7 \\
&\alpha \quad -5x^4 + 181 \\
&\beta \quad -5x^7 - 10x^6 - 25x^5 - 5x^4y^2 - 5x^4y - 85x^4 + 31x^3 - 28x^2 - 25x + 181y^2 \\
&\quad + 181y + 827 \\
&\gamma \quad 5x^7 + 10x^6 + 25x^5 + 60x^4 - 31x^3 + 28x^2 + 25x + 78
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{4, 35\}, k_{ij} = 2^3 \cdot 5^2 \\
&\alpha \quad -3x^4 + 53 \\
&\beta \quad -3x^7 - 3x^6 - 9x^5 - 3x^4y^2 - 3x^4y - 30x^4 + 17x^3 - 31x^2 + 3x + 53y^2 + 53y \\
&\quad + 206 \\
&\gamma \quad 3x^7 + 3x^6 + 9x^5 + 15x^4 - 17x^3 + 31x^2 - 3x + 59 \\
&\{i, j\} = \{5, 6\}, k_{ij} = 3 \\
&\alpha \quad 3 \\
&\beta \quad -3x^4 - 3x^3y - 3x^2y^2 - 3xy^3 - y^4 + 2y^2 + 7 \\
&\gamma \quad y^4 + y^2 + 5 \\
&\{i, j\} = \{5, 13\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad -2 \\
&\beta \quad 2x^4 + 2x^3y + 2x^2y^2 + 2xy^3 + y^4 + y^3 - 2y^2 + 2y - 5 \\
&\gamma \quad -y^4 + y^3 - 2y^2 + 2y - 7 \\
&\{i, j\} = \{5, 14\}, k_{ij} = 2^4 \\
&\alpha \quad 2 \\
&\beta \quad -2x^4 - 2x^3y - 2x^2y^2 - 2xy^3 - y^4 + y^3 + 2y^2 + 2y + 5 \\
&\gamma \quad y^4 + y^3 + 2y^2 + 2y + 7 \\
&\{i, j\} = \{5, 20\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad -7x^5 + 49y + 36 \\
&\beta \quad 7x^9 + 7x^8y + 7x^7y^2 + 7x^6y^2 - 14x^6y + 14x^6 + 7x^5y^3 - 14x^5y^2 \\
&\quad + 14x^5y + 7x^5 - 29x^4 - 29x^3y - 29x^2y^2 + 20xy^2 + 16xy + 229x + 20y^3 \\
&\quad + 16y^2 + 180y - 36 \\
&\gamma \quad 7x^5y^3 - 7x^5y^2 + 14x^5y - 14x^5 - 29y^3 - 20y^2 - 16y - 229 \\
&\{i, j\} = \{5, 21\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad 7x^5 - 49y + 48 \\
&\beta \quad -7x^9 - 7x^8y - 7x^7y^2 + 7x^6y^2 + 14x^6y + 14x^6 + 7x^5y^3 \\
&\quad + 14x^5y^2 + 14x^5y - 7x^5 - 55x^4 - 55x^3y - 55x^2y^2 + 6xy^2 + 54xy - 191x \\
&\quad + 6y^3 + 54y^2 - 142y - 48 \\
&\gamma \quad -7x^5y^3 - 7x^5y^2 - 14x^5y - 14x^5 - 55y^3 - 6y^2 - 54y + 191 \\
&\{i, j\} = \{5, 23\}, k_{ij} = 3^2 \cdot 7^2 \\
&\alpha \quad -25x^5 + 625y + 441 \\
&\beta \quad 25x^9 + 25x^8y + 25x^7y^2 + 50x^6y^2 - 125x^6y + 250x^6 + 25x^5y^3 \\
&\quad - 75x^5y^2 + 125x^5y + 275x^5 - 416x^4 - 416x^3y - 416x^2y^2 - 207xy^2 \\
&\quad + 855xy + 1415x + 209y^3 + 648y^2 + 1645y + 974 \\
&\gamma \quad 25x^5y^3 - 50x^5y^2 + 125x^5y - 250x^5 - 416y^3 + 207y^2 - 855y - 1415 \\
&\{i, j\} = \{5, 26\}, k_{ij} = 3 \cdot 5 \cdot 7 \\
&\alpha \quad 4x^5 - 16y + 21 \\
&\beta \quad -4x^9 - 4x^8y - 4x^7y^2 + 4x^6y^2 + 4x^6y + 4x^5y^3 + 4x^5y^2 - 4x^5 \\
&\quad - 25x^4 - 25x^3y - 25x^2y^2 + 9xy^2 + 5xy - 84x + 9y^3 + 5y^2 - 68y - 21 \\
&\gamma \quad -4x^5y^3 - 4x^5y^2 - 4x^5y - 25y^3 - 9y^2 - 5y + 84 \\
&\{i, j\} = \{5, 32\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad x^5 - 3y - 2 \\
&\beta \quad -x^9 - x^8y - x^7y^2 - x^6y^2 + 2x^6 - x^5y^3 + 2x^5y - x^5 + 5x^4 \\
&\quad + 5x^3y + 5x^2y^2 + 2xy^2 + 6xy - 25x + 2y^3 + 6y^2 - 22y + 2 \\
&\gamma \quad -x^5y^3 + x^5y^2 - 2x^5 + 5y^3 - 2y^2 - 6y + 25 \\
&\{i, j\} = \{5, 41\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7 \\
&\alpha \quad -13x^5 + 507y - 154 \\
&\beta \quad 13x^9 + 13x^8y + 13x^7y^2 - 26x^6y^2 - 78x^6y - 182x^6 - 26x^5y^3 \\
&\quad - 78x^5y^2 - 182x^5y + 13x^5 + 349x^4 + 349x^3y + 349x^2y^2 - 191xy^2 - 885xy \\
&\quad + 574x - 191y^3 - 885y^2 + 67y + 154 \\
&\gamma \quad 13x^5y^3 + 26x^5y^2 + 78x^5y + 182x^5 + 349y^3 + 191y^2 + 885y - 574 \\
&\{i, j\} = \{6, 8\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \\
&\alpha \quad 16 \\
&\beta \quad 8x^4 - 4x^3y + 2x^2y^2 - xy^3 + y^4 + y^3 - 14y^2 - 28y - 104 \\
&\gamma \quad -y^4 - 2y^3 + 28y^2 + 56y + 224 \\
&\{i, j\} = \{6, 9\}, k_{ij} = 2 \cdot 13 \\
&\alpha \quad -2 \\
&\beta \quad -x^4 - x^3 - x^2 - x - 2 \\
&\gamma \quad x^4 + x^3 + x^2 + x - 2y^2 + 4y - 14
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{6, 19\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{6, 20\}, k_{ij} = 2^3 \cdot 3^3 \cdot 5 \cdot 13 \\
&\alpha \quad -184x^5 + 2116y + 14040 \\
&\beta \quad -92x^9 + 46x^8y - 23x^7y^2 + 46x^6y^2 + 184x^6y - 736x^6 - 23x^5y^3 \\
&\quad -92x^5y^2 + 368x^5y - 92x^5 + 6928x^4 - 3464x^3y + 1732x^2y^2 - 7696xy^2 \\
&\quad + 2704xy - 25628x + 3848y^3 - 1352y^2 + 13872y + 7020 \\
&\gamma \quad -23x^5y^3 + 92x^5y^2 + 368x^5y - 1472x^5 + 1732y^3 - 15392y^2 + 5408y - 51256 \\
&\{i, j\} = \{6, 22\}, k_{ij} = 2^3 \cdot 7 \\
&\alpha \quad 96x^5 - 228y^2 + 2528y + 56 \\
&\beta \quad 48x^9 - 100x^8y + 50x^7y^2 - 150x^6y^2 + 100x^6y - 600x^6 + 50x^5y^3 \\
&\quad -52x^5y^2 + 276x^5y + 648x^5 + 3990x^4 + 27550x^3y - 13775x^2y^2 \\
&\quad + 40525xy^2 - 21950xy - 4822x - 228y^4 - 12159y^3 + 23688y^2 + 8354y + 936 \\
&\gamma \quad -152x^7y - 152x^6y + 50x^5y^3 - 148x^5y^2 + 48x^5y - 1200x^5 + 456x^3y^2 \\
&\quad -1064x^3y + 7828x^3 + 456x^2y^2 + 58026x^2y + 7828x^2 - 456xy^3 + 1520xy^2 \\
&\quad + 50198xy + 7828x - 14687y^3 + 24544y^2 - 1530y - 1816 \\
&\{i, j\} = \{6, 23\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 13 \\
&\alpha \quad -112x^5 + 784y + 1560 \\
&\beta \quad -56x^9 + 28x^8y - 14x^7y^2 + 14x^6y^2 + 196x^6y - 392x^6 - 14x^5y^3 \\
&\quad -84x^5y^2 + 392x^5y - 448x^5 + 724x^4 - 362x^3y + 181x^2y^2 - 1749xy^2 \\
&\quad + 378xy - 6244x + 965y^3 - 1938y^2 + 3892y - 5464 \\
&\gamma \quad -14x^5y^3 + 28x^5y^2 + 392x^5y - 784x^5 + 181y^3 - 3498y^2 + 756y - 12488 \\
&\{i, j\} = \{6, 27\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad -96x^5 - 228y^2 - 2528y + 40 \\
&\beta \quad -48x^9 + 100x^8y - 50x^7y^2 - 150x^6y^2 - 100x^6y - 600x^6 + 50x^5y^3 \\
&\quad + 52x^5y^2 + 276x^5y - 648x^5 + 3278x^4 + 19850x^3y - 9925x^2y^2 \\
&\quad - 28975xy^2 - 15850xy + 2510x - 228y^4 + 8309y^3 + 16684y^2 - 7330y - 776 \\
&\gamma \quad 152x^7y - 152x^6y - 50x^5y^3 - 148x^5y^2 - 48x^5y - 1200x^5 + 456x^3y^2 \\
&\quad + 1064x^3y + 6612x^3 - 456x^2y^2 + 41914x^2y - 6612x^2 + 456xy^3 + 1520xy^2 \\
&\quad - 35302xy + 6612x - 10837y^3 - 17556y^2 - 3010y - 1592 \\
&\{i, j\} = \{6, 30\}, k_{ij} = 2^5 \cdot 5 \\
&\alpha \quad 64x^5 - 72y^2 + 1312y + 968 \\
&\beta \quad 32x^9 + 8x^8y - 4x^7y^2 + 12x^6y^2 - 48x^6 - 4x^5y^3 + 36x^5y^2 \\
&\quad + 48x^5y + 80x^5 + 3096x^4 - 1362x^3y + 681x^2y^2 - 2555xy^2 + 2816xy + 2224x \\
&\quad - 72y^4 + 1705y^3 - 1769y^2 - 1154y + 888 \\
&\gamma \quad 48x^7y + 48x^6y - 4x^5y^3 - 24x^5y^2 - 96x^5 + 144x^3y^2 - 1392x^3y \\
&\quad + 5256x^3 + 144x^2y^2 - 1020x^2y + 5256x^2 - 144xy^3 + 1392xy^2 - 4884xy \\
&\quad + 393y^3 - 2842y^2 - 3488y - 808 \\
&\{i, j\} = \{6, 33\}, k_{ij} = 2^4 \cdot 3 \cdot 7^2 \\
&\alpha \quad 32x^5 - 128y + 24 \\
&\beta \quad 16x^9 - 8x^8y + 4x^7y^2 - 4x^6y^2 - 64x^6y + 144x^6 + 4x^5y^3 \\
&\quad + 28x^5y^2 - 136x^5y + 160x^5 + 172x^4 - 86x^3y + 43x^2y^2 + 213xy^2 - 560xy \\
&\quad + 1164x - 85y^3 + 493y^2 - 1206y + 1176 \\
&\gamma \quad 4x^5y^3 - 8x^5y^2 - 128x^5y + 288x^5 + 43y^3 + 426y^2 - 1120y + 2328 \\
&\{i, j\} = \{6, 38\}, k_{ij} = 2^4 \cdot 3^3 \\
&\alpha \quad -2024x^5 - 3168y^2 - 91256y - 54416 \\
&\beta \quad -1012x^9 + 858x^8y - 429x^7y^2 - 1287x^6y^2 - 1716x^6y - 15444x^6 \\
&\quad + 429x^5y^3 + 275x^5y^2 + 7062x^5y - 16456x^5 + 247176x^4 + 272532x^3y \\
&\quad - 136266x^2y^2 - 333294xy^2 - 37128xy + 752432x - 3168y^4 + 57682y^3 + 75174y^2 \\
&\quad - 695332y - 574536 \\
&\gamma \quad 704x^7y - 704x^6y - 429x^5y^3 - 1870x^5y^2 - 2024x^5y - 30888x^5 \\
&\quad + 6336x^3y^2 + 49984x^3y + 519904x^3 - 6336x^2y^2 + 742256x^2y - 519904x^2 \\
&\quad + 6336xy^3 + 62656xy^2 - 172368xy + 1039808x - 148938y^3 + 6676y^2 + 320464y - 54848 \\
&\{i, j\} = \{7, 10\}, k_{ij} = 2^3 \\
&\alpha \quad -2 \\
&\beta \quad -2x^4 + 2x^3y + 2x^3 - 2x^2y^2 - 4x^2y - 2x^2 + 2xy^3 + 6xy^2 + 6xy + 2x - y^4 \\
&\quad - 4y^3 - 7y^2 - 3y - 7 \\
&\gamma \quad y^4 + 4y^3 + 5y^2 + 5y - 3
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{7, 11\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y + x^3 - x^2y^2 - 2x^2y - x^2 + xy^3 + 3xy^2 + 3xy + x + y^4 + 4y^3 \\
&\quad + 4y^2 + 6y - 8 \\
&\gamma \quad -y^4 - 4y^3 - 5y^2 - 5y + 3 \\
&\{i, j\} = \{7, 12\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y + x^3 - x^2y^2 - 2x^2y - x^2 + xy^3 + 3xy^2 + 3xy + x + y^4 - 2y - 4 \\
&\gamma \quad -y^4 - 2y^3 - 3y^2 - y + 1 \\
&\{i, j\} = \{7, 25\}, k_{ij} = 2^4 \\
&\alpha \quad 3x^5 - 9y - 13 \\
&\beta \quad 3x^9 - 3x^8y - 3x^8 + 3x^7y^2 + 6x^7y + 3x^7 + 3x^6y^2 + 6x^6y + 12x^6 \\
&\quad - 3x^5y^3 - 9x^5y^2 - 18x^5y - 9x^5 - 16x^4 + 16x^3y + 16x^3 - 16x^2y^2 \\
&\quad - 32x^2y - 16x^2 - 7xy^2 - 44xy - 25x + 7y^3 + 51y^2 + 60y + 12 \\
&\gamma \quad 3x^5y^3 + 12x^5y^2 + 15x^5y + 15x^5 - 16y^3 - 55y^2 - 92y - 41 \\
&\{i, j\} = \{7, 30\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y + x^3 - x^2y^2 - 2x^2y - x^2 + 2xy^2 + xy + 2x - y^3 - 2y^2 - y - 4 \\
&\gamma \quad -y^3 - y^2 - 2y + 1 \\
&\{i, j\} = \{7, 32\}, k_{ij} = 3^2 \cdot 7 \cdot 13 \\
&\alpha \quad 10x^5 - 100y - 191 \\
&\beta \quad 10x^9 - 10x^8y - 10x^8 + 10x^7y^2 + 20x^7y + 10x^7 - 10x^6y^2 - 30x^6y \\
&\quad + 30x^6 + 10x^5y^3 + 40x^5y^2 - 20x^5 - 81x^4 + 81x^3y + 81x^3 - 81x^2y^2 \\
&\quad - 162x^2y - 81x^2 + 181xy^2 + 53xy + 327x - 181y^3 - 234y^2 - 480y - 518 \\
&\gamma \quad 10x^5y^3 + 20x^5y^2 + 40x^5 - 81y^3 - 62y^2 - 190y + 246 \\
&\{i, j\} = \{7, 39\}, k_{ij} = 2^2 \cdot 13 \\
&\alpha \quad 3x^5 - 18y - 8 \\
&\beta \quad 3x^9 - 3x^8y - 3x^8 + 3x^7y^2 + 6x^7y + 3x^7 + 6x^6y^2 + 15x^6y + 12x^6 \\
&\quad - 3x^5y^3 - 6x^5y^2 - 3x^5y + 6x^5 + 4x^4 - 4x^3y - 4x^3 + 4x^2y^2 \\
&\quad + 8x^2y + 4x^2 + 26xy^2 + 32xy + 52x - 22y^3 - 20y^2 - 58y - 4 \\
&\gamma \quad 3x^5y^3 + 15x^5y^2 + 24x^5y + 15x^5 + 4y^3 + 38y^2 + 44y + 56 \\
&\{i, j\} = \{8, 13\}, k_{ij} = 3 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{8, 15\}, k_{ij} = 3 \cdot 5 \cdot 17 \\
&\alpha \quad 32 \\
&\beta \quad 32x^4 - 32x^3y + 32x^3 + 32x^2y^2 - 64x^2y + 32x^2 - 32xy^3 + 96xy^2 - 96xy \\
&\quad + 32x + 16y^4 - 40y^3 + 4y^2 + 126y - 95 \\
&\gamma \quad -16y^4 + 88y^3 - 188y^2 + 254y - 159 \\
&\{i, j\} = \{8, 17\}, k_{ij} = 2 \cdot 5 \cdot 7 \cdot 11 \\
&\alpha \quad 243 \\
&\beta \quad 243x^4 - 243x^3y + 243x^3 + 243x^2y^2 - 486x^2y + 243x^2 - 243xy^3 + 729xy^2 \\
&\quad - 729xy + 243x + 81y^4 - 216y^3 + 252y^2 + 246y + 568 \\
&\gamma \quad -81y^4 + 378y^3 - 603y^2 + 609y + 41 \\
&\{i, j\} = \{8, 18\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad 3 \\
&\beta \quad 3x^4 - 3x^3y + 3x^3 + 3x^2y^2 - 6x^2y + 3x^2 - 3xy^3 + 9xy^2 - 9xy + 3x + y^4 \\
&\quad - 4y^3 + 8y^2 - 2y + 12 \\
&\gamma \quad -y^4 + 4y^3 - 5y^2 + 5y + 3 \\
&\{i, j\} = \{8, 23\}, k_{ij} = 5^3 \\
&\alpha \quad 4x^5 - 16y + 31 \\
&\beta \quad 4x^9 - 4x^8y + 4x^8 + 4x^7y^2 - 8x^7y + 4x^7 - 8x^6y - 4x^6 + 4x^5y^3 \\
&\quad - 4x^5y^2 - 8x^5 + 27x^4 - 27x^3y + 27x^3 + 27x^2y^2 - 54x^2y + 27x^2 \\
&\quad + 16xy^2 - 26xy + 45x + 11y^3 - 23y^2 - 32y + 94 \\
&\gamma \quad 4x^5y^3 - 12x^5y^2 + 4x^5y - 8x^5 + 27y^3 - 65y^2 + 55y + 18 \\
&\{i, j\} = \{8, 25\}, k_{ij} = 2^3 \cdot 3^3 \\
&\alpha \quad x^5 - y + 71 \\
&\beta \quad x^9 - x^8y + x^8 + x^7y^2 - 2x^7y + x^7 + x^6y^2 - x^5y^3 + x^5y^2
\end{aligned}$$

$$\begin{aligned}
& +x^5 + 74x^4 - 74x^3y + 74x^3 + 74x^2y^2 - 148x^2y + 74x^2 + 75xy^2 + 6xy + 21x \\
& -75y^3 + 69y^2 - 16y + 92 \\
\gamma \quad & x^5y^3 - 2x^5y^2 + 3x^5y - x^5 + 74y^3 - 147y^2 + 228y - 53 \\
\{i, j\} = \{8, 27\}, k_{ij} = 7 \cdot 17^2 \\
\alpha \quad & 12x^5 + 144y + 425 \\
\beta \quad & 12x^9 - 12x^8y + 12x^8 + 12x^7y^2 - 24x^7y + 12x^7 + 24x^6y^2 + 60x^6 \\
& -12x^5y^3 + 12x^5y^2 - 24x^5y + 120x^5 + 557x^4 - 557x^3y + 557x^3 \\
& +557x^2y^2 - 1114x^2y + 557x^2 + 970xy^2 - 444xy + 865x - 413y^3 + 713y^2 + 62y \\
& +1598 \\
\gamma \quad & 12x^5y^3 - 12x^5y^2 + 36x^5y + 48x^5 + 557y^3 - 701y^2 + 1227y + 308 \\
\{i, j\} = \{8, 30\}, k_{ij} = 2^3 \cdot 5 \\
\alpha \quad & 3x^5 - 18y + 10 \\
\beta \quad & 3x^9 - 3x^8y + 3x^8 + 3x^7y^2 - 6x^7y + 3x^7 - 6x^6y^2 + 15x^6y - 12x^6 \\
& +3x^5y^3 - 6x^5y^2 + 3x^5y + 6x^5 + 4x^4 - 4x^3y + 4x^3 + 4x^2y^2 \\
& -8x^2y + 4x^2 + 10xy^2 + 14xy + 20x - 14y^3 - 2y^2 - 50y + 14 \\
\gamma \quad & 3x^5y^3 - 15x^5y^2 + 24x^5y - 15x^5 + 4y^3 - 2y^2 + 26y + 16 \\
\{i, j\} = \{8, 34\}, k_{ij} = 11 \\
\alpha \quad & 1 \\
\beta \quad & x^4 - x^3y + x^3 + x^2y^2 - 2x^2y + x^2 - xy^2 + 3xy - 7x - y^3 + 4y^2 - 2y + 10 \\
\gamma \quad & y^3 - 4y^2 + 6y - 8 \\
\{i, j\} = \{8, 37\}, k_{ij} = 3^3 \cdot 13 \\
\alpha \quad & 2x^5 + 12y + 69 \\
\beta \quad & 2x^9 - 2x^8y + 2x^8 + 2x^7y^2 - 4x^7y + 2x^7 + 2x^6y^2 + 2x^6y + 2x^6 \\
& -2x^5y^3 + 4x^5 + 75x^4 - 75x^3y + 75x^3 + 75x^2y^2 - 150x^2y + 75x^2 \\
& +63xy^2 + 45xy - 63x - 63y^3 + 18y^2 + 120y + 6 \\
\gamma \quad & 2x^5y^3 - 4x^5y^2 + 8x^5y + 75y^3 - 162y^2 + 270y - 138 \\
\{i, j\} = \{8, 42\}, k_{ij} = 3^2 \cdot 11 \\
\alpha \quad & 4x^5 - 48y - 3 \\
\beta \quad & 4x^9 - 4x^8y + 4x^8 + 4x^7y^2 - 8x^7y + 4x^7 - 8x^6y^2 + 16x^6y + 4x^6 \\
& +8x^5y^3 - 24x^5y^2 + 12x^5y + 8x^5 + 9x^4 - 9x^3y + 9x^3 + 9x^2y^2 \\
& -18x^2y + 9x^2 + 30xy^2 + 48xy + 57x - 30y^3 - 18y^2 - 57y + 54 \\
\gamma \quad & 4x^5y^3 - 20x^5y^2 + 28x^5y + 9y^3 + 3y^2 + 75y + 48 \\
\{i, j\} = \{9, 20\}, k_{ij} = 5 \cdot 7 \\
\alpha \quad & -x - 3 \\
\beta \quad & x^4 + 3x^3 + 2x^2 - xy^2 + 2xy - 2x - 3y^2 + 6y - 19 \\
\gamma \quad & x^4 + 3x^3 + 2x^2 + 6x + 5 \\
\{i, j\} = \{9, 23\}, k_{ij} = 5 \cdot 7^2 \cdot 13 \\
\alpha \quad & 12x^4 - 257 \\
\beta \quad & -12x^7 + 12x^6 - 36x^5 + 12x^4y^2 - 24x^4y + 156x^4 + 113x^3 + 187x^2 + 39x \\
& -257y^2 + 514y - 1721 \\
\gamma \quad & -12x^7 + 12x^6 - 36x^5 + 60x^4 + 113x^3 + 187x^2 + 39x + 335 \\
\{i, j\} = \{9, 24\}, k_{ij} = 3^2 \\
\alpha \quad & -1 \\
\beta \quad & -x^3 - x^2 + x - y^2 + 2y - 5 \\
\gamma \quad & x^3 + x^2 - x - 3 \\
\{i, j\} = \{9, 27\}, k_{ij} = 3 \cdot 7 \\
\alpha \quad & -1 \\
\beta \quad & -x^3 + x^2 + x - y^2 + 2y - 11 \\
\gamma \quad & x^3 - x^2 - x + 3 \\
\{i, j\} = \{9, 30\}, k_{ij} = 2 \cdot 3^2 \\
\alpha \quad & -x^4 - 1 \\
\beta \quad & x^7 + x^6 + 2x^5 - x^4y^2 + 2x^4y - 5x^4 + 7x^3 - 5x^2 + 2x - y^2 + 2y - 11 \\
\gamma \quad & x^7 + x^6 + 2x^5 + 3x^4 + 7x^3 - 5x^2 + 2x - 3 \\
\{i, j\} = \{9, 35\}, k_{ij} = 2 \cdot 7 \\
\alpha \quad & -1 \\
\beta \quad & -x^3 - x^2 - y^2 + 2y - 7 \\
\gamma \quad & x^3 + x^2 - 1 \\
\{i, j\} = \{9, 36\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
\alpha \quad & x^4 - 1
\end{aligned}$$

β	$x^7 + x^6 - 2x^5 + x^4y^2 - 2x^4y + 3x^4 + x^3 + x^2 - 2x - y^2 + 2y - 13$
γ	$-x^7 - x^6 + 2x^5 + 5x^4 - x^3 - x^2 + 2x + 5$
$\{i, j\}$	$\{9, 38\}, k_{ij} = 2^4$
α	-1
β	$-x^3 + x^2 - y^2 + 2y - 9$
γ	$x^3 - x^2 + 1$
$\{i, j\}$	$\{9, 39\}, k_{ij} = 2 \cdot 3^5 \cdot 5$
α	$-x^4 - 74$
β	$-x^7 + x^6 + 2x^5 - x^4y^2 + 2x^4y - 13x^4 - 76x^3 + 106x^2 + 122x - 74y^2$ $+ 148y - 1032$
γ	$x^7 - x^6 - 2x^5 + 5x^4 + 76x^3 - 106x^2 - 122x + 440$
$\{i, j\}$	$\{9, 41\}, k_{ij} = 3 \cdot 5 \cdot 7$
α	$-2x - 3$
β	$-4x^4 - 6x^3 + 12x^2 - 2xy^2 + 4xy + 2x - 3y^2 + 6y - 64$
γ	$2x^4 + 3x^3 - 6x^2 - 9x + 20$
$\{i, j\}$	$\{10, 12\}, k_{ij} = 2^3 \cdot 13$
α	243
β	$-243x^4 - 243x^3y - 243x^3 - 243x^2y^2 - 486x^2y - 243x^2 - 243xy^3$ $- 729xy^2 - 729xy - 243x - 81y^4 - 216y^3 + 72y^2 + 138y + 764$
γ	$81y^4 + 378y^3 + 765y^2 + 555y + 625$
$\{i, j\}$	$\{10, 23\}, k_{ij} = 3 \cdot 17$
α	$-32x^5 - 264y^2 - 144y - 39$
β	$-64x^9 - 16x^8y - 16x^8 - 16x^7y^2 - 32x^7y - 16x^7 - 32x^6y^2 + 32x^6y$ $- 208x^6 - 16x^5y^3 + 16x^5y^2 - 432x^5y - 224x^5 + 387x^4 + 885x^3y$ $+ 885x^3 + 885x^2y^2 + 1770x^2y + 885x^2 + 1002xy^2 + 870xy + 399x - 264y^4$ $+ 477y^3 - 21y^2 + 318y - 90$
γ	$96x^8 - 48x^7y - 144x^7 - 48x^6y + 144x^6 - 16x^5y^3 + 32x^5y^2 + 112x^5y$ $+ 48x^5 - 264x^3y^2 - 432x^3y - 540x^3 + 264x^2y^2 - 66x^2y + 42x^2$ $+ 264xy^3 + 168xy^2 + 606xy - 42x + 357y^3 + 1551y^2 + 1137y + 528$
$\{i, j\}$	$\{10, 26\}, k_{ij} = 2^5 \cdot 3$
α	$x^5 - 3y + 3$
β	$-x^9 - x^8y - x^8 - x^7y^2 - 2x^7y - x^7 + x^6y^2 + 2x^6y - 4x^6$ $+ x^5y^3 + 3x^5y^2 - 2x^5y - 5x^5 - 12x^4 - 12x^3y - 12x^3 - 12x^2y^2$ $- 24x^2y - 12x^2 + 9xy^2 + 18xy - 51x + 9y^3 + 27y^2 - 30y - 54$
γ	$-x^5y^3 - 4x^5y^2 - 5x^5y + 3x^5 - 12y^3 - 45y^2 - 54y + 39$
$\{i, j\}$	$\{10, 27\}, k_{ij} = 5 \cdot 7 \cdot 13$
α	$8x^5 - 64y - 71$
β	$-8x^9 - 8x^8y - 8x^8 - 8x^7y^2 - 16x^7y - 8x^7 - 24x^6 + 8x^5y^3$ $+ 24x^5y^2 - 48x^5 + 15x^4 + 15x^3y + 15x^3 + 15x^2y^2 + 30x^2y + 15x^2$ $- 64xy^2 + 72xy - 291x - 79y^3 - 101y^2 - 128y - 526$
γ	$-8x^5y^3 - 24x^5y^2 - 24x^5y + 16x^5 + 15y^3 + 109y^2 - 27y + 306$
$\{i, j\}$	$\{10, 33\}, k_{ij} = 2^5 \cdot 3^4$
α	$17x^5 - 578y - 1226$
β	$-17x^9 - 17x^8y - 17x^8 - 17x^7y^2 - 34x^7y - 17x^7 - 34x^6y^2 + 17x^6y$ $- 170x^6 - 17x^5y^3 - 85x^5y - 340x^5 + 716x^4 + 716x^3y + 716x^3$ $+ 716x^2y^2 + 1432x^2y + 716x^2 + 854xy^2 + 1086xy - 1204x + 138y^3 + 646y^2 - 602y$ $- 1898$
γ	$-17x^5y^3 - 17x^5y^2 - 68x^5y + 153x^5 + 716y^3 + 1294y^2 + 1062y + 1920$
$\{i, j\}$	$\{10, 36\}, k_{ij} = 2$
α	-1
β	$x^4 + x^3y + x^3 + x^2y^2 + 2x^2y + x^2 - 2xy^2 - 7xy - 4x - y^3 - 4y^2 - y + 2$
γ	$y^3 + 5y^2 + 10y + 5$
$\{i, j\}$	$\{10, 39\}, k_{ij} = 2^2 \cdot 3^2 \cdot 17$
α	$5x^5 - 50y - 44$
β	$-5x^9 - 5x^8y - 5x^8 - 5x^7y^2 - 10x^7y - 5x^7 - 5x^6y - 20x^6 + 5x^5y^3$ $+ 10x^5y^2 - 15x^5y - 40x^5 - 16x^4 - 16x^3y - 16x^3 - 16x^2y^2 - 32x^2y$ $- 16x^2 - 50xy^2 + 24xy - 244x - 34y^3 - 28y^2 - 98y - 428$
γ	$-5x^5y^3 - 15x^5y^2 - 10x^5y + 15x^5 - 16y^3 + 2y^2 - 72y + 228$
$\{i, j\}$	$\{10, 40\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5$
α	$37x^5 - 1369y + 203$

$$\begin{aligned}
\beta & -37x^9 - 37x^8y - 37x^8 - 37x^7y^2 - 74x^7y - 37x^7 + 111x^6y^2 + 518x^6y \\
& + 962x^6 + 74x^5y^3 + 407x^5y^2 + 851x^5y - 74x^5 - 314x^4 - 314x^3y \\
& - 314x^3 - 314x^2y^2 - 628x^2y - 314x^2 - 427xy^2 + 2916xy - 161x - 741y^3 \\
& + 1974y^2 + 266y - 517 \\
\gamma & -37x^5y^3 - 222x^5y^2 - 629x^5y - 999x^5 - 314y^3 - 515y^2 - 3858y - 153 \\
\{i, j\} & = \{11, 14\}, k_{ij} = 2^6 \cdot 5^2 \\
\alpha & -16 \\
\beta & -8x^4 + 4x^3y + 4x^3 - 2x^2y^2 - 4x^2y - 2x^2 + xy^3 + 3xy^2 + 3xy + x - y^4 \\
& - 7y^3 - 17y^2 - 77y - 330 \\
\gamma & y^4 + 10y^3 + 28y^2 + 150y + 643 \\
\{i, j\} & = \{11, 16\}, k_{ij} = 2^6 \cdot 3 \\
\alpha & 48 \\
\beta & 24x^4 - 12x^3y - 12x^3 + 6x^2y^2 + 12x^2y + 6x^2 - 3xy^3 - 9xy^2 - 9xy - 3x \\
& + y^4 + 3y^3 + 17y^2 + 9y + 82 \\
\gamma & -y^4 - 6y^3 + 16y^2 + 6y + 113 \\
\{i, j\} & = \{11, 24\}, k_{ij} = 2^3 \cdot 3 \\
\alpha & 16x^5 - 16y + 8 \\
\beta & 8x^9 - 4x^8y - 4x^8 + 2x^7y^2 + 4x^7y + 2x^7 + 2x^6y^2 - 28x^6y - 6x^6 \\
& - 2x^5y^3 + 8x^5y^2 + 42x^5y + 16x^5 + 4x^4 - 2x^3y - 2x^3 + x^2y^2 \\
& + 2x^2y + x^2 + 33xy^2 - 14xy + 13x - 17y^3 - 44y^2 + 5y - 16 \\
\gamma & 2x^5y^3 + 10x^5y^2 - 50x^5y - 10x^5 + y^3 + 69y^2 - 25y + 27 \\
\{i, j\} & = \{11, 25\}, k_{ij} = 2^6 \cdot 3^2 \cdot 7 \\
\alpha & -40x^5 + 100y - 908 \\
\beta & -20x^9 + 10x^8y + 10x^8 - 5x^7y^2 - 10x^7y - 5x^7 - 10x^6y^2 + 20x^6y \\
& - 50x^6 + 5x^5y^3 - 5x^5y^2 + 15x^5y + 5x^5 - 424x^4 + 212x^3y + 212x^3 \\
& - 106x^2y^2 - 212x^2y - 106x^2 - 412xy^2 + 144xy - 2920x + 206y^3 + 134y^2 + 1438y \\
& + 1006 \\
\gamma & -5x^5y^3 - 35x^5y^2 + 25x^5y - 105x^5 - 106y^3 - 1142y^2 - 30y - 5946 \\
\{i, j\} & = \{11, 26\}, k_{ij} = 2^4 \cdot 3 \cdot 5^2 \\
\alpha & 24x^5 - 108y - 28 \\
\beta & 12x^9 - 6x^8y - 6x^8 + 3x^7y^2 + 6x^7y + 3x^7 + 6x^6y^2 - 24x^6y - 78x^6 \\
& - 3x^5y^3 + 9x^5y^2 + 51x^5y + 51x^5 - 104x^4 + 52x^3y + 52x^3 - 26x^2y^2 \\
& - 52x^2y - 26x^2 + 164xy^2 + 280xy + 496x - 82y^3 - 222y^2 - 442y - 262 \\
\gamma & 3x^5y^3 + 21x^5y^2 - 39x^5y - 153x^5 - 26y^3 + 250y^2 + 482y + 966 \\
\{i, j\} & = \{11, 28\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
\alpha & 2472x^5 - 4224y^2 + 334704y + 177536 \\
\beta & 1428x^9 - 810x^8y - 810x^8 + 405x^7y^2 + 810x^7y + 405x^7 - 1620x^6y^2 \\
& + 4050x^6y - 39690x^6 + 405x^5y^3 + 234x^5y^2 + 11193x^5y + 97752x^5 \\
& + 96968x^4 + 712620x^3y + 712620x^3 - 356310x^2y^2 - 712620x^2y - 356310x^2 \\
& + 1075320xy^2 - 258300xy - 2685700x - 4224y^4 - 51174y^3 + 474804y^2 + 2122354y \\
& + 177056 \\
\gamma & 384x^8 - 192x^7y + 576x^7 - 768x^6y + 1152x^6 + 405x^5y^3 - 1833x^5y^2 \\
& + 7203x^5y - 76671x^5 + 8448x^3y^2 - 158976x^3y + 111680x^3 + 16896x^2y^2 \\
& + 1204256x^2y + 1745568x^2 - 8448xy^3 + 192768xy^2 + 2296832xy + 3491136x \\
& - 390102y^3 + 262990y^2 + 1262566y + 1254562 \\
\{i, j\} & = \{11, 33\}, k_{ij} = 2^5 \cdot 3^3 \cdot 11 \\
\alpha & 64x^5 - 1024y - 3400 \\
\beta & 32x^9 - 16x^8y - 16x^8 + 8x^7y^2 + 16x^7y + 8x^7 - 8x^6y^2 - 128x^6y \\
& + 488x^6 + 8x^5y^3 + 72x^5y^2 - 296x^5y + 280x^5 - 1316x^4 + 658x^3y \\
& + 658x^3 - 329x^2y^2 - 658x^2y - 329x^2 + 2377xy^2 - 1392xy + 7579x - 1353y^3 \\
& + 1391y^2 - 5491y + 1925 \\
\gamma & 8x^5y^3 + 8x^5y^2 - 232x^5y + 984x^5 - 329y^3 + 3767y^2 - 3771y + 14829 \\
\{i, j\} & = \{11, 36\}, k_{ij} = 2^5 \cdot 7 \cdot 11 \\
\alpha & 24x^5 + 72y - 8 \\
\beta & 12x^9 - 6x^8y - 6x^8 + 3x^7y^2 + 6x^7y + 3x^7 + 3x^6y^2 - 48x^6y - 39x^6 \\
& - 3x^5y^3 + 15x^5y^2 + 87x^5y + 69x^5 - 64x^4 + 32x^3y + 32x^3 - 16x^2y^2 \\
& - 32x^2y - 16x^2 - 160xy^2 + 16xy + 712x + 88y^3 + 256y^2 - 320y - 1064 \\
\gamma & 3x^5y^3 + 15x^5y^2 - 87x^5y - 75x^5 - 16y^3 - 368y^2 - 16y + 1408 \\
\{i, j\} & = \{11, 41\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \cdot 11 \cdot 17 \\
\alpha & -248x^5 - 15376y - 80368
\end{aligned}$$

$$\begin{aligned}
\beta & -124x^9 + 62x^8y + 62x^8 - 31x^7y^2 - 62x^7y - 31x^7 - 124x^6y^2 - 682x^6y \\
& -5022x^6 + 62x^5y^3 + 403x^5y^2 + 2852x^5y + 2387x^5 - 36464x^4 \\
& +18232x^3y + 18232x^3 - 9116x^2y^2 - 18232x^2y - 9116x^2 - 5712xy^2 - 1160xy \\
& -80552x + 2856y^3 + 3436y^2 + 33168y + 92 \\
\gamma & -31x^5y^3 - 341x^5y^2 - 1457x^5y - 10075x^5 - 9116y^3 - 38772y^2 - 29668y \\
& -170220 \\
\{i, j\} & = \{12, 13\}, k_{ij} = 2^5 \cdot 3 \cdot 17 \\
\alpha & 16 \\
\beta & 8x^4 - 4x^3y + 4x^3 + 2x^2y^2 - 4x^2y + 2x^2 - xy^3 + 3xy^2 - 3xy + x + y^4 \\
& -7y^3 + 17y^2 - 77y + 330 \\
\gamma & -y^4 + 10y^3 - 28y^2 + 150y - 643 \\
\{i, j\} & = \{12, 17\}, k_{ij} = 2 \cdot 7^2 \\
\alpha & 64 \\
\beta & 32x^4 - 16x^3y + 16x^3 + 8x^2y^2 - 16x^2y + 8x^2 - 4xy^3 + 12xy^2 - 12xy + 4x \\
& +y^4 - 3y^3 + 34y^2 + 2y + 141 \\
\gamma & -y^4 + 5y^3 + 22y^2 + 10y + 107 \\
\{i, j\} & = \{12, 23\}, k_{ij} = 2^3 \cdot 7 \\
\alpha & 16x^5 - 16y + 8 \\
\beta & 8x^9 - 4x^8y + 4x^8 + 2x^7y^2 - 4x^7y + 2x^7 - 2x^6y^2 - 28x^6y + 6x^6 \\
& +2x^5y^3 + 8x^5y^2 - 42x^5y + 16x^5 + 20x^4 - 10x^3y + 10x^3 + 5x^2y^2 \\
& -10x^2y + 5x^2 + 27xy^2 - 6xy + 31x - 11y^3 + 36y^2 - 25y + 48 \\
\gamma & 2x^5y^3 - 10x^5y^2 - 50x^5y + 10x^5 + 5y^3 + 39y^2 + 3y + 57 \\
\{i, j\} & = \{12, 27\}, k_{ij} = 13 \cdot 17 \\
\alpha & 3660x^5 + 6105y^2 + 150620y - 60296 \\
\beta & 1830x^9 + 1120x^8y - 1120x^8 - 560x^7y^2 + 1120x^7y - 560x^7 - 1680x^6y^2 \\
& +1120x^6y + 7280x^6 + 560x^5y^3 + 1830x^5y^2 - 4835x^5y + 10995x^5 \\
& -426768x^4 + 545496x^3y - 545496x^3 - 272748x^2y^2 + 545496x^2y - 272748x^2 \\
& -907844xy^2 - 162344xy - 78363x + 6105y^4 + 405053y^3 + 413427y^2 + 410774y + 60517 \\
\gamma & 4070x^7y - 4070x^7 - 4070x^6y + 4070x^6 - 560x^5y^3 + 2390x^5y^2 + 560x^5y \\
& +11050x^5 - 12210x^3y^2 - 138380x^3y - 807895x^3 + 12210x^2y^2 \\
& +802604x^2y + 143671x^2 - 12210xy^3 - 150590xy^2 - 1610499xy - 143671x - 248328y^3 \\
& -44250y^2 + 611238y + 259693 \\
\{i, j\} & = \{12, 35\}, k_{ij} = 2^7 \cdot 7^3 \\
\alpha & -152x^5 + 2888y - 18912 \\
\beta & -76x^9 + 38x^8y - 38x^8 - 19x^7y^2 + 38x^7y - 19x^7 - 19x^6y^2 + 228x^6y \\
& +931x^6 + 19x^5y^3 - 133x^5y^2 - 551x^5y - 551x^5 - 10748x^4 + 5374x^3y \\
& -5374x^3 - 2687x^2y^2 + 5374x^2y - 2687x^2 - 8463xy^2 + 3972xy - 30521x \\
& +5575y^3 - 1785y^2 + 18749y + 4461 \\
\gamma & -19x^5y^3 + 19x^5y^2 + 399x^5y + 1881x^5 - 2687y^3 - 8865y^2 - 117y - 58355 \\
\{i, j\} & = \{12, 41\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
\alpha & 35000x^5y^2 - 35000x^5y + 31878x^5 + 3182500y^2 - 3833456y - 1403187 \\
\beta & -35000x^{11} + 17500x^{10}y - 17500x^{10} - 5936x^9 + 2968x^8y - 2968x^8 \\
& -1484x^7y^2 + 2968x^7y - 1484x^7 - 5936x^6y^2 - 20776x^6y - 564472x^6 \\
& +35000x^5y^4 - 312032x^5y^3 - 123830x^5y^2 + 201848x^5y - 218172x^5 \\
& -6689448x^4 + 3344724x^3y - 3344724x^3 - 1672362x^2y^2 + 3344724x^2y \\
& -1672362x^2 - 4362536xy^2 + 520884xy + 1787502x + 3182500y^4 + 2693768y^3 \\
& +23108915y^2 + 22616213y + 6650595 \\
\gamma & -70000x^{10} - 17500x^8y^2 + 175000x^8y - 96250x^8 - 35000x^6y^3 \\
& +332500x^6y^2 - 17500x^6y - 96250x^6 - 1484x^5y^3 - 7420x^5y^2 \\
& -46004x^5y - 1127460x^5 - 1591250x^3y^2 + 1335000x^3y - 12916875x^3 \\
& -3182500xy^3 + 1078750xy^2 - 24498750xy - 12916875x - 1672362y^3 - 3707986y^2 \\
& -3975318y + 5247366 \\
\{i, j\} & = \{13, 26\}, k_{ij} = 3^4 \cdot 7 \\
\alpha & 2x^5 - 12y - 15 \\
\beta & 2x^9 - 2x^8y - 4x^8 + 2x^7y^2 + 8x^7y + 8x^7 + 2x^6y^2 + 2x^6y - 4x^6 \\
& -2x^5y^3 - 6x^5y^2 + 10x^5 - 21x^4 + 21x^3y + 42x^3 - 21x^2y^2 - 84x^2y \\
& -84x^2 - 9xy^2 - 63xy + 144x + 9y^3 + 81y^2 - 30y - 303 \\
\gamma & 2x^5y^3 + 14x^5y^2 + 26x^5y + 12x^5 - 21y^3 - 135y^2 - 315y - 24 \\
\{i, j\} & = \{13, 27\}, k_{ij} = 5^2 \cdot 17 \\
\alpha & 9x^5 - 81y - 107
\end{aligned}$$

$$\begin{aligned}
\beta \quad & 9x^9 - 9x^8y - 18x^8 + 9x^7y^2 + 36x^7y + 36x^7 + 18x^6y^2 + 81x^6y \\
& + 126x^6 - 9x^5y^3 - 45x^5y^2 - 99x^5y - 45x^5 - 8x^4 + 8x^3y + 16x^3 \\
& - 8x^2y^2 - 32x^2y - 32x^2 + 65xy^2 - 135xy + 95x - 73y^3 + 22y^2 - 137y - 266 \\
\gamma \quad & 9x^5y^3 + 72x^5y^2 + 189x^5y + 198x^5 - 8y^3 + 17y^2 - 231y + 31 \\
\{i, j\} = \{13, 28\}, k_{ij} = 2^9 \cdot 5 \cdot 11 \\
\alpha \quad & 19x^5 + 1805y - 776 \\
\beta \quad & 19x^9 - 19x^8y - 38x^8 + 19x^7y^2 + 76x^7y + 76x^7 - 57x^6y^2 - 38x^6y \\
& - 418x^6 + 19x^5y^3 + 38x^5y^2 + 114x^5y + 1387x^5 - 3531x^4 + 3531x^3y \\
& + 7062x^3 - 3531x^2y^2 - 14124x^2y - 14124x^2 + 8788xy^2 + 16942xy + 17547x \\
& - 1726y^3 - 9722y^2 + 1234y - 14468 \\
\gamma \quad & 19x^5y^3 + 57x^5y^2 + 190x^5y - 266x^5 - 3531y^3 - 12398y^2 - 25430y - 10701 \\
\{i, j\} = \{13, 31\}, k_{ij} = 2^4 \cdot 13 \\
\alpha \quad & 3x^5 + 9y + 14 \\
\beta \quad & 3x^9 - 3x^8y - 6x^8 + 3x^7y^2 + 12x^7y + 12x^7 - 3x^6y^2 + 6x^6 \\
& + 3x^5y^3 + 6x^5y^2 - 6x^5y - 9x^5 - 25x^4 + 25x^3y + 50x^3 - 25x^2y^2 \\
& - 100x^2y - 100x^2 + 16xy^2 + 6xy - 89x - 16y^3 - 38y^2 + 86y + 192 \\
\gamma \quad & 3x^5y^3 + 15x^5y^2 + 36x^5y + 30x^5 - 25y^3 - 134y^2 - 294y - 289 \\
\{i, j\} = \{13, 35\}, k_{ij} = 2^2 \cdot 5^3 \cdot 7 \\
\alpha \quad & 13x^5 - 169y - 464 \\
\beta \quad & 13x^9 - 13x^8y - 26x^8 + 13x^7y^2 + 52x^7y + 52x^7 - 26x^6y + 26x^6 \\
& - 13x^5y^3 - 52x^5y^2 - 104x^5y - 169x^5 - 243x^4 + 243x^3y + 486x^3 \\
& - 243x^2y^2 - 972x^2y - 972x^2 + 169xy^2 + 31xy + 996x + 74y^3 + 920y^2 + 1658y \\
& - 1508 \\
\gamma \quad & 13x^5y^3 + 78x^5y^2 + 130x^5y + 130x^5 - 243y^3 - 1289y^2 - 2885y - 948 \\
\{i, j\} = \{13, 39\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \\
\alpha \quad & x^5 - 15y - 14 \\
\beta \quad & x^9 - x^8y - 2x^8 + x^7y^2 + 4x^7y + 4x^7 + 2x^6y^2 + 8x^6y + 8x^6 \\
& - x^5y^3 - 4x^5y^2 - 4x^5y + x^5 + x^4 - x^3y - 2x^3 + x^2y^2 + 4x^2y \\
& + 4x^2 + 17xy^2 - 7xy + 38x - 16y^3 - 4y^2 - 34y - 44 \\
\gamma \quad & x^5y^3 + 8x^5y^2 + 20x^5y + 16x^5 + y^3 + 23y^2 + 5y + 46 \\
\{i, j\} = \{14, 28\}, k_{ij} = 2^5 \cdot 5 \cdot 7 \\
\alpha \quad & -27x^5 - 729y + 896 \\
\beta \quad & -27x^9 + 27x^8y - 54x^8 - 27x^7y^2 + 108x^7y - 108x^7 + 81x^6y^2 - 486x^6y \\
& + 1134x^6 - 27x^5y^3 + 162x^5y^2 - 486x^5y - 459x^5 - 49x^4 + 49x^3y \\
& - 98x^3 - 49x^2y^2 + 196x^2y - 196x^2 + 876xy^2 - 1098xy + 4029x - 778y^3 \\
& + 510y^2 - 3582y + 112 \\
\gamma \quad & -27x^5y^3 + 243x^5y^2 - 810x^5y + 1350x^5 - 49y^3 + 1170y^2 - 1686y + 4421 \\
\{i, j\} = \{14, 38\}, k_{ij} = 2^6 \cdot 5^2 \cdot 17 \\
\alpha \quad & 27x^5 + 729y + 2644 \\
\beta \quad & 27x^9 - 27x^8y + 54x^8 + 27x^7y^2 - 108x^7y + 108x^7 + 54x^6y^2 - 54x^6y \\
& + 216x^6 - 27x^5y^3 + 54x^5y^2 - 54x^5y + 459x^5 + 3589x^4 - 3589x^3y \\
& + 7178x^3 + 3589x^2y^2 - 14356x^2y + 14356x^2 + 6449xy^2 - 10607xy + 9596x \\
& - 2860y^3 + 8420y^2 + 2380y + 2720 \\
\gamma \quad & 27x^5y^3 - 108x^5y^2 + 270x^5y + 3589y^3 - 15085y^2 + 32461y - 19116 \\
\{i, j\} = \{14, 39\}, k_{ij} = 2^6 \cdot 7 \cdot 17 \\
\alpha \quad & 3x^5 - 9y + 662 \\
\beta \quad & 3x^9 - 3x^8y + 6x^8 + 3x^7y^2 - 12x^7y + 12x^7 + 6x^6y^2 - 12x^6y + 12x^6 \\
& - 3x^5y^3 + 12x^5y^2 - 12x^5y + 15x^5 + 683x^4 - 683x^3y + 1366x^3 \\
& + 683x^2y^2 - 2732x^2y + 2732x^2 + 1375xy^2 - 2657xy + 3146x - 692y^3 + 2684y^2 \\
& - 2930y + 4636 \\
\gamma \quad & 3x^5y^3 - 12x^5y^2 + 24x^5y - 12x^5 + 683y^3 - 2723y^2 + 5539y - 2318 \\
\{i, j\} = \{14, 40\}, k_{ij} = 2^2 \cdot 3 \cdot 13 \\
\alpha \quad & -x^5 - 3y + 22 \\
\beta \quad & -x^9 + x^8y - 2x^8 - x^7y^2 + 4x^7y - 4x^7 - x^6y^2 + 2x^6 + 2x^5y^3 \\
& - 7x^5y^2 + 10x^5y - 7x^5 + 7x^4 - 7x^3y + 14x^3 + 7x^2y^2 - 28x^2y + 28x^2 \\
& + 10xy^2 - 11x - 17y^3 + 52y^2 - 76y + 67 \\
\gamma \quad & -x^5y^3 + 5x^5y^2 - 12x^5y + 10x^5 + 7y^3 - 32y^2 + 84y - 67 \\
\{i, j\} = \{14, 42\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 17 \\
\alpha \quad & 7x^5 - 147y - 116 \\
\beta \quad & 7x^9 - 7x^8y + 14x^8 + 7x^7y^2 - 28x^7y + 28x^7 - 14x^6y^2 + 42x^6y
\end{aligned}$$

$$\begin{aligned}
& +28x^6 + 14x^5y^3 - 70x^5y^2 + 56x^5y + 63x^5 - 11x^4 + 11x^3y - 22x^3 \\
& - 11x^2y^2 + 44x^2y - 44x^2 + 169xy^2 + 333xy + 460x - 169y^3 + 5y^2 + 59y + 804 \\
\gamma \quad & 7x^5y^3 - 56x^5y^2 + 126x^5y - 28x^5 - 11y^3 + 235y^2 + 201y + 548 \\
\{i, j\} = \{15, 17\}, k_{ij} = 2^3 \cdot 5 \\
\alpha \quad & -1 \\
\beta \quad & x^4 + x^3y + 2x^3 + x^2y^2 + 4x^2y + 4x^2 + xy^3 + 6xy^2 + 12xy + 8x - y^4 \\
& - 6y^3 - 16y^2 - 8y - 33 \\
\gamma \quad & y^4 + 7y^3 + 20y^2 + 20y + 25 \\
\{i, j\} = \{15, 26\}, k_{ij} = 3^5 \cdot 5 \\
\alpha \quad & 2x^5 - 20y - 25 \\
\beta \quad & -2x^9 - 2x^8y - 4x^8 - 2x^7y^2 - 8x^7y - 8x^7 + 2x^6y^2 + 6x^6y - 16x^6 \\
& + 2x^5y^3 + 10x^5y^2 - 4x^5y - 34x^5 - 45x^4 - 45x^3y - 90x^3 - 45x^2y^2 \\
& - 180x^2y - 180x^2 + 25xy^2 + 125xy - 330x + 25y^3 + 175y^2 - 60y - 635 \\
\gamma \quad & -2x^5y^3 - 14x^5y^2 - 30x^5y - 45y^3 - 295y^2 - 665y - 30 \\
\{i, j\} = \{15, 30\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5 \\
\alpha \quad & 19x^5 - 361y - 1100 \\
\beta \quad & -19x^9 - 19x^8y - 38x^8 - 19x^7y^2 - 76x^7y - 76x^7 + 38x^6y - 38x^6 \\
& - 19x^5y^3 - 76x^5y^2 - 228x^5y - 209x^5 + 777x^4 + 777x^3y + 1554x^3 \\
& + 777x^2y^2 + 3108x^2y + 3108x^2 - 361xy^2 - 433xy - 3652x + 416y^3 + 3868y^2 \\
& + 5600y + 3664 \\
\gamma \quad & -19x^5y^3 - 114x^5y^2 - 266x^5y - 114x^5 + 777y^3 + 5023y^2 + 9757y + 9868 \\
\{i, j\} = \{15, 33\}, k_{ij} = 2^2 \cdot 7 \cdot 17^2 \\
\alpha \quad & 9x^5 - 567y - 1808 \\
\beta \quad & -9x^9 - 9x^8y - 18x^8 - 9x^7y^2 - 36x^7y - 36x^7 - 18x^6y^2 - 18x^6y \\
& - 144x^6 - 9x^5y^3 - 18x^5y^2 - 90x^5y - 369x^5 + 989x^4 + 989x^3y \\
& + 1978x^3 + 989x^2y^2 + 3956x^2y + 3956x^2 + 1411xy^2 + 4561xy + 1460x + 422y^3 \\
& + 2860y^2 + 3842y - 1724 \\
\gamma \quad & -9x^5y^3 - 36x^5y^2 - 90x^5y + 72x^5 + 989y^3 + 4523y^2 + 7307y + 6452 \\
\{i, j\} = \{15, 34\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7 \\
\alpha \quad & -37x^5 + 1369y + 1196 \\
\beta \quad & 37x^9 + 37x^8y + 74x^8 + 37x^7y^2 + 148x^7y + 148x^7 - 111x^6y^2 - 740x^6y \\
& - 1480x^6 - 37x^5y^3 - 296x^5y^2 - 592x^5y + 629x^5 + 99x^4 + 99x^3y \\
& + 198x^3 + 99x^2y^2 + 396x^2y + 396x^2 + 1072xy^2 - 2054xy + 2959x + 1270y^3 \\
& - 866y^2 + 3966y + 388 \\
\gamma \quad & 37x^5y^3 + 333x^5y^2 + 1184x^5y + 1776x^5 + 99y^3 - 478y^2 + 3242y - 2167 \\
\{i, j\} = \{16, 17\}, k_{ij} = 2^3 \cdot 7^2 \\
\alpha \quad & 1 \\
\beta \quad & -x^4 - x^3y + 2x^3 - x^2y^2 + 4x^2y - 4x^2 - xy^3 + 6xy^2 - 12xy + 8x + y^4 \\
& - 2y^3 + 8y^2 + 32y + 153 \\
\gamma \quad & -y^4 + 5y^3 - 16y^2 - 85 \\
\{i, j\} = \{16, 23\}, k_{ij} = 3 \cdot 5 \cdot 7 \\
\alpha \quad & x^5 - 3y + 13 \\
\beta \quad & -x^9 - x^8y + 2x^8 - x^7y^2 + 4x^7y - 4x^7 - 2x^6y^2 + 11x^6y - 10x^6 \\
& - x^5y^3 + 7x^5y^2 - 9x^5y + x^5 - 16x^4 - 16x^3y + 32x^3 - 16x^2y^2 \\
& + 64x^2y - 64x^2 - 35xy^2 + 167xy - 187x - 19y^3 + 106y^2 - 159y + 46 \\
\gamma \quad & -x^5y^3 + 8x^5y^2 - 23x^5y + 18x^5 - 16y^3 + 131y^2 - 359y + 315 \\
\{i, j\} = \{16, 28\}, k_{ij} = 2^8 \cdot 3 \cdot 11 \\
\alpha \quad & 7x^5 - 147y + 472 \\
\beta \quad & -7x^9 - 7x^8y + 14x^8 - 7x^7y^2 + 28x^7y - 28x^7 + 7x^6y^2 - 28x^6y \\
& + 56x^6 - 7x^5y^3 + 28x^5y^2 - 119x^5 - 241x^4 - 241x^3y + 482x^3 \\
& - 241x^2y^2 + 964x^2y - 964x^2 + 94xy^2 - 1468xy - 151x - 388y^3 + 1048y^2 + 84y \\
& + 3988 \\
\gamma \quad & -7x^5y^3 + 35x^5y^2 - 56x^5y - 241y^3 + 1352y^2 - 1424y + 2079 \\
\{i, j\} = \{16, 33\}, k_{ij} = 2^2 \cdot 3^4 \\
\alpha \quad & -5x^5 - 25y + 8 \\
\beta \quad & 5x^9 + 5x^8y - 10x^8 + 5x^7y^2 - 20x^7y + 20x^7 + 10x^6y^2 - 50x^6y \\
& + 20x^6 + 5x^5y^3 - 30x^5y^2 + 10x^5y + 25x^5 + 7x^4 + 7x^3y - 14x^3 \\
& + 7x^2y^2 - 28x^2y + 28x^2 - 11xy^2 - 105xy - 92x - 18y^3 - 52y^2 - 46y + 140 \\
\gamma \quad & 5x^5y^3 - 40x^5y^2 + 110x^5y - 60x^5 + 7y^3 - 31y^2 + 189y + 36 \\
\{i, j\} = \{17, 20\}, k_{ij} = 2^6 \cdot 7
\end{aligned}$$

α	$162y^2 + 1080y + 790$
β	$972x^6 + 486x^5y + 486x^5 - 1506x^3 - 753x^2y - 753x^2 + 1474x + 162y^4 + 918y^3$ $- 233y^2 + 197y + 342$
γ	$-1944x^5 + 324x^3y^2 + 1836x^3y - 304x^3 + 3012x^2 - 324xy^3 - 1836xy^2$ $+ 304xy - 753y^2 - 4518y - 3701$
$\{i, j\} = \{17, 23\}, k_{ij} = 2 \cdot 17^2$	
α	$26080x^5 + 133000y^2 - 357600y - 163546$
β	$42960x^9 + 480x^8y + 480x^8 + 240x^7y^2 + 480x^7y + 240x^7 + 720x^6y^2$ $- 7200x^6y + 58320x^6 + 240x^5y^3 - 16880x^5y^2 + 207000x^5y - 2680x^5$ $- 278407x^4 - 226966x^3y - 226966x^3 - 113483x^2y^2 - 226966x^2y - 113483x^2$ $+ 120351xy^2 - 101430xy - 373784x + 133000y^4 - 72083y^3 + 257861y^2 - 729650y - 164124$
γ	$-112000x^8 + 42000x^7y + 154000x^7 + 70000x^6y - 154000x^6 + 240x^5y^3$ $- 42720x^5y^2 - 208880x^5y + 37600x^5 + 266000x^3y^2 + 420000x^3y$ $+ 691600x^3 - 266000x^2y^2 - 244475x^2y - 516075x^2 - 266000xy^3 - 154000xy^2$ $- 447125xy + 516075x + 418517y^3 - 182676y^2 + 825611y + 118010$
$\{i, j\} = \{17, 31\}, k_{ij} = 2^7 \cdot 7 \cdot 11$	
α	$-78y^2 - 312y + 1622$
β	$-468x^6 - 234x^5y - 234x^5 + 750x^3 + 375x^2y + 375x^2 + 630x - 78y^4 - 234y^3$ $+ 1919y^2 - 1745y - 464$
γ	$936x^5 - 156x^3y^2 - 1092x^3y + 16x^3 - 1500x^2 + 156xy^3 + 780xy^2 - 2200xy$ $+ 32x + 375y^2 + 2250y - 3885$
$\{i, j\} = \{17, 39\}, k_{ij} = 2^6 \cdot 5 \cdot 7$	
α	$24x^5 + 72y + 8$
β	$-12x^9 - 6x^8y - 6x^8 - 3x^7y^2 - 6x^7y - 3x^7 + 3x^6y^2 + 48x^6y - 39x^6$ $+ 3x^5y^3 + 33x^5y^2 + 57x^5y - 69x^5 - 88x^4 - 44x^3y - 44x^3 - 22x^2y^2$ $- 44x^2y - 22x^2 + 166xy^2 + 16xy - 646x + 94y^3 + 290y^2 - 302y - 962$
γ	$-3x^5y^3 - 15x^5y^2 - 105x^5y + 75x^5 - 22y^3 - 398y^2 - 98y + 1270$
$\{i, j\} = \{18, 22\}, k_{ij} = 2^3 \cdot 3^2$	
α	$16x^5 - 16y + 8$
β	$-8x^9 - 4x^8y + 4x^8 - 2x^7y^2 + 4x^7y - 2x^7 - 2x^6y^2 + 36x^6y + 6x^6$ $- 2x^5y^3 + 24x^5y^2 - 54x^5y - 16x^5 - 4x^4 - 2x^3y + 2x^3 - x^2y^2$ $+ 2x^2y - x^2 - 33xy^2 + 18xy - 45x - 17y^3 + 60y^2 - 43y + 64$
γ	$-2x^5y^3 + 10x^5y^2 - 78x^5y - 10x^5 - y^3 + 69y^2 - 39y + 91$
$\{i, j\} = \{18, 29\}, k_{ij} = 2^4 \cdot 3 \cdot 11$	
α	$-16x^5 + 96y - 8$
β	$8x^9 + 4x^8y - 4x^8 + 2x^7y^2 - 4x^7y + 2x^7 + 2x^6y^2 - 40x^6y + 14x^6$ $+ 2x^5y^3 - 26x^5y^2 + 70x^5y - 14x^5 + 4x^4 + 2x^3y - 2x^3 + x^2y^2$ $- 2x^2y + x^2 + 193xy^2 - 20xy + 1063x + 97y^3 - 301y^2 + 515y - 1591$
γ	$2x^5y^3 - 10x^5y^2 + 86x^5y - 30x^5 + y^3 - 389y^2 + 43y - 2127$
$\{i, j\} = \{18, 33\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5 \cdot 7$	
α	$88x^5 - 2904y - 6176$
β	$-44x^9 - 22x^8y + 22x^8 - 11x^7y^2 + 22x^7y - 11x^7 - 33x^6y^2 + 396x^6y$ $- 1551x^6 - 11x^5y^3 + 165x^5y^2 - 561x^5y - 825x^5 + 1900x^4 + 950x^3y$ $- 950x^3 + 475x^2y^2 - 950x^2y + 475x^2 - 4383xy^2 + 1380xy - 15657x - 2429y^3$ $- 789y^2 - 6399y - 4503$
γ	$-11x^5y^3 + 99x^5y^2 - 825x^5y + 3113x^5 + 475y^3 + 7341y^2 - 1335y + 30839$
$\{i, j\} = \{18, 35\}, k_{ij} = 2^9$	
α	$-8x^5 + 1112y - 4448$
β	$4x^9 + 2x^8y - 2x^8 + x^7y^2 - 2x^7y + x^7 - 3x^6y^2 - 32x^6y - 265x^6$ $- x^5y^3 - 13x^5y^2 - 83x^5y + 401x^5 + 2292x^4 + 1146x^3y - 1146x^3$ $+ 573x^2y^2 - 1146x^2y + 573x^2 + 505xy^2 - 272xy + 4899x + 539y^3 - 1753y^2$ $+ 3161y - 5411$
γ	$x^5y^3 + 3x^5y^2 + 67x^5y + 529x^5 + 573y^3 - 2729y^2 + 2263y - 10371$
$\{i, j\} = \{19, 21\}, k_{ij} = 2^3 \cdot 5^2$	
α	$-24x^5 + 36y + 128$
β	$-12x^9 + 6x^8y + 12x^8 - 3x^7y^2 - 12x^7y - 12x^7 - 6x^6y^2 + 12x^6y$ $- 48x^6 + 3x^5y^3 + 12x^5y + 36x^5 + 16x^4 - 8x^3y - 16x^3 + 4x^2y^2$ $+ 16x^2y + 16x^2 - 64xy^2 - 208xy + 100x + 32y^3 + 168y^2 + 176y - 36$
γ	$-3x^5y^3 - 30x^5y^2 - 12x^5y - 120x^5 + 4y^3 - 104y^2 - 368y + 232$
$\{i, j\} = \{19, 27\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 17$	

α	$-128x^5 - 1024y - 2168$
β	$-64x^9 + 32x^8y + 64x^8 - 16x^7y^2 - 64x^7y - 64x^7 - 48x^6y^2 - 160x^6y$ $-1088x^6 + 16x^5y^3 + 32x^5y^2 + 448x^5y - 128x^5 - 1020x^4 + 510x^3y$ $+1020x^3 - 255x^2y^2 - 1020x^2y - 1020x^2 + 1283xy^2 + 1802xy - 1468x - 769y^3$ $-1666y^2 - 1308y - 2104$
γ	$-16x^5y^3 - 192x^5y^2 - 512x^5y - 2304x^5 - 255y^3 + 1036y^2 + 544y - 4976$
$\{i, j\} = \{19, 29\}, k_{ij} = 2^5 \cdot 3 \cdot 5$	
α	$672x^5 - 1302y^2 + 44408y + 18712$
β	$336x^9 + 266x^8y + 532x^8 - 133x^7y^2 - 532x^7y - 532x^7 + 399x^6y^2$ $+532x^6y - 2128x^6 - 133x^5y^3 + 203x^5y^2 + 1232x^5y + 2520x^5$ $+151099x^4 - 246468x^3y - 492936x^3 + 123234x^2y^2 + 492936x^2y + 492936x^2$ $-399494xy^2 - 203528xy + 56071x - 1302y^4 + 165038y^3 - 62941y^2 + 307907y - 285226$
γ	$868x^7y + 1736x^7 + 868x^6y + 1736x^6 - 133x^5y^3 - 868x^5y^2 - 532x^5y$ $-1848x^5 + 2604x^3y^2 - 40796x^3y + 286006x^3 + 2604x^2y^2 - 382633x^2y$ $-397668x^2 - 2604xy^3 + 46004xy^2 - 709435xy - 111662x + 118026y^3 + 371657y^2$ $+377352y + 588684$
$\{i, j\} = \{19, 33\}, k_{ij} = 2^4 \cdot 11$	
α	$8x^5 - 184y^2 - 1096y - 784$
β	$-28x^9 + 6x^8y + 12x^8 - 3x^7y^2 - 12x^7y - 12x^7 + 3x^6y^2 + 48x^6y$ $-264x^6 - 3x^5y^3 + 15x^5y^2 + 122x^5y - 24x^5 - 140x^4 - 992x^3y$ $-1984x^3 + 496x^2y^2 + 1984x^2y + 1984x^2 - 2080xy^2 + 512xy + 2212x - 184y^4$ $+504y^3 + 3884y^2 + 8132y + 4248$
γ	$-64x^8 - 16x^7y + 32x^7 + 80x^6y + 32x^6 - 3x^5y^3 + 4x^5y^2 - 36x^5y$ $-616x^5 + 368x^3y^2 + 1936x^3y + 1064x^3 - 368x^2y^2 - 4060x^2y - 5312x^2$ $-368xy^3 - 1936xy^2 + 1060xy + 4248x + 1232y^3 + 5180y^2 + 15288y + 9456$
$\{i, j\} = \{19, 38\}, k_{ij} = 2^2 \cdot 11^2$	
α	$2256x^5 + 924y^2 + 60208y + 72950$
β	$1128x^9 - 608x^8y - 1216x^8 + 304x^7y^2 + 1216x^7y + 1216x^7 + 912x^6y^2$ $+3648x^6y + 29184x^6 - 304x^5y^3 - 1000x^5y^2 - 13076x^5y + 2080x^5$ $-137899x^4 - 30210x^3y - 60420x^3 + 15105x^2y^2 + 60420x^2y + 60420x^2$ $-32509xy^2 - 100852xy - 275071x + 924y^4 + 39559y^3 + 101362y^2 + 335803y + 217928$
γ	$-88x^7y - 176x^7 + 88x^6y + 176x^6 + 304x^5y^3 + 3560x^5y^2 + 10592x^5y$ $+60448x^5 - 1848x^3y^2 - 22264x^3y - 286396x^3 + 1848x^2y^2 - 176055x^2y$ $-110242x^2 - 1848xy^3 - 25960xy^2 - 132605xy - 176154x + 18801y^3 - 122635y^2$ $-174136y - 363390$
$\{i, j\} = \{19, 41\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 11 \cdot 13$	
α	$-656x^5 - 26896y - 52136$
β	$-328x^9 + 164x^8y + 328x^8 - 82x^7y^2 - 328x^7y - 328x^7 - 328x^6y^2$ $-2132x^6y - 13448x^6 + 164x^5y^3 + 1394x^5y^2 + 8856x^5y + 13120x^5$ $-25740x^4 + 12870x^3y + 25740x^3 - 6435x^2y^2 - 25740x^2y - 25740x^2$ $+28052xy^2 + 49170xy - 1148x - 14026y^3 - 52637y^2 - 62044y - 24920$
γ	$-82x^5y^3 - 1148x^5y^2 - 5248x^5y - 27552x^5 - 6435y^3 + 17494y^2 + 21120y$ -53776
$\{i, j\} = \{20, 27\}, k_{ij} = 2^4 \cdot 5 \cdot 7$	
α	$792x^5y^2 + 198x^5y - 1584x^5 + 248y^2 + 6596y + 560$
β	$-297x^{11} - 594x^{10}y - 297x^{10} + 396x^8y^2 + 99x^8y + 1584x^8 - 396x^6y^3$ $-99x^6y^2 - 1584x^6y - 588x^6 - 396x^5y^4 - 99x^5y^3 - 792x^5y^2$ $-978x^5y - 2172x^5 + 124x^3y^2 + 3298x^3y + 1024x^3 - 124xy^3 - 3298xy^2$ $-1024xy - 9956x - 124y^4 - 3298y^3 - 776y^2 - 13316y - 9396$
γ	$297x^{11} + 594x^{10}y + 588x^6 + 396x^5y^4 + 99x^5y^3 + 792x^5y^2 + 978x^5y$ $+1584x^5 + 9956x + 124y^4 + 3298y^3 + 776y^2 + 13316y - 560$
$\{i, j\} = \{20, 28\}, k_{ij} = 2$	
α	2
β	$-2x^3 + xy^2 + 2xy + 4x - 3y^2 - 9$
γ	$-xy^2 - 4x + y^2 + 1$
$\{i, j\} = \{20, 30\}, k_{ij} = 2$	
α	-1
β	$-x^6 - x^3y + xy^2 - 4x - y^3 + 1$
γ	$x^6 + x^5 + 4x + y^3 + 3$
$\{i, j\} = \{20, 32\}, k_{ij} = 2$	

α	0
β	-1
γ	1
$\{i, j\}$	$\{20, 35\}, k_{ij} = 2 \cdot 5 \cdot 7$
α	$248y^2 + 186y - 97$
β	$31x^6 - 62x^5y - 62x^5 - 124x^3y^2 - 31x^3y - 28x^3 + 312x^2y + 312x^2$ $+ 124xy^3 + 31xy^2 - 128xy - 70x - 124y^4 - 31y^3 - 744y^2 - 826y + 81$
γ	$-31x^6 + 62x^5y + 31x^5 + 156x^3 - 312x^2y - 156x^2 + 70x + 124y^4 + 31y^3$ $+ 120y^2 + 46y - 167$
$\{i, j\}$	$\{20, 41\}, k_{ij} = 2^2 \cdot 5$
α	$81x + 27$
β	$-81x^4 - 27x^3 + 54x^2y + 9x^2 - 27xy^2 + 27xy - 108x - 9y^2 + 3y - 97$
γ	$27x^2y - 9x^2 + 27xy^2 + 108x + 9y^2 - 3y + 7$
$\{i, j\}$	$\{20, 42\}, k_{ij} = 2 \cdot 3$
α	$-x - 2$
β	$x^4 + 2x^3 - 2x^2y + x^2 - xy^2 - 3xy - 3x - 2y^2 + 2y - 6$
γ	$x^2y - x^2 + xy^2 + xy + 3x + 2y^2 - 2y + 8$
$\{i, j\}$	$\{21, 27\}, k_{ij} = 1$
α	1
β	$-x^3 + xy^2 - xy + 4x + 2y^2 + 5$
γ	$-xy^2 - 4x - y^2 - 1$
$\{i, j\}$	$\{21, 35\}, k_{ij} = 2^2$
α	1
β	$x^6 - 2x^5 - x^3y - xy^2 - 4x - y^3 + 7$
γ	$-x^6 + x^5 + 4x + y^3 - 3$
$\{i, j\}$	$\{21, 38\}, k_{ij} = 2^3$
α	1
β	$x^6 + 2x^5 + x^3y - 2x^3 + xy^2 + 6x - y^3 + 2y^2 + 4y + 13$
γ	$-x^6 - x^5 - 2xy - 6x + y^3 - 2y^2 - 2y - 7$
$\{i, j\}$	$\{22, 33\}, k_{ij} = 3^3 \cdot 5 \cdot 7$
α	$192y - 176$
β	$32x^9 + 64x^8 + 96x^7 - 64x^6y + 96x^6 - 32x^5y^2 - 32x^5y + 48x^5 + 180x^4$ $+ 128x^3y^2 - 160x^3y + 488x^3 + 128x^2y^2 - 160x^2y + 668x^2 - 128xy^3$ $+ 288xy^2 - 648xy + 1134x - 64y^4 + 16y^3 - 100y^2 - 228y + 479$
γ	$-32x^9 + 32x^5y^2 - 128x^5y + 48x^5 - 180x^4 - 466x + 64y^4 - 272y^3 + 548y^2$ $- 1088y + 1121$
$\{i, j\}$	$\{23, 26\}, k_{ij} = 2 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11$
α	$272x^5 - 1768y^2 - 3944y + 8994$
β	$-34x^9 - 238x^8 + 68x^7y + 204x^7 + 170x^6y - 204x^6 - 170x^5y^2 + 204x^5y$ $- 272x^5 - 3925x^4 + 884x^3y^2 + 4182x^3y - 5817x^3 - 884x^2y^2 + 3668x^2y$ $+ 1892x^2 - 884xy^3 - 3298xy^2 + 2149xy - 2347x + 884y^4 + 3298y^3 - 381y^2 + 6746y$ $- 9449$
γ	$34x^9 + 102x^5y^2 - 578x^5y + 476x^5 + 3925x^4 + 455x - 884y^4 - 1530y^3$ $+ 11y^2 - 10787y + 11796$
$\{i, j\}$	$\{23, 31\}, k_{ij} = 2^2 \cdot 7^3$
α	$-16y - 60$
β	$2x^9 - 2x^8 - 2x^7 + 2x^6y + 2x^6 - 2x^5y^2 - 2x^5y + 7x^4 + 4x^3y^2$ $+ 22x^3y + 49x^3 - 4x^2y^2 - 22x^2y - 63x^2 - 4xy^3 - 18xy^2 - 27xy - 530x + 4y^4$ $+ 18y^3 + 27y^2 - 47y + 1246$
γ	$-2x^9 + 2x^5y^2 + 2x^5y - 2x^5 - 7x^4 + 593x - 4y^4 - 10y^3 + 13y^2 + 137y - 716$
$\{i, j\}$	$\{23, 42\}, k_{ij} = 3^2 \cdot 17$
α	$8x^5 + 44y^2 - 104y - 213$
β	$8x^9 - 16x^8 + 8x^7y + 24x^7 + 8x^6y - 24x^6 - 8x^5y^2 + 24x^5y - 8x^5$ $+ 83x^4 + 44x^3y^2 + 72x^3y + 90x^3 - 44x^2y^2 + 11x^2y - 7x^2 - 44xy^3$ $- 28xy^2 - 101xy - 286x + 44y^4 + 28y^3 + 57y^2 - 196y - 80$
γ	$-8x^9 - 56x^5y + 32x^5 - 83x^4 + 293x - 44y^4 + 60y^3 - 40y^2 + 304y + 366$
$\{i, j\}$	$\{24, 25\}, k_{ij} = 2 \cdot 3 \cdot 7$
α	$-12y + 14$
β	$-2x^9 - 2x^8 - 2x^6y + 2x^5y^2 - 21x^4 - 4x^3y^2 + 10x^3y - 35x^3$ $- 4x^2y^2 + 10x^2y - 14x^2 - 4xy^3 + 6xy^2 - 25xy + 85x + 4y^4 - 6y^3 + 25y^2 + 2y$

$$\begin{aligned}
& +113 \\
\gamma \quad & 2x^9 - 2x^5y^2 - 2x^5y + 21x^4 - 99x - 4y^4 - 2y^3 - 9y^2 - 41y - 28 \\
\{i, j\} = \{24, 26\}, k_{ij} = 2 \cdot 3 \cdot 5^2 \\
\alpha \quad & -4y + 18 \\
\beta \quad & -2x^9 - 2x^8 - 4x^7 - 2x^6y - 4x^6 + 2x^5y^2 + 4x^5y + 7x^4 - 4x^3y^2 \\
& + 10x^3y - 7x^3 - 4x^2y^2 + 10x^2y - 4xy^3 + 6xy^2 + 3xy + 75x + 4y^4 - 6y^3 \\
& - 3y^2 - 4y - 57 \\
\gamma \quad & 2x^9 - 2x^5y^2 - 10x^5y - 4x^5 - 7x^4 - 75x - 4y^4 - 2y^3 + 19y^2 + 7y + 132 \\
\{i, j\} = \{24, 30\}, k_{ij} = 3 \cdot 5 \\
\alpha \quad & -16x - 8 \\
\beta \quad & 16x^4 + 24x^3 + 8x^2y + 4x^2 - 8xy^2 + 24xy - 20x - 4y^2 + 10y - 3 \\
\gamma \quad & 8x^2y + 20x^2 + 8xy^2 + 16xy + 44x + 4y^2 + 6y + 23 \\
\{i, j\} = \{24, 33\}, k_{ij} = 3^2 \cdot 11 \\
\alpha \quad & -64y - 40 \\
\beta \quad & -24x^4 + 64x^3y - 8x^3 + 16x^2y - 32x^2 + 64xy^2 + 8xy + 58x - 32y^3 + 36y^2 - 118y \\
& + 9 \\
\gamma \quad & 24x^4 - 90x + 32y^3 + 44y^2 + 94y + 139 \\
\{i, j\} = \{24, 34\}, k_{ij} = 2^3 \\
\alpha \quad & -8y + 8 \\
\beta \quad & 2x^9 - 2x^8 - 2x^7 - 2x^6y - 2x^6 - 2x^5y^2 - 10x^5y - 4x^5 + 11x^4 \\
& - 4x^3y^2 + 6x^3y - 19x^3 - 4x^2y^2 + 6x^2y - 19x^2 - 4xy^3 + 2xy^2 - 13xy \\
& - 62x - 4y^4 - 14y^3 + 3y^2 - 75y + 40 \\
\gamma \quad & -2x^9 + 2x^5y^2 + 6x^5y + 2x^5 - 11x^4 + 43x + 4y^4 + 6y^3 + 5y^2 + 43y - 16 \\
\{i, j\} = \{25, 33\}, k_{ij} = 2^3 \cdot 3^4 \cdot 7 \\
\alpha \quad & 536y^2 - 670y - 2340 \\
\beta \quad & -67x^6 + 134x^5y + 67x^5 - 268x^3y^2 + 1139x^3y + 281x^3 + 4520x^2y \\
& + 2260x^2 - 268xy^3 + 871xy^2 + 3680xy + 3839x + 268y^4 - 603y^3 + 5025y^2 + 6333y \\
& + 601 \\
\gamma \quad & 67x^6 - 134x^5y - 134x^5 + 2260x^3 - 4520x^2y - 4520x^2 - 1298x - 268y^4 \\
& + 603y^3 + 4015y^2 + 9487y + 7477 \\
\{i, j\} = \{25, 39\}, k_{ij} = 2^2 \cdot 3 \\
\alpha \quad & 2 \\
\beta \quad & -x^6 + x^5 - x^3y + x^3 - xy^2 - 2xy + 5x + y^3 + y^2 + y - 3 \\
\gamma \quad & x^6 - 2x^5 + 2xy - 4x - y^3 - y^2 - 3y + 7 \\
\{i, j\} = \{26, 36\}, k_{ij} = 2 \cdot 3 \\
\alpha \quad & 1 \\
\beta \quad & -x^3 - xy^2 - 2xy - 4x + 3y + 3 \\
\gamma \quad & xy^2 + xy + 5x + y^2 - 2y + 2 \\
\{i, j\} = \{26, 40\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \\
\alpha \quad & x^5 + 3y^2 - 18y - 29 \\
\beta \quad & -x^8 - x^6y + 2x^6 - x^5y^2 + x^5y - 4x^5 - 3x^3y^2 + 15x^3y + 32x^3 \\
& - 3xy^3 + 18xy^2 + 17xy - 136x - 3y^4 + 21y^3 - 4y^2 + 73y + 269 \\
\gamma \quad & -x^6 + x^5y^2 - x^5y + 3x^5 + 104x + 3y^4 - 21y^3 + 4y^2 - 73y - 165 \\
\{i, j\} = \{27, 33\}, k_{ij} = 3 \cdot 17 \\
\alpha \quad & -16x - 40 \\
\beta \quad & 16x^4 + 24x^3 + 8x^2y - 44x^2 - 8xy^2 + 8xy - 60x - 20y^2 - 30y - 119 \\
\gamma \quad & 8x^2y + 20x^2 + 8xy^2 + 84x + 20y^2 - 50y + 91 \\
\{i, j\} = \{27, 37\}, k_{ij} = 2^2 \cdot 5^2 \cdot 17 \\
\alpha \quad & -16y - 68 \\
\beta \quad & 2x^9 - 2x^8 - 2x^7 - 2x^6y + 2x^6 - 2x^5y^2 + 2x^5y + 135x^4 + 4x^3y^2 \\
& + 10x^3y - 71x^3 - 4x^2y^2 - 10x^2y - 199x^2 + 4xy^3 + 14xy^2 - 61xy - 418x \\
& + 4y^4 + 14y^3 - 61y^2 + 215y + 1302 \\
\gamma \quad & -2x^9 + 2x^5y^2 - 2x^5y - 2x^5 - 135x^4 + 617x - 4y^4 - 22y^3 + 37y^2 - 353y - 884 \\
\{i, j\} = \{28, 31\}, k_{ij} = 2^6 \cdot 5 \cdot 11 \\
\alpha \quad & -40y + 480 \\
\beta \quad & -x^9 - 2x^8 - 2x^7 + 2x^6y - 4x^6 + x^5y^2 - 2x^5y + 466x^4 - 8x^3y^2 \\
& + 64x^3y + 436x^3 - 16x^2y^2 + 128x^2y - 60x^2 + 8xy^3 - 96xy^2 - 180xy - 1160x \\
& + 4y^4 - 48y^3 - 90y^2 - 80y - 800 \\
\gamma \quad & x^9 - x^5y^2 + 8x^5y - 8x^5 - 466x^4 + 1040x - 4y^4 + 80y^3 - 230y^2 - 220y - 1520
\end{aligned}$$

$$\begin{aligned}
& \{i, j\} = \{28, 39\}, k_{ij} = 2^4 \cdot 5 \cdot 7 \\
& \alpha \quad -67944x^5 + 53580y^2 - 96672y + 108025 \\
& \beta \quad -28215x^9 + 2850x^8 + 28500x^7y - 8550x^7 + 54150x^6y - 17100x^6 \\
& \quad + 33687x^5y^2 + 21147x^5y - 35910x^5 + 200887x^4 + 52440x^3y^2 + 207024x^3y \\
& \quad + 426316x^3 + 104880x^2y^2 + 94104x^2y + 1012604x^2 - 52440xy^3 + 2736xy^2 \\
& \quad - 238108xy + 787447x - 26790y^4 - 97869y^3 - 207216y^2 - 655866y + 598703 \\
& \gamma \quad 28215x^9 - 19551x^8 + 27930x^7y + 5586x^7 - 47481x^6y + 13965x^6 \\
& \quad - 34257x^5y^2 + 120840x^5y - 17841x^5 - 200887x^4 + 27930x^3y^2 + 58653x^3y \\
& \quad + 154105x^3 - 27930x^2y^2 - 140483x^2y - 113190x^2 + 27930xy^3 + 58653xy^2 \\
& \quad + 235935xy + 1196846x + 26790y^4 - 167751y^3 - 52622y^2 - 1041596y - 107465 \\
& \{i, j\} = \{29, 32\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5 \cdot 7 \\
& \alpha \quad 2y + 1 \\
& \beta \quad -x^9 - x^8 - 4x^7 + x^6y - 5x^6 + x^5y^2 - 6x^5y + 5x^5 + 113x^4 - 2x^3y^2 \\
& \quad - 21x^3y + 145x^3 - 2x^2y^2 - 21x^2y + 484x^2 + 2xy^3 + 17xy^2 - 187xy + 3348x \\
& \quad + 2y^4 + 15y^3 - 204y^2 + 818y - 8785 \\
& \gamma \quad x^9 - x^5y^2 + 11x^5y - 14x^5 - 113x^4 - 2719x - 2y^4 - 11y^3 + 240y^2 - 1510y \\
& \quad + 12617 \\
& \{i, j\} = \{29, 35\}, k_{ij} = 2^2 \\
& \alpha \quad -2x^4 + 14 \\
& \beta \quad 2x^7 + 2x^6 - x^5y - x^5 + x^4y^2 - 2x^4y + 5x^4 - 2x^3 + 6x^2 + xy + x - 7y^2 \\
& \quad - 2y - 25 \\
& \gamma \quad -x^5y + 5x^5 - x^4y^2 - 2x^4y + x^4 + xy + 3x + 7y^2 - 2y + 35 \\
& \{i, j\} = \{29, 38\}, k_{ij} = 2^5 \cdot 3 \\
& \alpha \quad 14y - 2 \\
& \beta \quad 7x^4y - x^4 - 14x^2y^2 + 2x^2y - 14xy^2 + 2xy - 49x - 7y^2 - 41y - 50 \\
& \gamma \quad -7x^4y + x^4 + 49x + 14y^3 - 23y^2 + 45y - 48 \\
& \{i, j\} = \{29, 39\}, k_{ij} = 2^2 \cdot 3^2 \\
& \alpha \quad 8x^5 - 12y^2 + 16y + 2 \\
& \beta \quad 4x^9 - 8x^7y + 8x^7 - 8x^6y + 8x^6 - 4x^5y + 8x^5 - 29x^4 - 12x^3y^2 \\
& \quad - 32x^3y - 68x^3 - 12x^2y^2 + 26x^2y - 126x^2 + 12xy^3 + 8xy^2 + 62xy - 79x \\
& \quad + 6y^4 + 10y^3 + 6y^2 + 102y - 62 \\
& \gamma \quad -4x^9 + 8x^5y^2 - 20x^5y + 8x^5 + 29x^4 - 115x - 6y^4 + 14y^3 - 36y^2 + 112y - 28 \\
& \{i, j\} = \{30, 38\}, k_{ij} = 2^3 \cdot 5 \\
& \alpha \quad 8x^5 + 12y^2 + 4y - 18 \\
& \beta \quad 4x^9 - 8x^7y - 8x^7 - 8x^6y - 8x^6 - 4x^5y + 15x^4 + 12x^3y^2 - 20x^3y \\
& \quad + 40x^3 + 12x^2y^2 - 50x^2y + 10x^2 - 12xy^3 + 20xy^2 - 70xy + 23x - 6y^4 - 8y^3 \\
& \quad + 10y^2 - 10y + 62 \\
& \gamma \quad -4x^9 + 8x^5y^2 + 12x^5y - 15x^4 - 53x + 6y^4 - 16y^3 + 48y^2 - 20y + 4 \\
& \{i, j\} = \{31, 33\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \cdot 11 \\
& \alpha \quad -3 \\
& \beta \quad -x^6 + 2x^5 + x^3y + 10x^3 - xy^2 - 26xy + 266x - y^3 - 6y^2 + 72y - 535 \\
& \gamma \quad x^6 - 3x^5 + 18xy - 246x + y^3 + 6y^2 - 90y + 781 \\
& \{i, j\} = \{31, 37\}, k_{ij} = 2 \cdot 7^2 \\
& \alpha \quad 10x - 2 \\
& \beta \quad -10x^4 + 2x^3 + 5x^2y - 5xy^2 - xy - 20x + y^2 - 21 \\
& \gamma \quad 5x^2y - 20x^2 + 5xy^2 - xy + 24x - y^2 - 29 \\
& \{i, j\} = \{32, 33\}, k_{ij} = 2^2 \cdot 3^2 \\
& \alpha \quad -1 \\
& \beta \quad x^6 + 2x^5 - x^3y - 2x^3 + xy^2 - 2xy + 14x + y^3 + 6y^2 + 27 \\
& \gamma \quad -x^6 - x^5 + 6xy - 10x - y^3 - 6y^2 - 6y - 17 \\
& \{i, j\} = \{32, 37\}, k_{ij} = 2 \cdot 3^2 \cdot 5^2 \\
& \alpha \quad -2x - 30 \\
& \beta \quad 2x^4 + 30x^3 - x^2y - 2x^2 + xy^2 - 17xy - 22x + 15y^2 - 30y + 121 \\
& \gamma \quad -x^2y - 2x^2 - xy^2 - 13xy - 38x - 15y^2 + 30y - 119 \\
& \{i, j\} = \{32, 39\}, k_{ij} = 2^3 \cdot 3^3 \cdot 13 \\
& \alpha \quad -392y^2 - 1078y - 897 \\
& \beta \quad 147x^6 + 294x^5y + 294x^5 - 196x^3y^2 + 343x^3y - 768x^3 - 2448x^2y \\
& \quad - 2448x^2 + 196xy^3 + 49xy^2 - 1142xy - 1218x + 196y^4 + 441y^3 + 3460y^2 + 1234y + 339 \\
& \gamma \quad -147x^6 - 294x^5y - 147x^5 + 1224x^3 + 2448x^2y + 1224x^2 + 306x - 196y^4 \\
& \quad - 441y^3 + 1436y^2 + 4886y + 3027
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{32, 41\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
&\alpha \quad -3x - 6 \\
&\beta \quad 3x^4 + 6x^3 - 2x^2y - 3x^2 + xy^2 - 5xy - x + 2y^2 - 2y + 14 \\
&\gamma \quad -x^2y - 3x^2 - xy^2 - xy - 11x - 2y^2 + 2y - 8 \\
&\{i, j\} = \{32, 42\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
&\alpha \quad -5x + 3 \\
&\beta \quad 5x^4 - 3x^3 - 10x^2y - 35x^2 - 5xy^2 + xy - 4x + 3y^2 + 3y + 115 \\
&\gamma \quad 5x^2y + 25x^2 + 5xy^2 + 2xy + 10x - 3y^2 - 3y - 65 \\
&\{i, j\} = \{33, 34\}, k_{ij} = 2 \cdot 3 \cdot 7 \cdot 11 \\
&\alpha \quad 21432x^5 - 83524y^2 - 147160y - 150723 \\
&\beta \quad 2877x^9 - 33825x^8 - 6150x^7y + 30750x^7 + 39975x^6y + 3075x^6 - 7443x^5y^2 \\
&\quad -41032x^5y - 10127x^5 - 183426x^4 + 178350x^3y^2 + 857925x^3y - 443128x^3 \\
&\quad -178350x^2y^2 - 1700721x^2y + 21730x^2 - 178350xy^3 - 857925xy^2 + 1285924xy \\
&\quad +3471868x + 41762y^4 + 124921y^3 + 464727y^2 + 2335032y + 151185 \\
&\gamma \quad -2877x^9 + 3762x^8 + 396x^7y + 7722x^7 + 4554x^6y + 15444x^6 + 13989x^5y^2 \\
&\quad -6713x^5y + 7190x^5 + 183426x^4 - 53064x^3y^2 - 738144x^3y + 1132824x^3 \\
&\quad -106128x^2y^2 - 266640x^2y + 2870472x^2 - 53064xy^3 - 950400xy^2 + 599544xy \\
&\quad +2690474x - 41762y^4 + 19523y^3 + 104829y^2 + 2127595y + 1736165 \\
&\{i, j\} = \{33, 37\}, k_{ij} = 2 \cdot 3^6 \cdot 7 \cdot 11 \\
&\alpha \quad 824x^5 - 412y^2 - 11124y - 28599 \\
&\beta \quad -103x^9 - 721x^8 + 206x^7y + 1030x^7 + 515x^6y - 309x^6 - 515x^5y^2 \\
&\quad -206x^5y - 515x^5 + 5210x^4 + 206x^3y^2 + 5253x^3y + 22668x^3 - 206x^2y^2 \\
&\quad -15673x^2y - 38298x^2 - 206xy^3 - 5253xy^2 - 12248xy - 5412x + 206y^4 + 5459y^3 \\
&\quad +17913y^2 + 49826y + 76095 \\
&\gamma \quad 103x^9 + 309x^5y^2 - 1545x^5y - 206x^5 - 5210x^4 + 21042x - 206y^4 - 5047y^3 \\
&\quad +3219y^2 + 16393y - 32385 \\
&\{i, j\} = \{33, 40\}, k_{ij} = 2^7 \cdot 3^3 \cdot 17 \\
&\alpha \quad 11736x^5 + 18582y^2 - 87368y - 20726 \\
&\beta \quad -652x^9 - 10432x^8 + 1956x^7y + 11084x^7 + 8476x^6y - 652x^6 - 5216x^5y^2 \\
&\quad -14670x^5y - 10758x^5 + 12163x^4 - 12388x^3y^2 + 66504x^3y - 4578x^3 \\
&\quad +12388x^2y^2 - 102993x^2y - 7585x^2 + 12388xy^3 - 66504xy^2 + 41067xy - 39494x \\
&\quad -6194y^4 + 14670y^3 + 81982y^2 + 129310y + 79478 \\
&\gamma \quad 652x^9 + 3260x^5y^2 - 6846x^5y + 326x^5 - 12163x^4 + 51657x + 6194y^4 \\
&\quad -39446y^3 + 75127y^2 - 59799y + 19258 \\
&\{i, j\} = \{33, 41\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5 \cdot 11 \\
&\alpha \quad 612x^5 - 2040y^2 - 4046y - 2786 \\
&\beta \quad -68x^9 - 544x^8 + 204x^7y + 680x^7 + 340x^6y - 136x^6 - 340x^5y^2 - 204x^5y \\
&\quad -476x^5 + 149x^4 + 680x^3y^2 + 1122x^3y + 2433x^3 - 680x^2y^2 - 1569x^2y \\
&\quad -2731x^2 - 680xy^3 - 1122xy^2 - 1986xy - 443x + 680y^4 + 1802y^3 + 5148y^2 + 7957y \\
&\quad +3970 \\
&\gamma \quad 68x^9 + 136x^5y^2 - 1020x^5y - 68x^5 - 149x^4 + 741x - 680y^4 - 442y^3 \\
&\quad -1777y^2 - 1671y - 796 \\
&\{i, j\} = \{34, 35\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad -20y - 40 \\
&\beta \quad -x^9 - x^8 - x^7 - x^6y - 2x^6 + x^5y^2 + 5x^5y + 4x^5 - 84x^4 - 4x^3y^2 \\
&\quad +8x^3y - 52x^3 - 8x^2y^2 + 16x^2y - 20x^2 - 4xy^3 - 8xy^2 - 20xy + 4y^4 \\
&\quad +24y^3 + 20y^2 + 80y + 40 \\
&\gamma \quad x^9 - x^5y^2 - 8x^5y - 8x^5 + 84x^4 - 40x - 4y^4 - 40y^3 - 20y^2 - 140y - 80 \\
&\{i, j\} = \{34, 42\}, k_{ij} = 2^3 \cdot 5 \cdot 7 \cdot 11 \\
&\alpha \quad 6192x^5 - 17085792y^2 - 26567120y - 9489060 \\
&\beta \quad 172x^9 - 5848x^8 + 516x^7y - 11524x^7 - 4816x^6y + 3184752x^6 + 2408x^5y^2 \\
&\quad -4800176x^5y - 1600804x^5 - 3536595x^4 - 157552x^3y^2 + 4350224x^3y \\
&\quad -7893372x^3 - 315104x^2y^2 + 9559294x^2y - 208954x^2 - 157552xy^3 \\
&\quad +3720016xy^2 - 4066292xy - 6296840x - 5695264y^4 - 7377424y^3 + 2929388y^2 \\
&\quad -1211782y - 2971435 \\
&\gamma \quad -172x^9 - 3207800x^6 - 1892x^5y^2 + 4779020x^5y + 7970308x^5 + 3536595x^4 \\
&\quad +2887020x^3y^2 - 128312x^3y + 2332742x^3 - 11468631x^2y - 19114385x^2 \\
&\quad -5774040xy^3 - 2630396xy^2 + 10754336xy + 3546190x + 5695264y^4 + 6747216y^3 \\
&\quad +37007200y^2 + 42567645y + 9492140 \\
&\{i, j\} = \{35, 39\}, k_{ij} = 2^3
\end{aligned}$$

α	$4y - 2$
β	$x^9 + 2x^8 + 4x^7 + 2x^6y + 6x^6 - x^5y^2 - x^5y + 2x^5 + 7x^4 + 4x^3y^2$ $- 2x^3y + 10x^3 + 4x^2y^2 - 2x^2y + 24x^2 + 4xy^3 + 6xy^2 + 6xy + 21x - 2y^4$ $- y^3 - 2y^2 - 4y + 11$
γ	$-x^9 + x^5y^2 + 7x^5y + 8x^5 - 7x^4 + 13x + 2y^4 + 9y^3 + 10y^2 + 32y + 21$
$\{i, j\}$	$\{35, 40\}, k_{ij} = 2^3 \cdot 5^2$
α	$-3x^4 + 53$
β	$3x^7 + 3x^6 + 6x^5y + 24x^5 + 3x^4y^2 + 15x^4y + 45x^4 - 17x^3 + 31x^2 - 34xy$ $+ 8x - 53y^2 - 25y - 147$
γ	$-3x^5y - 18x^5 - 3x^4y^2 - 9x^4y - 36x^4 + 17xy + 6x + 53y^2 + 39y + 192$
$\{i, j\}$	$\{36, 39\}, k_{ij} = 2^7 \cdot 3$
α	$-16y + 80$
β	$-2x^9 - 4x^8 - 4x^7 - 4x^6y + 2x^5y^2 - 6x^5y + 2x^5 + 49x^4 - 8x^3y^2$ $+ 28x^3y + 14x^3 - 8x^2y^2 + 28x^2y + 14x^2 - 8xy^3 + 28xy^2 + 14xy + 159x + 4y^4$ $- 26y^3 + 39y^2 - y + 192$
γ	$2x^9 - 2x^5y^2 - 2x^5y + 2x^5 - 49x^4 - 159x - 4y^4 + 10y^3 + 25y^2 + y + 112$
$\{i, j\}$	$\{36, 41\}, k_{ij} = 2^2 \cdot 3 \cdot 11^2$
α	$-10x^5 + 10y^2 + 20y + 128$
β	$10x^9 + 20x^8 - 10x^7y + 10x^6y - 20x^6 - 10x^5y^2 + 30x^5y - 30x^5 - 213x^4$ $+ 10x^3y^2 - 30x^3y - 291x^3 + 10x^2y^2 + 183x^2y + 135x^2 + 10xy^3$ $- 30xy^2 - 78xy + 595x - 10y^4 + 40y^3 + 58y^2 - 315y + 892$
γ	$-10x^9 - 10x^5y + 10x^5 + 213x^4 - 169x + 10y^4 - 20y^3 + 85y^2 + 189y - 432$
$\{i, j\}$	$\{37, 39\}, k_{ij} = 2 \cdot 3^3 \cdot 7$
α	3
β	$-x^6 + 2x^5 - x^3y + 4x^3 - xy^2 + 2xy + 56x + y^3 - 12y - 109$
γ	$x^6 - 3x^5 - 48x - y^3 + 12y + 157$
$\{i, j\}$	$\{37, 41\}, k_{ij} = 2^2 \cdot 3 \cdot 11$
α	$5x - 3$
β	$-5x^4 + 3x^3 - 10x^2y - 35x^2 - 5xy^2 + xy - 4x + 3y^2 + 3y - 85$
γ	$5x^2y + 25x^2 + 5xy^2 + 2xy + 10x - 3y^2 - 3y + 35$
$\{i, j\}$	$\{39, 42\}, k_{ij} = 2^3 \cdot 3^2 \cdot 17^2$
α	$152352x^5 + 10156800y^2 + 2290570y + 1280126$
β	$-33856x^9 - 118496x^8 - 101568x^7y + 118496x^7 - 16928x^6y + 1251614x^6$ $- 16928x^5y^2 + 3771770x^5y - 152352x^5 + 86575x^4 - 80408x^3y^2$ $- 246514x^3y - 4499548x^3 + 80408x^2y^2 - 12794510x^2y + 65965x^2 - 80408xy^3$ $- 246514xy^2 - 325690xy - 3240230x + 3385600y^4 + 870734y^3 - 12130554y^2 - 12624774y$ $- 1363104$
γ	$33856x^9 - 1251614x^6 - 84640x^5y^2 - 3534778x^5y - 1217758x^5 - 86575x^4$ $+ 866502x^3y^2 - 607292x^3y + 4350605x^3 + 13300749x^2y + 4433583x^2$ $- 1733004xy^3 + 348082xy^2 + 773248xy + 3323208x - 3385600y^4 - 709918y^3$ $- 27099348y^2 + 3643024y + 103786$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 8$ and for $x \leq -9$.

i	k_i	$\varphi_i(x)$	$x \geq 8$	$x \leq -9$
0	-2	$-x^6 - 2x^5 - 9x^4 - 4x^3 - 7x^2 + 30x + 9$	+	+
2	-1	$-x^5 + x^3 + 3x - 1$	+	-
3	-1	$-x^5 - x + 4$	+	-
4	-1	$-x^5 - x - 6$	+	-
5	-1	$-x^5 - x^3 - 5x - 1$	+	-
6	-1	$-x^5 + 8x^3 + 7x - 1$	+	-
7	-1	$-x^5 + x^3 + 3x^2 + 6x + 4$	+	-
8	-1	$-x^5 + x^3 - 3x^2 + 6x - 6$	+	-
9	-1	$-x^5 - x + 15$	+	-
10	-1	$-x^5 - x^3 + 3x^2 - 8x + 4$	+	-
11	-1	$-x^5 + 8x^3 + 12x^2 + 13x + 4$	+	-
12	-1	$-x^5 + 8x^3 - 12x^2 + 13x - 6$	+	-
13	-1	$-x^5 + x^3 + 6x^2 + 15x + 15$	+	-
14	-1	$-x^5 + x^3 - 6x^2 + 15x - 17$	+	-

i	k_i	$\varphi_i(x)$	$x \geq 8$	$x \leq -9$
15	-1	$-x^5 - x^3 + 6x^2 - 17x + 15$	+	-
16	-1	$-x^5 - x^3 - 6x^2 - 17x - 17$	+	-
17	-1	$-x^5 - 8x^3 + 12x^2 - 15x + 4$	+	-
18	-1	$-x^5 - 8x^3 - 12x^2 - 15x - 6$	+	-
19	-1	$-x^5 + 8x^3 + 24x^2 + 31x + 15$	+	-
20	1	$-x^6 + x^5 - 4x^2 + x + 1$	+	+
21	-1	$-x^6 - x^5 - 4x^2 - x - 1$	+	+
22	1	$-x^6 + 4x^5 - 3x^4 + x^3 - 4x^2 + 5x + 1$	+	+
23	1	$-x^6 - 2x^5 - 3x^4 - x^3 - 4x^2 - 3x + 1$	+	+
24	-1	$-x^6 + 2x^5 - 3x^4 + x^3 - 4x^2 + 3x - 1$	+	+
25	-1	$-x^6 - x^5 + 3x^4 - 7x^2 - x + 4$	+	+
26	-1	$-x^6 - x^5 - 3x^4 - 7x^2 - x - 6$	+	+
27	-1	$-x^6 - 4x^5 - 3x^4 - x^3 - 4x^2 - 5x - 1$	+	+
28	1	$-x^6 + 7x^5 - 12x^4 + 8x^3 - 4x^2 + 9x + 1$	+	+
29	1	$-x^6 + 4x^5 - 5x^3 - 4x^2 + 8x + 6$	+	+
30	1	$-x^6 + 4x^5 - 6x^4 + 7x^3 - 10x^2 + 8x - 4$	+	+
31	1	$-x^6 + x^5 + 6x^4 - 16x^2 + x + 17$	+	+
32	1	$-x^6 + x^5 - 6x^4 - 16x^2 + x - 15$	+	+
33	1	$-x^6 - 2x^5 - 6x^4 - 7x^3 - 10x^2 - 6x - 4$	+	+
34	-1	$-x^6 + 5x^5 - 12x^4 + 8x^3 - 4x^2 + 7x - 1$	+	+
35	-1	$-x^6 + 2x^5 - 5x^3 - 4x^2 + 6x + 4$	+	+
36	-1	$-x^6 + 2x^5 - 6x^4 + 7x^3 - 10x^2 + 6x - 6$	+	+
37	-1	$-x^6 - x^5 + 6x^4 - 16x^2 - x + 15$	+	+
38	-1	$-x^6 - 4x^5 + 5x^3 - 4x^2 - 8x + 4$	+	+
39	-1	$-x^6 - 4x^5 - 6x^4 - 7x^3 - 10x^2 - 8x - 6$	+	+
40	-2	$-x^6 - 5x^5 - 3x^4 + x^3 - 16x^2 + 8x - 8$	+	+
41	-2	$-x^6 - 8x^5 + 3x^4 - 19x^2 - 8x + 9$	+	+
42	2	$-x^6 + 8x^5 - 3x^4 - 19x^2 + 8x - 9$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of $(B_p^{(g)})_g$.

p	k	value	classes (a, b)
2	5	(1, -1)	(0, 11), (8, 19), (16, 27), (24, 3)
2	6	(1, -1)	(4, 63), (12, 7), (20, 15), (28, 23), (36, 31), (44, 39), (52, 47), (60, 55)
2	6	(-1, 1)	(1, 9), (14, 5), (17, 41), (26, 1), (33, 9), (49, 41), (50, 25), (61, 17)
2	6	(-1, -1)	(15, 53), (19, 45), (31, 21), (43, 61), (47, 53), (51, 45), (63, 21)
2	7	(-1, 1)	(2, 41), (5, 65), (10, 49), (18, 57), (21, 33), (25, 25), (29, 81), (30, 21), (34, 73), (37, 1), (42, 81), (46, 37), (53, 97), (57, 89), (58, 97), (62, 53), (66, 105), (69, 65), (74, 113), (82, 121), (85, 33), (89, 25), (93, 81), (94, 85), (98, 9), (101, 1), (106, 17), (110, 101), (117, 97), (121, 89), (122, 33), (126, 117)
2	7	(-1, -1)	(3, 77), (7, 69), (11, 125), (23, 37), (35, 13), (39, 5), (55, 101), (67, 77), (71, 69), (75, 125), (87, 37), (99, 13), (103, 5), (119, 101)
2	8	(-1, 1)	(6, 61), (9, 57), (38, 93), (41, 249), (70, 125), (73, 185), (86, 141), (102, 157), (105, 121), (134, 189), (137, 57), (166, 221), (169, 249), (182, 237), (198, 253), (201, 185), (230, 29), (233, 121)
2	9	(-1, 1)	(13, 241), (22, 77), (45, 433), (54, 109), (109, 305), (118, 173), (150, 205), (173, 177), (205, 369), (214, 269), (237, 49), (246, 301), (278, 333), (301, 433), (310, 365), (365, 305), (374, 429), (406, 461), (429, 177), (470, 13), (493, 49), (502, 45)
2	9	(-1, -1)	(59, 285), (123, 157), (187, 29), (251, 413), (315, 285), (379, 157), (411, 349), (443, 29), (475, 221), (507, 413)
2	10	(-1, 1)	(77, 625), (141, 1009), (269, 753), (333, 113), (397, 497), (461, 881), (589, 625), (653, 1009), (781, 753), (845, 113), (909, 497), (973, 881)
2	10	(-1, -1)	(27, 605), (91, 989), (155, 349), (219, 733), (283, 93), (347, 477), (539, 605), (603, 989), (667, 349), (731, 733), (795, 93), (859, 477)
3	2	(1, 1)	(8, 5)

p	k	value	classes (a, b)
3	3	(1, 1)	(0, 4), (1, 6), (2, 14), (3, 19), (6, 7), (7, 15), (9, 22), (10, 6), (14, 23), (16, 15), (18, 13), (19, 6), (21, 1), (22, 6), (23, 23), (25, 15)
3	4	(1, 1)	(4, 33), (5, 23), (11, 14), (12, 10), (13, 6), (20, 14), (24, 43), (31, 33), (32, 23), (38, 14), (39, 64), (40, 6), (42, 52), (47, 14), (51, 16), (58, 33), (59, 23), (66, 37), (67, 6), (69, 25), (74, 14)
3	5	(1, 1)	(65, 176), (78, 70), (96, 160), (146, 176), (159, 232), (177, 79), (227, 176), (240, 151)
3	6	(1, 1)	(15, 241), (258, 727), (501, 484)
5	2	(1, 1)	(0, 13), (2, 6), (2, 9), (2, 10), (4, 7), (5, 8), (7, 15), (9, 2), (10, 3), (12, 1), (12, 4), (12, 20), (14, 22), (17, 0), (17, 11), (19, 17), (20, 18), (22, 21), (24, 12)
5	3	(1, 1)	(7, 19), (7, 91), (15, 123), (17, 114), (22, 49), (22, 55), (32, 16), (32, 69), (40, 98), (42, 39), (47, 80), (47, 99), (57, 119), (65, 73), (67, 89), (72, 24), (72, 105), (82, 44), (82, 116), (90, 48), (92, 14), (97, 5), (97, 74), (107, 41), (107, 94), (115, 23), (117, 64), (122, 30)
5	4	(1, 1)	(57, 441), (122, 374), (182, 66), (247, 624), (307, 316), (372, 249), (432, 566), (497, 499), (557, 191), (622, 124)
7	1	(1, 1)	(1, 3)
7	2	(1, 1)	(2, 28), (4, 0), (5, 1), (5, 8), (5, 15), (5, 22), (5, 29), (5, 33), (5, 36), (5, 43), (9, 21), (10, 5), (11, 0), (12, 19), (16, 14), (17, 5), (18, 0), (19, 5), (23, 7), (24, 5), (26, 40), (30, 0), (31, 5), (32, 0), (33, 26), (37, 42), (38, 5), (40, 12), (44, 35), (45, 5), (46, 0)
7	3	(1, 1)	(3, 152), (25, 147), (39, 49), (47, 96), (52, 152), (74, 147), (88, 49), (96, 341), (101, 152), (123, 147), (137, 49), (145, 243), (150, 152), (172, 147), (186, 49), (194, 145), (199, 152), (221, 147), (235, 49), (243, 47), (248, 152), (270, 147), (284, 49), (292, 292), (297, 152), (319, 147), (333, 49), (341, 194)
11	1	(1, 1)	(1, 7), (2, 6), (3, 3), (3, 9), (6, 2), (9, 0)
11	2	(1, 1)	(4, 8), (4, 35), (4, 78), (6, 12), (6, 118), (8, 26), (14, 87), (15, 1), (15, 24), (15, 96), (17, 8), (17, 78), (19, 103), (25, 54), (26, 13), (26, 45), (26, 63), (28, 19), (28, 23), (30, 59), (36, 21), (37, 2), (37, 30), (37, 89), (39, 30), (39, 89), (41, 15), (47, 109), (48, 12), (48, 112), (48, 118), (50, 34), (50, 41), (52, 92), (58, 76), (59, 56), (59, 85), (59, 101), (61, 52), (61, 100), (63, 48), (69, 43), (70, 52), (70, 90), (70, 100), (72, 45), (72, 63), (74, 4), (80, 10), (81, 19), (81, 23), (81, 79), (83, 111), (85, 81), (91, 98), (92, 67), (92, 68), (92, 107), (94, 56), (94, 85), (96, 37), (102, 65), (103, 57), (103, 74), (103, 111), (105, 1), (105, 96), (107, 114), (113, 32), (114, 34), (114, 41), (114, 46), (116, 67), (116, 107), (118, 70)
11	3	(1, 1)	(3, 725), (83, 921), (124, 362), (204, 1042), (245, 1330), (325, 1163), (366, 967), (446, 1284), (487, 604), (567, 74), (608, 241), (688, 195), (729, 1209), (809, 316), (850, 846), (930, 437), (971, 483), (1051, 558), (1092, 120), (1172, 679), (1213, 1088), (1293, 800)
13	1	(1, 1)	(3, 0), (3, 3), (3, 10), (4, 9), (5, 5), (7, 7), (9, 0), (9, 3), (9, 10), (10, 8), (10, 9), (11, 7)
13	2	(1, 1)	(1, 115), (4, 125), (5, 4), (5, 17), (5, 30), (5, 43), (5, 56), (5, 69), (5, 82), (5, 95), (5, 108), (5, 121), (5, 134), (5, 147), (5, 160), (14, 89), (17, 34), (27, 63), (30, 112), (40, 37), (43, 21), (53, 11), (56, 99), (66, 154), (69, 8), (79, 128), (82, 86), (92, 102), (95, 164), (105, 76), (108, 73), (118, 50), (121, 151), (131, 24), (134, 60), (144, 167), (147, 138), (157, 141), (160, 47)
17	1	(1, 1)	(0, 2), (1, 7), (2, 2), (4, 12), (5, 7), (6, 10), (7, 11), (8, 2), (9, 2), (10, 0), (11, 14), (11, 16), (12, 15), (16, 15)

p	k	value	classes (a, b)
17	2	(1, 1)	(3, 12), (10, 42), (10, 179), (11, 123), (15, 206), (20, 199), (27, 43), (27, 195), (32, 138), (37, 97), (44, 59), (44, 196), (45, 276), (49, 70), (54, 284), (61, 60), (61, 212), (62, 208), (66, 2), (71, 182), (78, 76), (78, 213), (79, 140), (83, 223), (88, 80), (95, 77), (95, 229), (96, 72), (100, 155), (105, 267), (112, 93), (113, 4), (117, 87), (122, 165), (129, 94), (129, 246), (130, 225), (134, 19), (139, 63), (146, 110), (146, 247), (147, 157), (151, 240), (156, 250), (163, 111), (163, 263), (164, 89), (168, 172), (173, 148), (180, 127), (180, 264), (181, 21), (185, 104), (190, 46), (197, 128), (197, 280), (198, 242), (202, 36), (207, 233), (214, 144), (214, 281), (215, 174), (219, 257), (224, 131), (231, 8), (231, 145), (232, 106), (236, 189), (241, 29), (248, 9), (248, 161), (249, 38), (253, 121), (258, 216), (265, 25), (265, 162), (266, 259), (270, 53), (275, 114), (282, 26), (282, 178), (283, 191), (287, 274)
17	3	(1, 1)	(28, 344), (112, 4276), (317, 4101), (401, 1964), (606, 2945), (690, 4565), (895, 1789), (979, 2253), (1184, 633), (1268, 4854), (1473, 4390), (1557, 2542), (1762, 3234), (1846, 230), (2051, 2078), (2135, 2831), (2340, 922), (2424, 519), (2629, 4679), (2713, 3120), (2918, 3523), (3002, 808), (3207, 2367), (3291, 3409), (3496, 1211), (3580, 1097), (3785, 55), (3869, 3698), (4074, 3812), (4158, 1386), (4363, 2656), (4447, 3987), (4652, 1500), (4736, 1675)